

To my family.

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Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). The same construction underlies the image and video generators deployed at scale by the large technology companies, from latent diffusion (Rombach et al., 2022) onwards, and reaches beyond media: structure prediction in AlphaFold 3 places atomic coordinates with a diffusion module (Abramson et al., 2024). Their central object is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x, t) = \nabla_x \log p_t(x), \tag{1.1}$$

which is the one object a model must learn in order to reverse a noising process: given the forward process, its schedule and its terminal law, the score determines the reverse drift (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of the function it has to learn remains poorly understood.

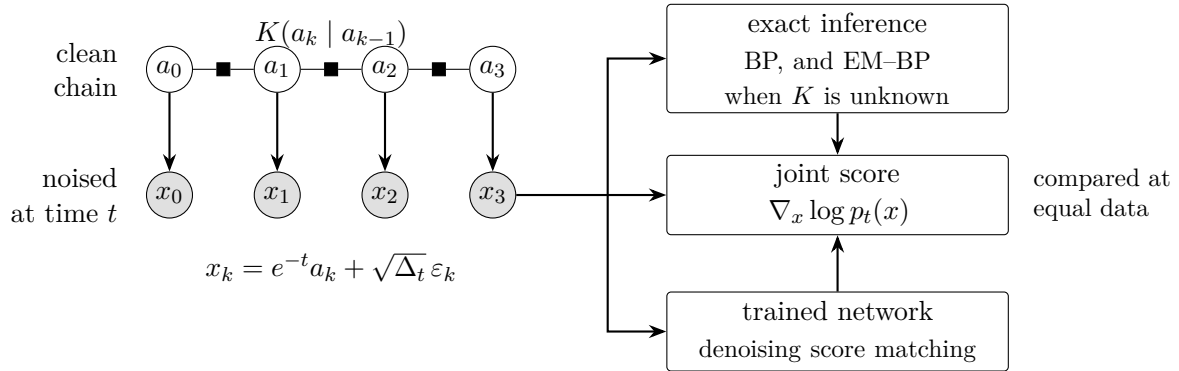


Figure 1.1: The object of the thesis and the two ways it is obtained. A Markov chain generates a clean trajectory; the Ornstein–Uhlenbeck channel corrupts every coordinate independently, so each observation attaches to the latent chain on its own and the posterior stays a chain. That is what makes the joint score exactly computable by message passing, with the transition kernel known (Chapters 5 and 7), or estimated from noised sequences alone when it is not (Chapter 8). The same score is what a denoising-score-matching network is trained to approximate without being given the chain structure. Chapter 10 measures what supplying that structure is worth, at equal training data, and where it stops paying.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems, such as speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024; Achilli et al., 2026). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis works inside that programme, in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \dots, a_{L-1}$ of a Markov process, the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Gaussian Markov chain is governed by the algebra of structured matrices: Toeplitz covariances, tridiagonal precisions, and their common eigenbasis. Second, by an identity that holds for any data distribution, computing a score is solving a Bayesian

inference problem, and for chain-structured data that problem lives on a tree-shaped factor graph, the domain of *belief propagation* (BP) and of the Kalman filter, together with the Gaussian message closures of statistical physics (TAP, AMP) (Pearl, 1988; Mézard and Montanari, 2009; Kalman, 1960; Donoho et al., 2009). The thesis develops both routes in full, proves where they coincide, and measures what is lost when they are forced apart.

1.2 Positioning and gap

Chapter 2 reviews the literature this work builds on; the gap it leaves is stated here.

Three strands meet. The statistical physics of generative diffusion has produced closed-form analyses of the backward dynamics and their transitions (Biroli and Mézard, 2023; Biroli et al., 2024; Achilli et al., 2026; Sclocchi et al., 2025b; Bonnaire et al., 2025). Exact posterior inference on Gauss–Markov chains is classical, through the Kalman filter and smoother and their formulation as sum–product message passing (Kalman, 1960; Rauch et al., 1965; Kschischang et al., 2001; Mézard and Montanari, 2009). And message passing with continuous variables closes its messages inside a parametric family, which is the step connecting belief propagation to AMP and to the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016).

Each strand stops short of the same place. The physics analyses treat data as unstructured collections of samples: the only time axis they carry is the diffusion time, and the *internal* time of a sequence, a frame correlated with its neighbours by a dynamical rule, does not appear. The inference literature has the machinery for that internal time but does not ask what generative modelling forces it to ask, which is how the exact posterior, and so the exact score, deforms as the noise level is swept from zero to saturation. And the standard justification for closing messages in a Gaussian family is a central-limit argument that requires each node to receive many messages; on a chain each node receives exactly two, so the argument fails by construction and we are not aware of a quantitative assessment of what the closure costs on chain-structured diffusion posteriors.

The gap this thesis addresses is the intersection: the joint score of a noised Markov

sequence, computed exactly where that is possible, approximated in a controlled and measured way where it is not, and learned from noised data when the dynamics are unknown.

One methodological choice follows from the same reasoning and is worth stating early. Where no closed form exists, the reference used throughout is deterministic quadrature and not Monte Carlo. A particle smoother carries sampling error, which would add a second error term to every comparison and force the cost of an approximation and the cost of the reference to be disentangled before either could be read. Grid-based inference carries a discretisation error instead, which can be bounded separately and reported, and that is what makes it usable as a reference and not merely as another competitor.

1.3 Research questions

The thesis addresses six questions. The first four concern scores under *known* dynamics and the last two concern learning dynamics that are unknown.

- RQ1.** For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?
- RQ2.** How can the same Gaussian-chain score be computed by belief propagation, and how does the resulting recursion relate to Kalman filtering and smoothing?
- RQ3.** For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?
- RQ4.** In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?

- RQ5.** Under what assumptions can an *unknown* local transition law be recovered from noised sequences alone, and which numerical, optimisation and statistical checks are required before such a recovery claim is reliable?
- RQ6.** On a family where the Markov assumption holds exactly, what is supplying that assumption worth, measured against denoising-score-matching networks trained on identical data and given none of it?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion scores, or about practical neural architectures. In particular RQ6 is a question about the value of a correct structural assumption on one family, not a comparison of learning algorithms in general: the two arms are asymmetric in supplied information by construction, which is what makes the measurement meaningful and also what bounds it.

Chapter 4 answers no numbered question. It comes first because it is where the work began, and because what it settles motivates the six: on a rotating-ring model, whose state space and estimand both differ from the chain's, it shows what joint observations can identify that no single-frame marginal can.

1.4 Contributions

Four contributions, in the order the body develops them. Each is stated here as a claim and a kind of claim; the derivations are in the chapters named and the measured values are reported there and in Chapter 11.

1. **The joint score of a noised Gaussian Markov chain, in closed form.** For the stationary AR(1) chain under coordinatewise OU corruption we derive the score, diagonalise it in a single eigenbasis valid at every diffusion time, and give the law by which its couplings fill in at small t and decay back to isotropy at large t , together with exact bulk formulas for the posterior variance and the correlation decay rate $q(\alpha, t)$ (Chapter 5). Computing the same score by message passing, we show

that belief propagation on the chain is the Kalman filter followed by Rauch–Tung–Striebel smoothing, update by update, and prove that the influence of evidence at distance d falls as $q(\alpha, t)^d$. That the radius- r windowed estimator’s error falls at the same rate is established separately, and by measurement, because windowing changes the boundary conditions of the recursion in a way the influence argument does not cover (Chapter 6).

2. **An exact reading of what Gaussian message closure computes.** For Laplace innovations the functional recursion stays exact on the chain but its messages no longer close in any finite-dimensional family. We build a numerically validated quadrature reference, and prove that moment-projected messages return exactly the LMMSE estimator of the covariance-matched Gaussian model (Theorem 7.6). The measured departure is therefore attributable: it is what discarding higher moments costs, for every innovation law with the same first two moments, and not what one implementation costs (Chapter 7).
3. **Recovery of an unknown transition law from noised sequences.** Dropping the assumption that the kernel is known, we show that Fisher’s identity makes one forward–backward pass yield the exact gradient of the marginal log-likelihood, so no derivative is taken through the recursion and the E-step stays structurally exact for any innovation law (Chapter 8). Measuring what convergence costs then produces a result rather than a diagnostic: the correlation and the innovation shape settle at very different rates, so a stopping rule read off the first certifies a kernel unconverged in the second (Chapter 9).
4. **The measured value of supplied Markov structure, and its limit.** At equal training data the structured estimator is substantially more accurate than networks trained by denoising score matching, and remains so against a baseline given locality and weight sharing by an architecture screen. The contribution is as much the boundary as the margin: controlled departures from the Markov assumption locate where the advantage ends and the networks overtake it (Chapter 10).

1.5 Structure of the thesis

Chapter 2 is the context: statistical mechanics and the variational principle behind the Boltzmann distribution, energy-based models as the same object with a learned energy, generative diffusion and the reverse-time SDE, and then the distinction the rest of the thesis turns on, between the exact, the empirical and the fitted score. It closes with the reading of U-Nets as belief propagation that makes exact inference a serious alternative to a trained network rather than a toy one. Chapter 3 fixes the model and the estimand: the Markov sequence, the Ornstein–Uhlenbeck channel, the argument that the joint score and not a per-frame score is the object worth studying, and the factor-graph language the computations are written in.

Chapter 4 begins the research where the research began, with a planar trajectory near a circle. It shows that the rotation rate carries zero Fisher information in every one-frame marginal and positive information in the trajectory. That is the argument for the joint score in a single exact statement, and it is what sends the following chapters to a model where the joint score can be computed in full.

Chapters 5 and 6 do that for the stationary Gaussian chain, first with matrices and then with messages: the score in closed form, the lifecycle of its couplings, the identification of belief propagation with the Kalman filter and the Rauch–Tung–Striebel smoother, and the geometric decay that says how local an accurate approximation can be (RQ1, RQ2, RQ4). Chapter 7 leaves Gaussianity in the most controlled way available, through Laplace innovations, builds a validated numerical reference, and measures exactly what a Gaussian message family discards (RQ3).

Chapters 8 and 9 drop the assumption that the transition law is known and recover it from noised sequences alone, first two parameters and then a whole kernel, with the convergence schedule of the two coordinates as a result in its own right (RQ5). Chapter 10 measures what supplying the Markov structure is worth against networks trained on identical data, and locates where it stops being worth anything (RQ6). Chapter 11 answers the six questions and states what is left open.

Each concept is developed once, in the chapter where it is used. The appendices carry the Gaussian identities, the ring derivations, the reproducibility protocol and the aggregation robustness of the reported ratios. Extended background, alternative derivations, the AMP/TAP analysis and the development record are in the companion compendium.

Chapter 2

Statistical Mechanics, Energy-Based Models, and Diffusion

2.1 What statistical mechanics is, and why it is here

Newton's second law is deterministic and complete. Given the positions and momenta of N particles and the forces between them, the trajectory of the system is fixed for all time. For N of order 10^{23} this completeness is useless: the initial condition cannot be written down, the equations cannot be integrated, and the answer, if it were available, would be a list of 6×10^{23} numbers and not an explanation of anything. Yet the gas in a room has a temperature, a pressure and a heat capacity, and these are stable, reproducible, and predictable from a handful of parameters.

Statistical mechanics is the resolution of that tension. It abandons the trajectory and keeps a probability distribution over microstates, and it fixes that distribution by a variational principle: among all distributions with a given mean energy, the one realised at equilibrium maximises the entropy

$$S[p] = - \sum_x p(x) \log p(x), \tag{2.1}$$

which is the unique measure of uncertainty consistent with a short list of consistency

axioms (Jaynes, 1957). Carrying out the maximisation subject to $\sum_x p(x)E(x) = \bar{E}$ produces the Boltzmann distribution

$$p(x) = \frac{1}{Z(\beta)} e^{-\beta E(x)}, \quad Z(\beta) = \sum_x e^{-\beta E(x)}, \quad (2.2)$$

with the inverse temperature β appearing as the Lagrange multiplier of the energy constraint (Landau and Lifshitz, 1980; Goldstein et al., 2002). Everything thermodynamic follows from the normalising constant: the free energy is $F = -\beta^{-1} \log Z$, and its derivatives give the energy, the entropy and the response functions.

Two features of this construction matter for what follows, and neither is specific to physics. The first is that the whole of the macroscopic behaviour is encoded in a single scalar function of the parameters, $\log Z$, which is a sum over exponentially many configurations and is therefore hard to compute. The second is that competition between energy, which orders, and entropy, which disorders, can produce *phase transitions*: the equilibrium state can change character abruptly as a parameter is varied, even though the microscopic law does not change at all. A system with a symmetry can settle into a state that lacks it, and which of the equivalent states it settles into is decided by fluctuations. The canonical example is the Ising model (Ising, 1925), whose two-dimensional case Onsager (1944) solved exactly, and which remains the reference picture for a transition driven by the competition between energy and entropy. Both features will reappear, unchanged in substance, in the generative models of the next sections.

2.2 Energy-based models: the same object, learned

Traffic between statistical mechanics and learning machines runs in both directions and is old. The Hopfield network stored memories as minima of an energy (Hopfield, 1982); its capacity was settled by spin-glass methods (Amit et al., 1985; Mézard et al., 1987; Gardner, 1988); and the Boltzmann machine and its restricted form made the Boltzmann distribution itself the model (Ackley et al., 1985; Smolensky, 1986). The 2024 Nobel

Prize in Physics recognised that lineage (The Royal Swedish Academy of Sciences, 2024). What follows takes (2.2) and reads it in the opposite direction. Rather than deriving a distribution from a known energy, it posits a parametric energy E_θ and fits it so that the induced distribution matches data (Ackley et al., 1985; LeCun et al., 2006):

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \quad Z(\theta) = \int_{\mathbb{R}^d} e^{-E_\theta(x)} dx. \quad (2.3)$$

The microstate becomes a data point, the energy becomes a learnable scoring function, and the partition function keeps its role and its difficulty. Fitting by maximum likelihood gives a gradient with the structure of a competition,

$$\nabla_\theta \mathbb{E}_{x \sim p_{\text{data}}} [-\log p_\theta(x)] = \mathbb{E}_{x^+ \sim p_{\text{data}}} [\nabla_\theta E_\theta(x^+)] - \mathbb{E}_{x^- \sim p_\theta} [\nabla_\theta E_\theta(x^-)], \quad (2.4)$$

in which the first term lowers the energy of observed data and the second raises the energy of the model's own samples. The second expectation is over p_θ , which cannot be sampled directly, so training requires Markov chain Monte Carlo, typically Langevin dynamics, whose stationary measure is $\propto e^{-E_\theta}$ (Hinton, 2002; Robert and Casella, 2004). This is expensive, and its cost is the reason the family was, for a long time, outperformed by models that give up the explicit density in exchange for cheap sampling (Goodfellow et al., 2014; Kingma and Welling, 2014; Papamakarios et al., 2021). Carbone (2025) surveys how the energy-based view relates to the rest of the generative family.

The learning dynamics of these models are themselves a statistical mechanics problem, and one with a structure worth carrying forward. The spectral dynamics of the weights (Decelle et al., 2017; Mézard, 2017) and the emergence of compositional features (Tubiana and Monasson, 2017) are both statistical-mechanics accounts of training. Bachtis et al. (2024) show that training a restricted Boltzmann machine proceeds through a *cascade of second-order phase transitions*: the model first encodes a coarse, centre-of-mass feature, and then resolves progressively finer sub-modes, each emergence accompanied by a diverging susceptibility. The features are learned in order, from the largest scale down. Section 2.5 finds the same ordering in a different process, and Chapter 3 onwards works

in a model where the ordering is analytic.

The connection to what follows is direct. Differentiating (2.3) gives

$$\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x), \quad (2.5)$$

and the intractable $Z(\theta)$, being independent of x , has vanished. The gradient of the log-density carries all the information needed to *sample* the model while requiring none of the information needed to *normalise* it. That observation is the whole of the following section.

2.3 Generative diffusion

A diffusion model destroys structure in a controlled way and then learns to run the destruction backwards (Sohl-Dickstein et al., 2015; Song and Ermon, 2019; Ho et al., 2020; Song et al., 2021). Its deterministic counterparts, neural ODEs and flow matching (Chen et al., 2018; Lipman et al., 2023), transport the same measures along a probability-flow trajectory instead of a stochastic one. The forward process is a stochastic differential equation

$$dX_t = f(X_t, t) dt + g(t) dW_t, \quad (2.6)$$

started at $X_0 \sim p_0$, the data law, and producing marginals p_t that approach a simple terminal measure. Throughout this thesis the choice is the Ornstein–Uhlenbeck channel, $f(x, t) = -x$ and $g = \sqrt{2}$, whose transition law is available in closed form,

$$x = e^{-t}a + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad \Delta_t = 1 - e^{-2t}, \quad (2.7)$$

and whose stationary law is the standard Gaussian (Uhlenbeck and Ornstein, 1930; Gardiner, 2009). Figure 2.1 shows what (2.7) does to a correlated sequence: the amplitude contracts, independent noise accumulates, and by $t \approx 2$ the trajectory is indistinguishable from white noise.

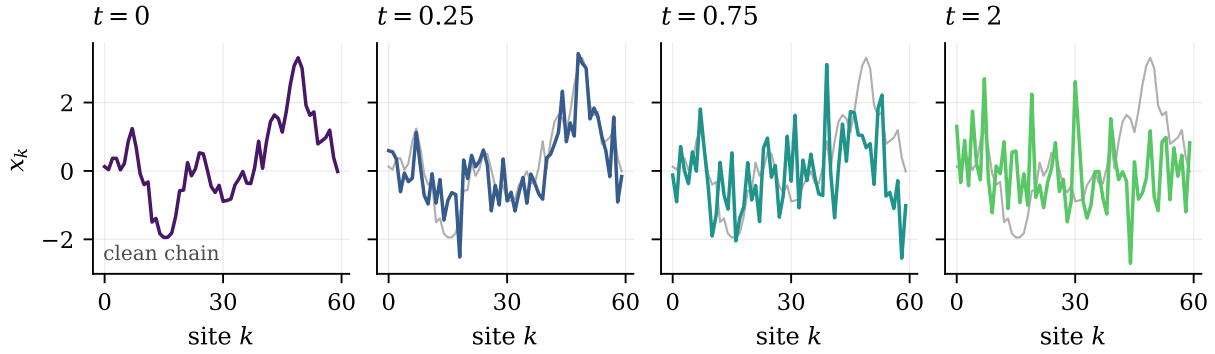


Figure 2.1: One clean stationary AR(1) trajectory with $\alpha = 0.85$, in grey, and the same trajectory after the Ornstein–Uhlenbeck channel (2.7) at four diffusion times. The signal contracts by e^{-t} while independent noise of variance Δ_t accumulates; correlation between neighbouring sites survives longer than amplitude. Generation runs this figure from right to left, and the score is the only quantity needed to do so.

The reversal is a classical result. [Anderson \(1982\)](#) shows that the process

$$dX_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t) \right] dt + g(t) d\bar{W}_t, \quad (2.8)$$

run backwards from T with $X_T \sim p_T$, reproduces the forward marginals. Every ingredient of (2.8) is known by construction except one. The drift correction is the *score* $\nabla_x \log p_t$, and supplying it is the entire modelling effort of a diffusion model. A Fokker–Planck argument for (2.8) and the probability-flow ODE it induces are in the companion compendium; a book-length treatment of the model family is [Lai et al. \(2025\)](#).

By (2.5) the score is the negative gradient of an energy, so a diffusion model is an energy-based model that never forms its partition function and never runs an inner Monte Carlo loop. What it does instead is train against a target that is available in closed form, which is the content of the identity in the next section.

2.4 Three scores

The word “score” is used for three different objects in this literature, and almost every difficulty in generative diffusion is a property of the gaps between them. The distinction is drawn sharply in [Biroli and Mézard \(2023\)](#) and organises what follows.

The exact score. The score of the true noised law, $\nabla_x \log p_t$ with $p_t = p_0 * \mathcal{N}(0, \Delta_t I)$ after contraction. It is the object (2.8) calls for. It has a form that will be used on every page of this thesis: writing p_t as the channel convolution and differentiating under the integral gives

$$\boxed{\nabla_x \log p_t(x) = \frac{e^{-t} \mathbb{E}[a | x] - x}{\Delta_t} \iff \mathbb{E}[a | x] = e^t (x + \Delta_t \nabla_x \log p_t(x)).} \quad (2.9)$$

We call (2.9) the *score-posterior identity*; in the empirical-Bayes literature it is Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011; Vincent, 2011). It holds for any p_0 whatever, and it says something stronger than an analogy: computing a score *is* solving a Bayesian inference problem. The exact score is unavailable in practice for the plain reason that p_0 is unknown.

For *vector-valued* data $a \in \mathbb{R}^L$ noised coordinatewise, (2.9) holds componentwise but with the *full* posterior:

$$\left(\nabla_x \log p_t(x) \right)_k = \frac{e^{-t} \mathbb{E}[a_k | x_0, \dots, x_{L-1}] - x_k}{\Delta_t}. \quad (2.10)$$

The conditioning is on *every* coordinate. The channel is independent across frames, but the prior is not, so the posterior mean of frame k , and with it the k -th component of the score, depends on all L observations. Equation (2.10) is the exact sense in which the joint score of a trajectory differs from a stack of per-frame scores, and the reason this thesis studies the former.

The empirical score. Replace p_0 by the empirical measure of a training set $\{a^{(i)}\}_{i=1}^N$, which puts mass $1/N$ on each example. The convolution is then a finite Gaussian mixture and the score is available in closed form,

$$\nabla_x \log p_t^{\text{emp}}(x) = \frac{1}{\Delta_t} \left(e^{-t} \sum_{i=1}^N w_i(x) a^{(i)} - x \right), \quad w_i(x) = \frac{e^{-\|x - e^{-t} a^{(i)}\|^2 / 2\Delta_t}}{\sum_j e^{-\|x - e^{-t} a^{(j)}\|^2 / 2\Delta_t}}, \quad (2.11)$$

a softmax-weighted average of training points. This is exactly computable and exactly useless as a generative model: as $t \rightarrow 0$ the weights concentrate on the nearest training example and the reverse dynamics deposit the trajectory on top of it. Running (2.8) with (2.11) returns the training set. Figure 2.2 shows the mechanism in one dimension.

The fitted score. A neural network $s_\phi(x, t)$ trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011) to minimise

$$\mathbb{E}_t \mathbb{E}_{a \sim p_{\text{data}}} \mathbb{E}_\varepsilon \left\| s_\phi(e^{-t}a + \sqrt{\Delta_t} \varepsilon, t) + \frac{\varepsilon}{\sqrt{\Delta_t}} \right\|^2. \quad (2.12)$$

The minimiser over all measurable functions is the exact score; the minimiser over a finite training set is the empirical score. Diffusion models work because the network reaches neither. It is constrained enough to be unable to represent (2.11), and that failure to fit is what produces novel samples (Kadkhodaie et al., 2024; Kamb and Ganguli, 2024). The mechanism of generalisation is therefore an approximation-theoretic property of an architecture, which is a considerably less comfortable foundation than the variational principle of Section 2.1.

2.5 Transitions along the generation trajectory

Because generation is a dynamical process, its failures are dated. Two crossover times structure the reverse trajectory, and both are statistical mechanics results in the sense of Section 2.1: a competition between an entropic and an energetic term, resolved abruptly.

Speciation. Run backwards from noise, the trajectory commits to a broad class of outcomes long before it commits to a specific sample. For data with a covariance whose top eigenvalue is λ , the commitment happens at $t_S \sim \frac{1}{2} \log \lambda$, when the contracted signal along that direction becomes comparable with the accumulated noise (Biroli et al., 2024; Achilli et al., 2026). In high dimension the top eigenvalue grows with the dimension, so t_S diverges logarithmically. When the data are hierarchical there is not one such time

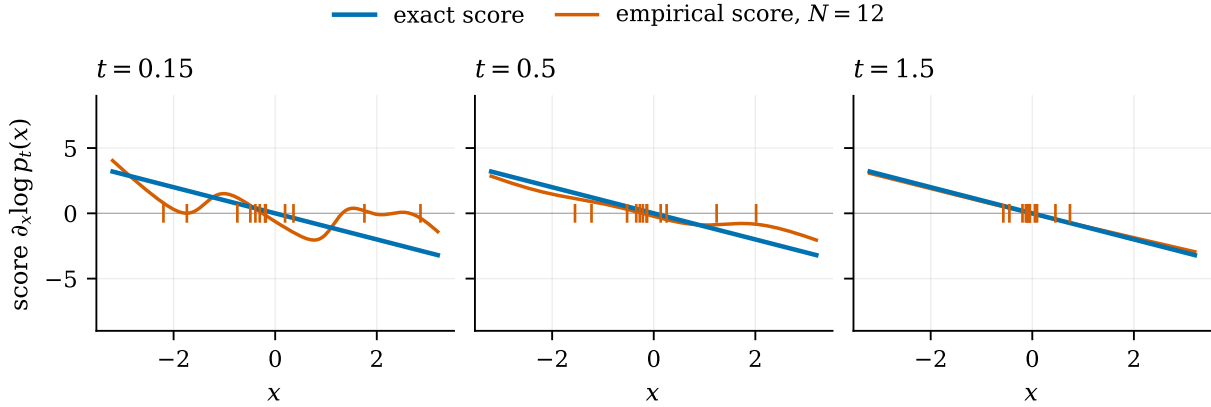


Figure 2.2: The exact score of a standard Gaussian, which is linear in x , against the empirical score (2.11) built from $N = 12$ samples of the same law; ticks mark the contracted training points $e^{-t}a^{(i)}$. At small t the empirical score is a sawtooth that points at whichever training example is nearest, and reverse dynamics driven by it terminate on one of them. As t grows the weights spread, the sawtooth flattens, and the two scores agree. The useful regime for a generative model lies between the two, and its width is what Biroli et al. (2024) quantify.

but a *ladder* of them, one per level of the hierarchy, and features are resolved coarse-to-fine (Sclocchi et al., 2025b,a; Favero et al., 2025a,b): the same ordering that Bachtis et al. (2024) observe in the training of an energy-based model, appearing here along the generation trajectory instead. Figure 2.3 shows the ladder in a Gaussian hierarchical model where every rung can be predicted in closed form from the covariance spectrum and then measured.

Collapse and memorisation. Later in the reverse trajectory a second transition occurs, and it is the one that limits the whole construction. Biroli et al. (2024) and Bonnaire et al. (2025) show that dynamics driven by the empirical score are captured by a single training example past a time t_C , and that avoiding this requires a training set exponentially large in the data dimension. Memorisation need not be a collapse onto one example: Achilli et al. (2024) identify a geometric form in which dimensions are lost without any single training point capturing the trajectory, and Garnier-Brun et al. (2026) show how the resulting generalisation can be biased. Solvable models of the learning problem itself (Cui et al., 2025) complete the picture. The criterion is entropic: collapse occurs when the

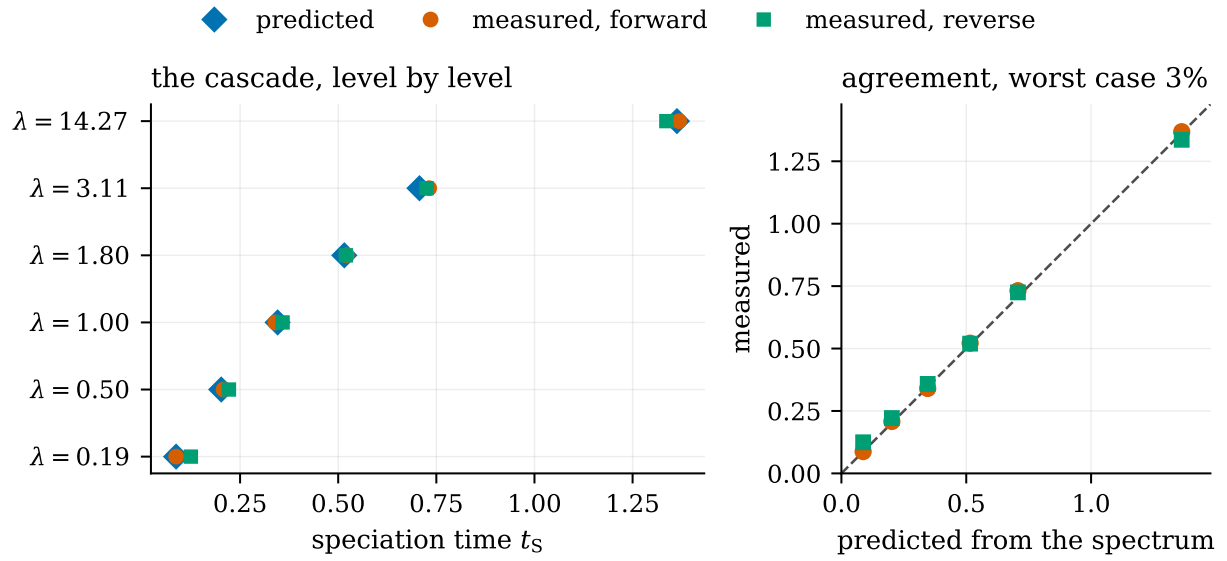


Figure 2.3: A Gaussian hierarchical prior has one covariance eigenvalue per level, and therefore a ladder of speciation times rather than a single one. *Left:* for each level, the time predicted from the eigenvalue alone (diamonds) against the times at which the forward and reverse processes are measured to cross (circles, squares); the grey bar spans the two measurements. *Right:* the same six levels as prediction against measurement, with the identity line. Worst-case relative disagreement across all twelve measurements is 3%. The ordering is coarse-to-fine: the largest eigenvalue resolves first.

excess entropy of the noised law relative to the terminal measure falls below $\log N$, so the dataset size needed to postpone it grows as $e^{(\text{excess entropy})}$. Figure 2.4 measures t_C against N in the chain model of this thesis and compares it with that closed-form criterion.

The two transitions bracket the useful part of the trajectory. Above t_S nothing has been decided; below t_C the empirical score reproduces its training data. A generative model earns its keep in between, and the fitted score of (2.12) is the only one of the three that operates there, for reasons, as noted above, that are properties of an architecture rather than of a principle.

2.6 Locality, architecture, and message passing

If the fitted score is doing the work, the natural question is what the architecture contributes, and here a second literature meets the first. The network in almost every image or video diffusion model is a U-Net (Ronneberger et al., 2015): a convolutional encoder

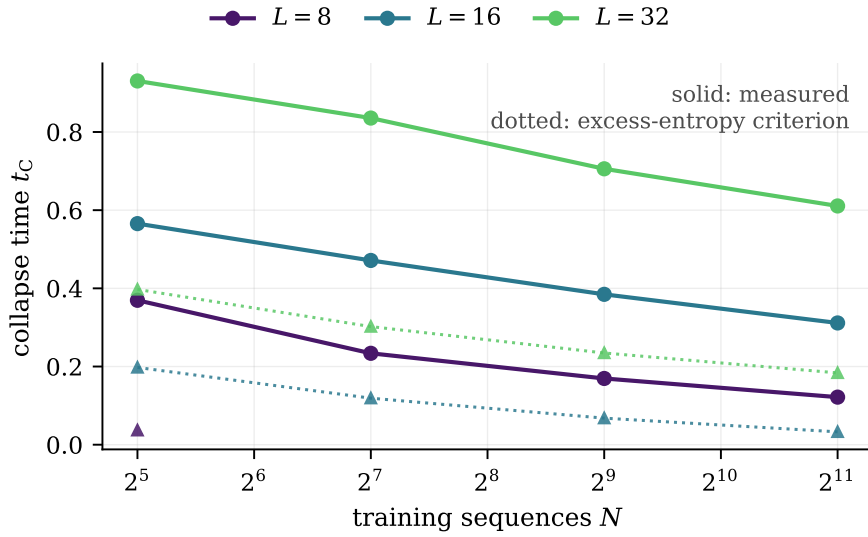


Figure 2.4: Measured collapse time for empirical-score dynamics on the stationary AR(1) chain, at three chain lengths L , against the number of training sequences. Solid: the time at which the generated trajectory’s nearest-training-neighbour distance falls below the distance to a fresh sample. Dotted: the excess-entropy criterion of [Biroli et al. \(2024\)](#), evaluated in closed form for this model, plotted where it has a solution. Postponing collapse costs exponentially many sequences, and the exponent is the excess entropy of the clean law.

that repeatedly downsamples, a decoder that upsamples back, and long skip connections joining each encoder stage to the decoder stage of the same resolution. The design is almost universal and was, for a long time, justified anecdotally.

[Mei \(2024\)](#) gives it a precise reading, and it is the reading this thesis works inside. For data generated by a tree-structured hierarchical model, the exact denoiser is computed by belief propagation, which consists of an upward pass that accumulates evidence from the leaves and a downward pass that redistributes it, the second reusing the messages formed by the first. A U-Net has exactly this shape: the encoder is the upward pass, the decoder is the downward pass, and the long skip connections are the reuse of the intermediate messages. On this account a U-Net is not merely a good inductive bias for denoising but an *approximate implementation of the exact inference algorithm* for hierarchically structured data, which yields sample complexity bounds for learning the denoiser. Convolutional classifiers admit the corresponding reading for the upward pass alone.

Two consequences frame this thesis. First, if the architecture is an inference algorithm, then running the inference algorithm directly is a legitimate alternative and not a toy substitute, and where the graph is a chain rather than a tree of higher degree, that algorithm is exact and cheap. Second, the question of *how large a receptive field a score network needs* becomes quantitative: it is set by how fast the influence of distant evidence decays in the corresponding inference problem. Chapter 6 answers that question in closed form for the chain. A related result on sequences is Garnier-Brun et al. (2024), who find that a transformer trained on hierarchically generated sequences reconstructs correlations sequentially, shortest length scale first, and who use exact belief propagation as the reference against which the network is measured.

2.7 Where this thesis sits

The three scores of Section 2.4 exhaust the options usually considered: one is unavailable, one memorises, and one works for reasons that are hard to state. This thesis studies a fourth.

For data with sequential structure, a trajectory of a Markov process noised frame by frame, the score-posterior identity (2.9) turns the score into a posterior expectation on a chain-structured factor graph, and that posterior is computable exactly by belief propagation. Call the result the *BP score*. It differs from the other three in a way that is structural rather than a matter of degree:

- Unlike the exact score, it is computable, because the model class is specified and the graph is a tree.
- Unlike the empirical score, its sufficient statistic is a transition kernel, two numbers for a Gaussian AR(1) chain or a fixed grid for a nonparametric one, and *not* the training set. The data enter only through an average. There is no training point for a trajectory to collapse onto, so the memorisation axis of Section 2.5 does not exist for it at any N .

- Unlike the fitted score, it is transparent: every quantity in it has a closed form or a validated numerical reference, so its errors can be attributed rather than observed.

The price is an assumption. The BP score is only correct if the data really are Markov, and the value of the assumption is exactly its correctness: a claim that has to be measured, not asserted. Chapter 4 begins where the research began, with a two-dimensional dynamic object whose behaviour motivated studying the joint score at all; Chapters 5 to 7 compute the BP score with the transition kernel known; Chapters 8 and 9 learn the kernel from noised sequences alone; and Chapter 10 measures what the assumption is worth against fitted scores trained on identical data, and locates where it stops being worth anything.

Chapter 3

The Model, and the Object of Study

This chapter fixes what the thesis computes and the language it computes in. A Markov chain generates a clean trajectory; an Ornstein–Uhlenbeck channel corrupts every frame independently; the object of study is the joint score of the corrupted sequence. Two justifications come before any analysis. The first is that the quantity worth studying is the joint score of a whole trajectory and not a per-frame score; the second is that the stationary AR(1) chain is the right model in which to study it. Both are settled here. The chapter closes with the representation the rest of the thesis reasons in: a distribution written as a product of local factors, and the recursion that returns its marginals exactly when those factors form a tree.

3.1 Markov sequences

Two notions of time homogeneity are used constantly below and have to be kept apart (Brockwell and Davis, 1991). A process (X_k) is *strictly stationary* if every finite-dimensional law is shift-invariant, and *weakly stationary* if its mean is constant and its autocovariance depends only on the lag, $\text{Cov}(X_k, X_{k+d}) = \gamma(d)$. Strict implies weak whenever second moments exist, and for Gaussian processes the two coincide. The distinction earns its place because Chapter 7 replaces the Gaussian innovation with a Laplace one, at which point they separate and only the strict notion survives the construction. Weak stationarity also

has the matrix consequence the research chapters exploit: the covariance of L consecutive observations, $(\Sigma)_{ij} = \gamma(i - j)$, is constant along diagonals. Wherever a covariance in this thesis is Toeplitz, stationarity is the reason.

Definition 3.1 (Markov chain). A sequence $(X_k)_{k \geq 0}$ on a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}(X_{k+1} \in A \mid X_k, X_{k-1}, \dots, X_0) = \mathbb{P}(X_{k+1} \in A \mid X_k). \quad (3.1)$$

The conditional law is encoded by a *transition kernel* $K(x' \mid x)$, with $\int K(x' \mid x) dx' = 1$ for every x .

Example 3.2 (The Gaussian AR(1) chain). The running example throughout is the scalar Gaussian autoregression of order one,

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad k = 0, \dots, L-2, \quad (3.2)$$

with kernel $K(a' \mid a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$. Taking $\sigma_\eta^2 = 1 - \alpha^2$ and $a_0 \sim \mathcal{N}(0, 1)$ makes it stationary with unit marginal variance and $\gamma(d) = \alpha^{|d|}$, so its covariance is Toeplitz. Every chapter from Chapter 5 onwards concerns (3.2), one of its non-Gaussian variants, or what an estimator can recover of it through noise.

The Markov property (3.1) is exactly a statement about *factorisation*: the joint law of the first L states is

$$p(x_0, x_1, \dots, x_{L-1}) = p_0(x_0) \prod_{k=0}^{L-2} K(x_{k+1} \mid x_k). \quad (3.3)$$

This is the most heavily used equation of the thesis. It says the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables: the structure that makes the chain a tree in the factor-graph sense of Chapter 3, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 5, and that makes belief propagation exact. Marginals evolve by integrating the kernel, $p_{k+1}(x') = \int K(x' \mid x) p_k(x) dx$,

and a law π with $\pi = K\pi$ is *stationary*: a chain started in π remains in it, and its trajectory is then strictly stationary.

One change of perspective sets up the research problem. Markov chains are classically studied as *algorithms*, where the stationary law is the target and the trajectory a means to reach it. From Chapter 5 onwards the chain (3.2) plays the opposite role: it is the *data-generating process*, one trajectory $(a_0, \dots, a_{L-1})$ is one data point, and the object of interest is the law of the whole trajectory after every coordinate has been independently corrupted. The factorisation (3.3) then stops being a computational convenience and becomes the structure to be recovered.

3.2 The Ornstein–Uhlenbeck corruption channel

The noising process used by diffusion models, and by every result of this thesis, is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 = a. \quad (3.4)$$

The drift and diffusion constants are the variance-preserving normalisation of the diffusion literature (Song et al., 2021); any other choice rescales time. The process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable. Solving (3.4) by integrating factor, the derivation is routine and is carried out in the companion compendium, gives a Gaussian transition kernel, $p(x_t | X_0 = a) = \mathcal{N}(x_t; ae^{-t}, \Delta_t)$ with $\Delta_t = 1 - e^{-2t}$. Corrupting a data point a for a time t therefore produces

$$x = ae^{-t} + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1) : \quad (3.5)$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$ the channel is the identity; as $t \rightarrow \infty$, $x \sim \mathcal{N}(0, 1)$ regardless of a . The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0 and diverges as $1/2t$ for small t , a factor that

reappears throughout the thesis as an error amplifier. The channel interpolates between the data distribution and a standard Gaussian, which is what a diffusion model needs.

Two structural remarks are used repeatedly later. First, the kernel is Gaussian *in a as well as in x*: read as a function of the clean variable, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$, a Gaussian likelihood factor of precision e^{-2t}/Δ_t centred at the deconvolved observation $e^t x$. This is what makes the posteriors of Chapter 5 jointly Gaussian and the belief-propagation messages closable in a finite-dimensional family. Second, if $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (3.5) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t}\mu, \quad \Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I, \quad (3.6)$$

since means and covariances transform linearly and the added noise is white. Equation (3.6), shrink the covariance, add a multiple of the identity, is the entire forward process of this thesis in one line.

3.3 How the object of study was arrived at

The estimand of this thesis was not chosen at the outset. It was reached by getting it wrong first, and the sequence is worth recording, both because it explains the shape of what follows and because the correction is the single most consequential step in the project.

The propagator hypothesis. The work began from a structural guess about sequences. If a trajectory is generated by a fixed dynamical rule, the score at one frame ought to be obtainable from the score at the previous one by a fixed operator. Writing $S_{:,u}$ for the block of the score at internal time u , the conjecture was that some *propagator* P_t satisfies

$$S_{:,u}(x) \approx P_t(S_{:,u-1}(x)). \quad (3.7)$$

If such an operator existed and were cheap, generation over a long sequence would cost one solve plus $L - 1$ applications of P_t . The programme adopted to test (3.7) was deliberately austere: exactly solvable models only, closed forms wherever possible, and no neural networks anywhere, so that any structure found could be attributed to the model and not to a fitting procedure.

Six toy models, and a mistake they shared. A sequence of solvable objects was built to probe (3.7): a static Gaussian mixture under the channel; a two-frame joint law; a two-dimensional rotating process $a_{u+1} = R_{\Delta\theta} a_u + \xi_u$; a billiard in a box; a scalar AR(1) chain examined on the whole (u, t) plane; and a diffusion on the circle. The rotating model and the circle are the direct ancestors of Chapter 4, and the AR(1) study is the ancestor of Chapters 5 and 6.

All but one of them computed the wrong object. What was evaluated was the *per-frame marginal* score

$$s(x_u, u, t) = \nabla_{x_u} \log p_t(x_u | u), \quad (3.8)$$

which integrates out every other frame. Doing so is not a mild approximation. It is equivalent to asserting that the clean law factorises as $P_0(a_0, \dots, a_{L-1}) = \prod_u p_0(a_u | u)$, which is false for any correlated process and, in particular, false for every model in the list. The quantity being studied had the frames' dependence removed from it before the study began, so no propagator could have been found in it: there was nothing left for P_t to propagate.

The correction. The object was corrected in a supervision meeting of March 2026 to the *joint* score of the whole trajectory, $S(x, t) = \nabla_{(x_0, \dots, x_{L-1})} \log P_t(x_0, \dots, x_{L-1})$, whose k -th block is the posterior expectation

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x_0, \dots, x_{L-1}] - x_k}{\Delta_t}, \quad (3.9)$$

taken over the full posterior of the clean trajectory given all L observations. Equation (3.9) is (2.10) of Chapter 2, and the conditioning on every coordinate is the whole of the difference between it and (3.8). The remainder of this chapter makes the case for it directly, without reference to how it was reached.

What the correction bought. It made the stationary Gaussian AR(1) chain the natural testbed rather than one model among six. Under (3.9) the posterior is Gaussian, so the conditional mean is available from a Kalman smoother, the joint score is linear in the whole of x , and the clean precision is tridiagonal. Each of these is a closed form that can be checked against another closed form, which is what Chapters 5 and 6 do.

The propagator question, answered. The conjecture that started the project was then settled rather than abandoned, and the rest of the thesis is organised around the answer. A propagator does exist for the Gaussian chain, and it is exact: the Rauch–Tung–Striebel smoother reproduces the precision-matrix score to machine precision, which Chapter 6 proves and measures. What is *not* exact is the naive reading of (3.7) in which P_t acts as a shift, one row of the precision matrix mapping to the next. That form holds only asymptotically: the precision is Toeplitz in the bulk but not at the boundaries, and its deviation from Toeplitz falls with both chain length and diffusion time, from a few per cent on short chains at low noise to a small fraction of one per cent on long chains at high noise. A related negative result closes the cheapest version of the idea: the joint law admits no exact factorisation into independent one-dimensional bonds for any $L \geq 2$, because the cross term in the two-frame characteristic function is $-\alpha\Delta_t$, which vanishes only when $\alpha = 0$ or $t = 0$.

So the propagator is a smoother, not a shift. That is a less convenient answer than the one hoped for, and a considerably more useful one, because a smoother is exactly the object that generalises: it survives the loss of Gaussianity as belief propagation (Chapter 7), and it survives the loss of a known kernel as EM (Chapter 8).

3.4 Why the joint score

Standard diffusion models generate *static* objects; the data of interest here are *dynamic*, a data point being a sequence of frames indexed by an internal time $u = 0, 1, \dots, L - 1$, a video, a rotating shape, a stochastic trajectory. Two times therefore appear from the outset, and keeping them apart is a permanent discipline of this thesis: the *internal* time u of the sequence, and the *diffusion* time t of the noising channel, which acts independently on every frame.

It is tempting to treat such a trajectory as L separate samples, computing each frame's noisy marginal score independently. In the Gaussian model that carries most of this thesis, that is not merely lossy, it is empty.

Proposition 3.3 (One-frame marginals in the stationary Gaussian chain). *Let the clean chain be stationary Gaussian $AR(1)$ with unit marginal variance, put through the variance-preserving channel $x = e^{-t}a + \sqrt{1 - e^{-2t}} z$, with $z \sim \mathcal{N}(0, 1)$ independent of a . Then every noisy frame is marginally $\mathcal{N}(0, 1)$ at every diffusion time, its per-frame marginal score is exactly $-x_k$, and the collection of one-frame marginals carries zero Fisher information about the coupling α .*

This is immediate: variance preservation gives $\text{Var}(x) = e^{-2t} \text{Var}(a) + (1 - e^{-2t}) = 1$, and a Gaussian a makes x Gaussian, so the marginal law is $\mathcal{N}(0, 1)$ whatever α is and cannot depend on it.

Remark 3.4 (The same statement is false off the Gaussian model). Variance preservation alone gives only $\text{Var}(x) = 1$; it does not give Gaussianity, and the distinction matters because the non-Gaussian chains of Chapters 7–10 are the ones the rest of this thesis measures. Writing φ for characteristic functions,

$$\varphi_x(u) = \varphi_a(e^{-t}u) \exp\left[-\frac{1-e^{-2t}}{2}u^2\right], \quad (3.10)$$

so if x were exactly $\mathcal{N}(0, 1)$ at some finite t , dividing by the Gaussian factor would force $\varphi_a(e^{-t}u) = \exp[-\frac{1}{2}e^{-2t}u^2]$, that is, a Gaussian. A genuinely non-Gaussian frame therefore

stays non-Gaussian under any finite amount of Gaussian corruption, and its one-frame score stays nonlinear; Section 7.2 computes that score for a Laplace prior and shows it approaching $-x$ only as $t \rightarrow \infty$.

So the correct general statement is narrower than “per-frame is empty”, and it is about *dependence* and not about information as such. One-frame marginals can see the innovation law, and can see a parameter that changes the marginals; what no collection of them can recover is how consecutive frames are coupled, since coupling is a property of the joint law and of no marginal of it. In the Gaussian case those two statements collapse into one, because the marginals are α -free by Proposition 3.3; off it they come apart, and Chapter 4 proves the sharper version for one specific parameter, that the rotation rate of the ring model has exactly zero Fisher information in every one-frame marginal, under that model’s rotational symmetry.

A sequence is *one* sample from a joint law, and the quantity that drives the reverse dynamics is the joint score of the whole noisy trajectory, the gradient of $\log P_t(x_0, \dots, x_{L-1})$, whose k -th component is a conditional expectation over the *full* posterior of the clean frames, as in (2.10). That is the object in which the temporal structure lives. Chapter 5 exhibits the contrast in closed form at $L = 2$.

This also settles the shape of the structural question that opened Section 3.3. The propagator (3.7) was hoped for so that a learning system could hold one block and one map instead of L blocks. What the joint view shows is that the structure worth studying is not a map between per-frame scores at all, but the coupling pattern of the joint score: a statement about a matrix in the Gaussian case, and about a posterior in general. That is the object the rest of this thesis analyses. Figure 3.1 shows it in the smallest case where it exists: the diagonal tilt of the joint score field is the whole of the coupling, and it is exactly what a per-frame treatment throws away.

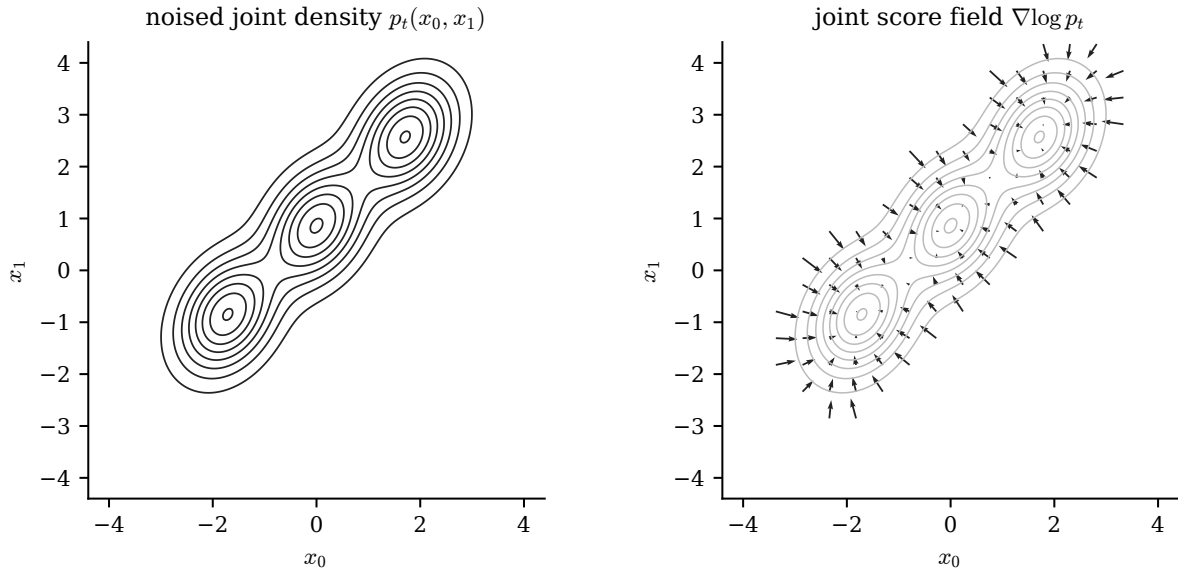


Figure 3.1: The coupling made visible, in the smallest case where it exists: $x_1 = x_0 + c + \eta$ with a trimodal Gaussian-mixture prior on x_0 , drift $c = 1$, innovation standard deviation $\sigma = 0.45$, after diffusion time $t = 0.15$. Left: the noised joint density $p_t(x_0, x_1)$. Right: the joint score field $\nabla \log p_t$ (arrows) over the same density (contours). Both panels are computed from the closed-form mixture expressions on a grid. The arrows point towards the modes and tilt along the diagonal, and that tilt is the entire effect of the coupling between frames, it is exactly what any per-frame treatment discards. Note that this prior is *not* Gaussian, so by Remark 3.4 its noised one-frame marginals are Gaussian mixtures with nonlinear scores rather than $\mathcal{N}(0, 1)$ with score $-x_k$; what those marginals cannot see is the coupling, which is the point the figure makes.

3.5 The minimal model

Several dynamic objects were candidates: rotations, coupled phases, bounded trajectories with reflections, and curved data manifolds. The stationary Gaussian AR(1) chain under coordinatewise OU noise is the minimal one in which the joint score is nontrivial yet exactly computable. one coupling parameter α for the internal dynamics against one noise parameter t for the corruption, linear algebra available at every step, and every structural question (coupling range, locality, message passing) posable in closed form.

The requirement doing the work here is *exact computability*. The whole programme of this thesis is to measure what approximations cost, Gaussian message closure in Chapter 7, an estimated transition kernel in Chapters 8 and 10, a finite receptive field in Chapter 5,

and every one of those measurements needs a reference that is known rather than approximated. A model in which the answer is available in closed form is not a simplification of the question; it is what makes the question answerable at all.

The nonlinear models are set aside at that point deliberately, as targets for after the solvable case is understood. Chapter 4 takes up the first of them.

3.6 Markov random fields and factor graphs

Graphical models assign structure to distributions the way a precision matrix assigns it to a Gaussian, and for arbitrary variables. A collection $x = (x_i)_{i \in V}$ on an undirected graph G is a *Markov random field* if each variable is conditionally independent of the rest given its neighbours, $p(x_i \mid x_{V \setminus \{i\}}) = p(x_i \mid x_{\partial i})$. For a strictly positive density the Hammersley–Clifford theorem (Besag, 1974) makes that equivalent to factorisation over the cliques of G ,

$$p(x) = \frac{1}{Z} \prod_{C \in \text{cliques}(G)} \psi_C(x_C) = \frac{1}{Z} \exp \left[- \sum_C E_C(x_C) \right], \quad (3.11)$$

so conditional independence, graphical locality and additive local energies are one thing seen three ways. For Gaussian fields it becomes matrix-concrete: the energy is $\frac{1}{2}a^\top J a - h^\top a$, and the graph is exactly the sparsity pattern of J . A Gaussian Markov chain therefore has a tridiagonal precision, which Chapter 5 derives and uses throughout.

One consequence carries the whole thesis. If each variable is observed through its own independent channel, the likelihood factors touch one variable each and no clique is enlarged: *the posterior of an MRF under per-node observations is an MRF on the same graph*. Observation changes the potentials, never the graph.

An MRF records which variables interact but not how the joint factorises: a triangle of pairwise terms and one triple term have the same graph. The *factor graph* makes the factorisation itself graphical (Kschischang et al., 2001). Given $p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a})$, it is the bipartite graph with a variable node per x_i , a factor node per ψ_a , and an edge

whenever $x_i \in x_{\partial a}$. A connected factor graph with one fewer edge than nodes is a *tree*, and trees are the tractable case: on them marginals and partition functions are exact under the recursion of the next section, whatever the alphabet. The factor graph of a Markov chain observed through per-node noise is a tree of a particularly simple shape, a spine of variables and transition factors with one evidence leaf per variable, as Chapter 5 verifies by counting.

3.7 The sum–product algorithm

On a factor graph, sum–product belief propagation (BP) passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). Variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages:

$$\nu_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(x_i), \quad \hat{\nu}_{a \rightarrow i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(x_j) \, dx_{\partial a \setminus i}, \quad (3.12)$$

with the belief at node i proportional to $\prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(x_i)$. The “exclude the recipient” rule is the heart of the algorithm: the message $i \rightarrow a$ summarises everything node i knows *except* what it learned from a , so that no piece of evidence is ever counted twice along a path.

Theorem 3.5 (Tree exactness of sum–product). *On a tree-shaped factor graph, one inward and one outward pass of (3.12) compute the exact node and factor marginals of p (Mézard and Montanari, 2009, Theorem 14.1). The resulting beliefs are unique; the individual unnormalised messages are defined only up to positive multiplicative constants.*

Proof. Each edge of a tree splits the graph into two disjoint halves, so the message crossing it summarises exactly the half behind it and no evidence reaches a node twice. The recursion terminates with every marginal having absorbed each factor once. $\square$

On graphs with cycles the same updates, iterated, define *loopy* BP, which is extraordinarily useful but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none

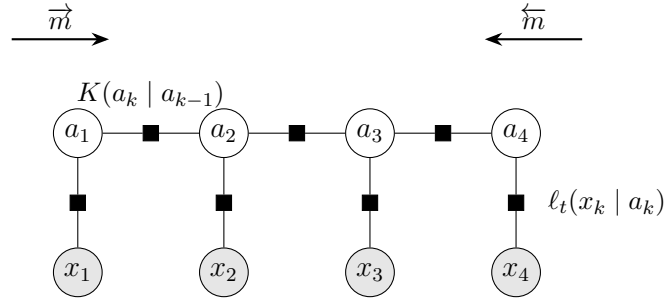


Figure 3.2: The posterior factor graph of a Markov chain observed through the coordinatewise channel. Open circles are latent variables, shaded circles observations, squares factors. Each transition factor $K(a_k | a_{k-1})$ joins two adjacent latent variables; each observation enters through a *unary* factor attached to a single latent variable, because the channel corrupts every coordinate independently. Conditioning therefore reweights node potentials without creating any edge among the a_k , so the posterior graph is still a chain and Theorem 3.5 applies to it. Arrows show the two message sweeps of (3.12). The construction fails as soon as the forward noise is correlated across coordinates: the likelihood then touches several a_k at once and the graph acquires edges.

of that is needed in this thesis, because our graphs are trees.

The representation problem. For *discrete* variables each message is a finite probability vector and each update in (3.12) is finite arithmetic. For *continuous* variables each message is a probability *density*, an infinite-dimensional object, and the update integrals must be carried out in function space. Tree exactness (a property of the graph) says nothing about representability of the messages (a property of the model’s function class); keeping the two apart is essential, and the distinction is where all the difficulty of the non-Gaussian research chapter lives. There are three ways to make continuous BP computable: restrict to models whose messages close in a finite-dimensional family (the Gaussian route, next section); represent messages numerically on a grid (the route our non-Gaussian reference solver takes, Chapter 7); or project each message onto a tractable family and accept the error (the route whose cost this thesis measures).

3.8 The Gaussian toolkit

Five standard facts about Gaussian densities, collected here because Chapters 5 to 7 use them repeatedly and each use is cleaner for pointing at a numbered identity than for restating it. Nothing in this section is original.

This appendix collects, with proofs, the handful of Gaussian identities on which Chapter 5 relies. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

3.8.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any h ,

$$\int_{\mathbb{R}^L} \exp\left(-\frac{1}{2}a^\top J a + h^\top a\right) da = \frac{(2\pi)^{L/2}}{\sqrt{\det J}} \exp\left(\frac{1}{2}h^\top J^{-1}h\right), \quad (3.13)$$

by completing the square, $-\frac{1}{2}a^\top J a + h^\top a = -\frac{1}{2}(a - J^{-1}h)^\top J(a - J^{-1}h) + \frac{1}{2}h^\top J^{-1}h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\top J a + h^\top a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance J^{-1} : *reading off (J, h) from a log-density is the fastest route to posterior parameters*, used in Proposition 5.5.

3.8.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1) \mathcal{N}(a; m_2, v_2) \propto \mathcal{N}(a; m, v), \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \quad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad (3.14)$$

Proof. Add the exponents; the sum of two quadratics in a is a quadratic in a whose a^2 -coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (3.13) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$, which is why the BP updates of Chapter 5 are two additions per message. $\square$

3.8.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2) \mathcal{N}(a; m, v) da = \mathcal{N}(a'; \alpha m, \alpha^2 v + \sigma^2). \quad (3.15)$$

Proof. a' is a sum of independent Gaussians, hence Gaussian; its mean is αm and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for a given a Gaussian summary of a'), the same computation gives $\mathcal{N}(a; m'/\alpha, (v' + \sigma^2)/\alpha^2)$, the backward half-step of the BP sweep. $\square$

3.8.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\begin{pmatrix} J_{11} & J_{12} \\ J_{12}^\top & J_{22} \end{pmatrix}$, the conditional of block 1 given block 2 is Gaussian with precision J_{11} (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12} a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality (conditioning is local in J , marginalising is local in Σ) is the algebraic engine behind the cavity recursion recorded in the companion compendium: integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \quad (3.16)$$

3.8.5 Woodbury and the two routes to the score

The Woodbury identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision Q_t and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t} \left(I - \frac{e^{-2t}}{\Delta_t} J^{-1} \right), \quad (3.17)$$

which is exactly the operator form of the score-posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 5.3) is the numerical shadow of (3.17).

3.9 Locality, and what this thesis can check

Section 2.6 set out the reading of U-Nets as belief propagation, and the cross-architecture evidence for it. One consequence of that reading belongs here, because it is what makes the chain worth studying rather than a convenient simplification.

Those results concern hierarchical data, where the graphical model is a balanced tree and the depth an architecture needs is logarithmic in the sequence length. A chain is a degenerate tree, of depth equal to its length and width two, so the same argument predicts a depth linear in the correlation length instead. That is a sharply different architectural prediction, on a model where it can be checked.

This thesis works one step below the correspondence, on a model small enough that the locality of the score is a computable quantity and not an architectural hypothesis. For the Gaussian chain the exact score at frame k depends on the observation at frame $k \pm d$ with a weight decaying geometrically in d , at a rate $q(\alpha, t)$ available in closed form (Chapter 6); the error of the radius- r windowed estimator is measured to decay at the same rate, which is a separate and weaker statement. Comparing truncation of the inference against truncation of the score matrix then quantifies, in one solvable case, how a local computation should be organised. All of this describes the function a network would have to learn and not the dynamics of learning it; the trained baselines of Chapter 10 supply the other half.

Chapter 4

The Rotating Ring: A Two-Dimensional Dynamic Object

This is where the work started, and it is placed first because what it settles is the reason for everything that follows.

The object is a planar trajectory near a circle: a point that rotates by a fixed angle each step, held near the unit radius by a confining drift, and observed through the Ornstein–Uhlenbeck channel of Section 2.3. It is a dynamic object in the sense of Chapter 3, but a nonlinear one, and it was chosen before the scalar chain precisely because rotation is the simplest dynamics whose signature is not visible in a single frame.

That last point turns out to be a theorem. The rotation rate carries *zero* Fisher information in every one-frame marginal, and positive information in the trajectory. A model that sees frames one at a time cannot estimate it at all, however many frames it sees and however long it is trained; a model that sees the trajectory can. Chapter 3 gives the corresponding statement for the stationary Gaussian chain, where the marginals are α -free outright (Proposition 3.3), but the argument there does not extend beyond Gaussian innovations. Here the conclusion is exact, and it holds because of a symmetry rather than a calculation that happens to work out.

This is the argument for studying the *joint* score, and it is what sends the rest of the thesis to a model in which the joint score can be computed in full. The ring itself is

only partly tractable: a Cartesian surrogate is exactly solvable, and the model as drawn is solvable numerically to a measured accuracy. Chapters 5 onwards trade this model's realism for a scalar chain in which every quantity has a closed form.

The question the chapter answers is narrower than the models it uses.

How can the Markov structure of the clean rotating trajectory be exploited to compute and interpret the joint diffusion score?

Four measurable subquestions make this operational: how far the joint score differs from a collection of one-frame marginal scores; how strongly the observation at frame j influences the score block at frame k ; how that influence decays with the temporal lag $|j - k|$; and how many neighbouring frames are needed to reproduce the score block to a stated accuracy. Every answer below is a statement about one model at one parameter set, in the sense fixed in Section 1.3.

4.1 Model and notation

4.1.1 The clean process

The clean state is a radius and a phase, evolving in the internal time u as two independent diffusions,

$$\mathrm{d}R_u = -\kappa(R_u - 1) \mathrm{d}u + \sqrt{2D_r} \mathrm{d}B_u^{(r)}, \quad (4.1)$$

$$\mathrm{d}\Theta_u = \omega \mathrm{d}u + \sqrt{2D_\theta} \mathrm{d}B_u^{(\theta)}, \quad (4.2)$$

observed through the polar embedding

$$a_u = R_u \begin{pmatrix} \cos \Theta_u, & \sin \Theta_u \end{pmatrix} \in \mathbb{R}^2. \quad (4.3)$$

We use the simplest constant-angular-drift reading of the model; Section 4.7 records the alternatives.

A point of $\mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis, and at each point the plane carries the orthonormal moving frame

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-\sin \theta, \cos \theta), \quad (4.4)$$

in which e_θ is e_r turned by a quarter revolution. Moving along e_r changes the radius at fixed angle, moving along e_θ changes the angle at fixed radius, and the area element is $dx dy = r dr d\theta$. The decomposition matters because the two directions are treated very differently by (4.1) and (4.2): the radius is pulled back towards one, the phase drifts and diffuses freely. Measuring the consequence of that asymmetry for the score is one purpose of the chapter.

4.1.2 The radial drift is a confining force

Equation (4.1) is an overdamped Langevin equation, $dx_u = -\nabla V(x_u) du + \sqrt{2D} dB_u$, for the quadratic potential

$$V(r) = \frac{\kappa}{2} (r - 1)^2. \quad (4.5)$$

A potential stores energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa (r - 1). \quad (4.6)$$

The sign carries the content: a system driven by $-V'$ moves downhill in V , since $\frac{d}{du} V(r_u) = V'(r) \dot{r} = -V'(r)^2 \leq 0$. Reading the two cases of (4.6), if $r > 1$ then $F < 0$ and the force points inward, and if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards one from either side at a rate proportional to the displacement, with κ setting the stiffness. Overdamped Langevin dynamics equilibrates to $p_\infty \propto e^{-V/D}$, so the stationary radial law is $\mathcal{N}(1, D_r/\kappa)$.

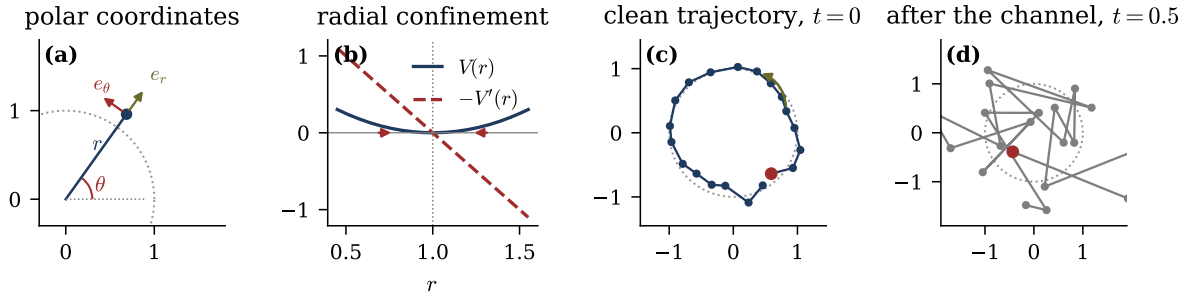


Figure 4.1: The rotating-ring model, at the parameters used throughout this chapter ($L = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, $\omega = 1$, $h = 2\pi/20$). (a) Polar coordinates and the moving frame (4.4). (b) The potential (4.5) and the force (4.6) it generates; the force changes sign at $r = 1$, which is therefore the stable radius. (c) One clean trajectory, a noisy coherently turning path near the unit circle; the arrow marks the sense of rotation. (d) The same trajectory after the channel (4.8) at $t = 0.5$, from a single standardised noise realisation.

4.1.3 One trajectory is one data point

The trajectory is sampled at internal times $u_k = kh$ for $k = 0, \dots, L-1$, and the complete datum is the stacked vector

$$a = (a_0, a_1, \dots, a_{L-1}) \in \mathbb{R}^{2L}. \quad (4.7)$$

The L frames are correlated coordinates of one data point, not L samples. Relative to Chapter 5 the only change of setting is that each frame is now two-dimensional, so the score has L blocks of size two rather than L scalar components; the forward channel is unchanged,

$$x = e^{-t}a + \sqrt{\Delta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_{2L}), \quad (4.8)$$

which is the channel of Section 3.2 acting on all $2L$ coordinates at once. Figure 4.1 shows the model and the two clocks: internal time orders the frames along the path, diffusion time corrupts them simultaneously.

4.2 The gauge, and the exact score

4.2.1 The surrogate, and what it keeps

The surrogate retains a ring, a coherent rotation, and dependence along the trajectory, and drops the rest:

$$z_0 \sim p_\lambda(z) \propto \exp\left[-\frac{(\|z\| - 1)^2}{2\lambda}\right], \quad (4.9)$$

$$z_{k+1} = R_\psi z_k + \sigma \eta_{k+1}, \quad \eta_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad (4.10)$$

with R_ψ the planar rotation through the one-frame angle $\psi = \omega h$.

Remark 4.1 (The surrogate is not a discretisation of the polar model). In (4.10) the ring acts only on z_0 . Later frames form a rotating Cartesian random walk and their radial variance grows without bound, whereas in (4.1) a confining force acts at every internal time. The surrogate is a solvable companion whose role is to exhibit the mechanism in closed form, not to approximate the model of Section 4.1.

4.2.2 The known rotation is an orthogonal change of frame

The rotation is the object to be removed, so we write it out:

$$R_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}. \quad (4.11)$$

Its columns are the standard basis vectors, each turned through ψ . They are unit vectors, since $\cos^2 \psi + \sin^2 \psi = 1$, and orthogonal, since $\cos \psi(-\sin \psi) + \sin \psi \cos \psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \quad \det R_\psi = 1, \quad R_\psi^\top = R_{-\psi}, \quad R_\psi R_\varphi = R_{\psi+\varphi}, \quad (4.12)$$

and orthogonality gives $\|R_\psi z\|^2 = z^\top R_\psi^\top R_\psi z = \|z\|^2$: lengths and inner products are preserved.

Lemma 4.2 (Rotations preserve the standard isotropic Gaussian). *If $\eta \sim \mathcal{N}(0, I_2)$ and R is orthogonal, then $R\eta \sim \mathcal{N}(0, I_2)$.*

Proof. Appendix 4.6. □

The isotropy is essential. A rotation would not preserve $\mathcal{N}(0, \Sigma)$ for general Σ , and the reduction below is exact only because both the innovation in (4.10) and the channel (4.8) are isotropic.

Lemma 4.3 (Gauge reduction). *Define the de-rotated frame $y_k = R_{-k\psi} z_k$ and stack the per-frame rotations into $U_\psi = \text{diag}(I_2, R_{-\psi}, \dots, R_{-(L-1)\psi})$. Then*

$$y_{k+1} = y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} \stackrel{iid}{\sim} \mathcal{N}(0, I_2), \quad (4.13)$$

U_ψ is orthogonal, and the noised density and score satisfy

$$P_t(z | \psi) = P_t^{(0)}(U_\psi z), \quad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t), \quad (4.14)$$

where the superscript zero denotes the quantities of the non-rotating model.

Proof. Appendix 4.6. □

The term *gauge* is used in the sense standard in physics, of a change of description that leaves the content untouched and may be chosen to simplify the equations. Equation (4.14) says that a known rotation adds no statistical difficulty: de-rotate the data, solve the non-rotating problem, rotate the score back. Since $U_\psi^\top = U_\psi^{-1}$, the operation is exactly invertible and destroys no information. It would not hold for an *unknown* ψ , which is a genuine latent variable (Section 4.7). Figure 4.2 shows the two frames.

4.2.3 The clean law after the gauge

Iterating (4.13) and cancelling the intermediate terms gives the anchored representation (4.20), whose derivation, together with the second-difference identity it satisfies, is in Appendix 4.6. The identity is verified to machine zero (Table 4.1).

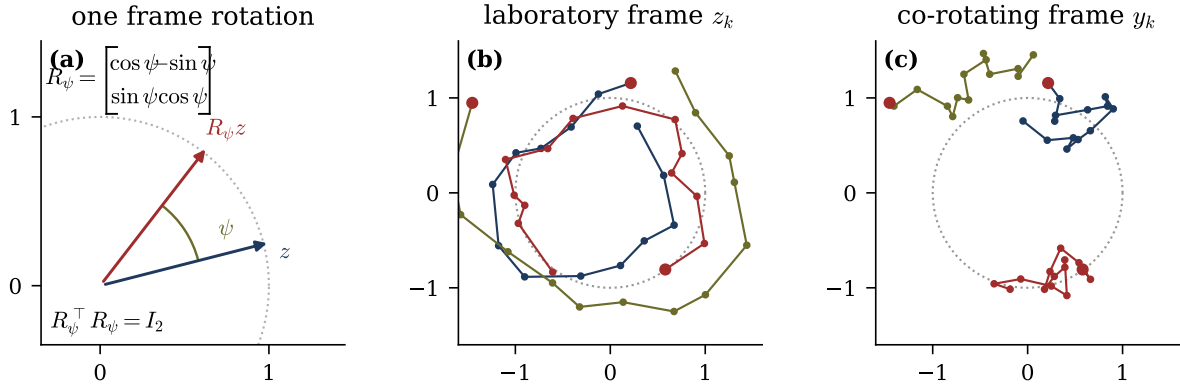


Figure 4.2: The rotation and the gauge. (a) R_ψ turns every vector through the same angle without changing its length, which is the statement $R_\psi^\top R_\psi = I_2$. (b) Three surrogate trajectories in the laboratory frame ($L = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; the marked point is $k = 0$), coherently circulating. (c) The same trajectories after U_ψ : a random walk anchored on the ring. Because U_ψ is orthogonal the reduction is exact at every diffusion time; the numerical round trip agrees to machine precision (Table 4.1).

4.2.4 The noised law and the exact score

Conditioning on the anchor, (4.20) gives $\mathbb{E}[y_i | A = \alpha] = \alpha$ and

$$\text{Cov}(y_i, y_j | A) = \sigma^2 \mathbb{E} \left[\left(\sum_{r \leq i} \tilde{\eta}_r \right) \left(\sum_{s \leq j} \tilde{\eta}_s \right)^\top \right] = \sigma^2 \min(i, j) I_2, \quad (4.15)$$

since $\mathbb{E}[\tilde{\eta}_r \tilde{\eta}_s^\top] = \delta_{rs} I_2$ leaves only the $\min(i, j)$ indices common to both sums. Compactly, $y | A = \alpha \sim \mathcal{N}(\mathbf{1} \otimes \alpha, G \otimes I_2)$ with $G_{ij} = \sigma^2 \min(i, j)$, where $\mathbf{1} \in \mathbb{R}^L$ is the all-ones vector. The two Kronecker products separate the temporal index from the spatial one: $\mathbf{1} \otimes \alpha$ repeats the same two-dimensional anchor in every frame, and $(G \otimes I_2)_{(i, \alpha'), (j, \beta')} = G_{ij} \delta_{\alpha' \beta'}$, so G carries all the temporal correlation while I_2 states that the two spatial coordinates are uncorrelated and share the same temporal covariance. Applying the channel and using $I_{2L} = I_L \otimes I_2$ to combine the covariances,

$$x | A = \alpha \sim \mathcal{N}(e^{-t} \mathbf{1} \otimes \alpha, C_t \otimes I_2), \quad C_t = e^{-2t} G + \Delta_t I_L, \quad (4.16)$$

in which C_t does not depend on α : the noised law is a two-parameter location mixture of one fixed Gaussian.

Proposition 4.4 (Exact surrogate score and response). *Let $q = C_t^{-1}\mathbf{1}$, $\beta = e^{-2t}\mathbf{1}^\top C_t^{-1}\mathbf{1}$ and $h = e^{-t}\sum_j q_j x_j \in \mathbb{R}^2$. Then the anchor posterior is*

$$p(\alpha | x) \propto \exp \left[-\frac{(\|\alpha\| - 1)^2}{2\lambda} - \frac{\beta}{2} \|\alpha\|^2 + h^\top \alpha \right], \quad (4.17)$$

and the score and response blocks are

$$S_k(x, t) = -\left(C_t^{-1}x\right)_k + e^{-t}q_k \mathbb{E}[A | x], \quad J_{kj}(x, t) = -\left(C_t^{-1}\right)_{kj}I_2 + e^{-2t}q_k q_j \text{Cov}(A | x). \quad (4.18)$$

Proof. Appendix 4.6. □

The $2L$ -dimensional integral over clean trajectories has collapsed to a two-dimensional posterior over the shared anchor, which is the whole simplification and a direct consequence of (4.20). Its two moments are cheap: the prior and the $\|\alpha\|^2$ term are rotationally symmetric, so the posterior depends on the direction of α only through $h^\top \alpha$, and $\mathbb{E}[A | x] = \mu(\|h\|)\hat{h}$ with $\text{Cov}(A | x) = v_\perp I_2 + (v_\parallel - v_\perp)\hat{h}\hat{h}^\top$, three scalar functions of one scalar argument, tabulated once per diffusion time by one-dimensional quadrature.

The structural content of (4.18) is that the temporal coefficient of the second term is the outer product $qq^\top$, which is of rank one. The nonlocal part of the response is therefore not a chain of local couplings but a single shared latent variable seen by every frame. Figure 4.3 separates the two terms.

Proposition 4.5 (What the surrogate establishes). *Clean Markov locality does not survive marginalisation through the OU channel as locality of the noised score: by (4.18) the response has nonzero blocks at every lag for every $t > 0$. The nonlocal part is nonetheless structured, decomposing exactly into a near-diagonal chain term and a rank-one correction generated by the shared ring anchor.*

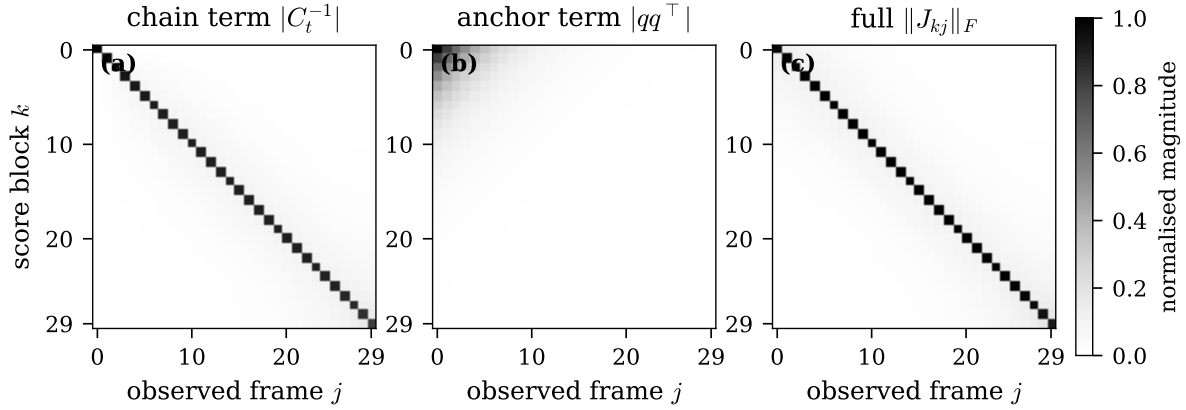


Figure 4.3: Exact decomposition of the surrogate response (4.18) at $t = 0.20$, $L = 30$, $\lambda = 0.05$, $\sigma = 0.15$, averaged over 250 noisy trajectories, each panel normalised by its own maximum. (a) The chain term $|C_t^{-1}|$ is concentrated near the diagonal. (b) The anchor coefficient $|qq^\top|$ is rank one and spread over all lags. (c) The mean norm of the complete 2×2 blocks. The analytic Jacobian agrees with central finite differences to 9.2×10^{-6} (Table 4.1).

Table 4.1: Validation of the rotating-ring computations. The response identity and the analytic surrogate Jacobian are compared with central finite differences of the score; the grid check compares two polar resolutions.

check	discrepancy	tolerance
posterior response identity vs finite differences	6.63×10^{-9}	10^{-7}
analytic surrogate Jacobian (4.18) vs finite differences	9.18×10^{-6}	10^{-4}
41×128 against 51×192 polar response grid	3.04×10^{-11}	10^{-7}
rotation gauge round trip (Lemma 4.3)	2.22×10^{-16}	10^{-12}
clean score on an exact co-rotating ring path	0 (exact)	10^{-12}

4.3 Numerical validation

Every identity of this chapter is checked against an independently computed reference, in the protocol of Appendix A.

Beyond Table 4.1 the checks include row normalisation of the transition kernels, global rotation equivariance of the score, stability under halving the finite-difference step, and the exact vanishing of the clean surrogate score on a perfect co-rotating ring path. One caveat on grid convergence: the difference between the 41×128 and 51×192 grids is small because the baseline likelihoods and kernels are already smooth at those resolutions,

and a deliberately coarse 11×24 grid does show visible deviations. The production grid is converged for these parameters rather than trivially grid-independent.

4.4 Marginal blindness

Everything above concerns how far influence *reaches* in the noised score. This section establishes something stronger and of a different kind: there is a parameter of this model about which the per-frame marginals carry not less information, but none.

Theorem 4.6 (Marginal blindness). *Give the initial phase the uniform law on the circle. Then every single-frame marginal is independent of the rotation, at every noise level. This holds for the polar model (4.1)–(4.3) in its parameter ω , and for the surrogate (4.10) in its parameter ψ . Consequently a model built from per-frame marginals carries exactly zero Fisher information about the rotation.*

Proof. The polar model. The radial and angular diffusions (4.1)–(4.2) are independent, and integrating the second gives $\Theta_u = \Theta_0 + \omega u + \sqrt{2D_\theta} B_u^{(\theta)}$. The initial phase is uniform on the circle and independent of everything added to it, and convolving the uniform law on a compact group with any independent law returns the uniform law; so Θ_u is uniform for every u , whatever ω is. Since R_u carries no ω and is independent of Θ_u , the law of $a_u = R_u(\cos \Theta_u, \sin \Theta_u)$ does not depend on ω . The confining drift is irrelevant to the argument precisely because it acts on the radius alone.

The surrogate. Here the dynamics are linear and the same conclusion follows by unrolling rather than by symmetry of the phase. Unrolling the recursion, $z_u = R_\psi^u z_0 + \sigma \sum_{k \leq u} R_\psi^{u-k} \eta_k$. Each η_k is isotropic Gaussian, and a rotation of an isotropic Gaussian has the same law, so $R_\psi^{u-k} \eta_k \stackrel{d}{=} \eta_k$ independently across k . The initial phase θ_0 is uniform on the circle, so z_0 has a rotationally invariant law and $R_\psi^u z_0 \stackrel{d}{=} z_0$. The two terms are independent, hence

$$z_u \stackrel{d}{=} z_0 + \sigma \sqrt{u} \xi, \quad \xi \sim \mathcal{N}(0, I_2), \quad (4.19)$$

in which ψ does not appear. The channel acts coordinatewise and independently of ψ ,

so the noised frame is likewise ψ -free. Therefore $\partial_\psi \log P_t(x_u) \equiv 0$ for every frame and every noise level, and a sum of per-frame log-densities cannot contain information no term contains. $\square$

Remark 4.7 (What the theorem does not say). The marginals are not uninformative in general. They determine the width and centre of the confining well and the walk noise perfectly well, and by (4.19) the per-frame law is a visible, non-Gaussian ring whose radius grows as $\sqrt{u}$. They are blind to the *rotation* specifically. Nor is the joint easy: the joint carries the information, and extracting it is what the estimator does.

The sharpness is what matters. A per-frame model is not less informative about the dynamics, nor informative only at low noise. It carries zero, exactly, at every noise level. Radii are not evidence.

4.5 Three estimation targets

The theorem furnishes an unusually clean experiment, because this model carries both a parameter the marginals can see, the confining potential $V(r) = (r - r_*)^2/2\lambda$, and one they provably cannot. Running both through the same estimator isolates the effect of the structure and not of the method. The marginal-visible potential is the control that makes the contrast readable; no measurement of it is reported here. What follows is the two targets the marginals cannot see: a single rotation ψ , and a law over rotations $p(\psi)$.

For (c) the estimator is expectation-maximisation over a grid of angles, with responsibilities $r_\mu(\psi) \propto p(\psi)P_t(x^\mu \mid \psi)$ and the update $p(\psi) \leftarrow M^{-1} \sum_\mu r_\mu(\psi)$. The same sufficiency that makes the expected-transition statistic a one-shot quantity on the chain (Chapter 8) appears here: the ψ -conditional log-likelihood does not depend on $p(\psi)$, so the matrix of log-densities is computed once and every iteration reweights it.

The blind arm is not a baseline chosen to lose. It is the theorem's own control: its likelihood is constant in ψ , so its responsibilities equal $p(\psi)$, its update is the identity, and $p(\psi)$ cannot leave its uniform initialisation. If it moves, the implementation is wrong rather than the theorem.

The blind arm returns exactly the null. Over sixteen seeds, seven budgets and twelve noise levels, 1344 cells, the per-frame arm’s estimate moves 0.0000 in total variation from its uniform initialisation, and over the 480 cells of target (b) its profile likelihood in ψ has range 0.00×10^0 nats. This is a machine-precision zero rather than a small number, which is what a theorem predicting exactly no information requires.

It is worth being precise about what that zero does and does not evidence. The per-frame likelihood is implemented by a function that takes no ψ argument, so its independence of ψ is a property of the code path, not a measurement: the zero confirms that the estimator is wired to the marginal it is supposed to use, and would catch a leak, but it cannot corroborate the theorem against an independently constructed alternative.

The independent check is a paired one. Trajectories are simulated separately at $\psi \in \{0, \pi/6, \pi/3, 1.1\}$, sharing no code with the estimator, and two statistics are computed on the *samples*. As a negative control, the energy distance between one-frame clouds at different ψ is compared against the same distance between two independent draws at *equal* ψ , which is the sampling floor: the floor is 1.5×10^{-2} and the cross- ψ distances are 2.6 to 4.7×10^{-3} — strictly *below* the noise of having drawn twice, both on clean trajectories and after OU noising at $t = 0.5$. As a positive control, the lag-one cross-covariance of those same samples must equal $s^2 R_\psi$, so its antisymmetric part is $\sin \psi$; measured, it is $-0.001, 0.503, 0.865, 0.893$ against $\sin \psi = 0, 0.500, 0.866, 0.891$.

The pairing is the point. A null result on its own cannot distinguish a true invariance from a statistic too blunt to detect anything; here the same samples, at the same size, move by 0.9 in the joint statistic and by less than their own resampling noise in the marginal one. Both controls are in `tests/test_ring_blindness_controls.py`.

The joint arm recovers the rotation, and loses it to the channel. Estimating a single angle, the error is $2.1 \pm 0.7^\circ$ pooled over the whole schedule. Recovering the two-species law $p^*(\psi) = \frac{1}{2}\delta_{+\omega} + \frac{1}{2}\delta_{-\omega}$ against a truth of $\omega = 30^\circ$ and a null of 90° , Table 4.2 reports:

Table 4.2: Recovery of the two-species rotation law from the joint likelihood, against a truth of $\omega = 30^\circ$ and a null of 90° . The rotation survives the channel until $t \approx 1$ and has returned to the null by $t = 3$; the blind arm never leaves the null at any noise level.

t	0.05	0.32	0.68	0.98	1.43	3.00
median $ \hat{\omega} - \omega $	0.15°	0.24°	0.90°	2.86°	7.09°	58.8°

Information about the rotation survives until $t \approx 1$ and is gone by $t = 3$, where the estimate has returned to the null. That is the channel doing what a channel does, and it is legible here in a way it is not on the chain, because the null is known exactly rather than estimated.

Remark 4.8 (A normalisation that fails silently). The radial density of the confining well is naturally written unnormalised, and its normaliser depends on both λ and r_* . While only ψ is estimated the term is constant and cancels in every responsibility, which is why it can be absent for a long time without anything appearing wrong. The moment λ is estimated it stops being constant: a wider well contains more mass, the likelihood grows without bound in λ , and the estimate pins to the top of whatever grid it is given. The failure produces a smooth, confident, wrong answer and not an error. It is recorded here because the same shape of mistake, a quantity that is constant in the case one has been testing, and is not constant in the case one moves to, is the most common way a correct derivation becomes an incorrect measurement.

4.6 Derivations

The four calculations the chapter relies on, in the order it uses them. The mechanics behind Chapter 4. The chapter states the gauge reduction, the exact surrogate score and their consequences, which is what the argument needs; the algebra establishing them is here, where it can be checked without interrupting that argument.

4.6.1 Rotations preserve the standard isotropic Gaussian

Proof of Lemma 4.2.

Proof. A linear map of a Gaussian is Gaussian, so $R\eta$ is fixed by its first two moments: $\mathbb{E}[R\eta] = R\mathbb{E}[\eta] = 0$ and $\text{Cov}(R\eta) = RI_2R^\top = RR^\top = I_2$. Equivalently, in terms of densities, $(2\pi)^{-1}e^{-\|z\|^2/2}$ depends on z only through $\|z\|$, and the change of variables $z \mapsto Rz$ changes neither $\|R^{-1}z\| = \|z\|$ nor $|\det R^{-1}| = 1$: the level sets are circles centred at the origin and a rotation maps each circle to itself. $\square$

4.6.2 The gauge reduction

Proof of Lemma 4.3.

Proof. Substituting (4.10) and using the composition rule of (4.12),

$$y_{k+1} = R_{-(k+1)\psi} \left(R_\psi z_k + \sigma \eta_{k+1} \right) = \underbrace{R_{-(k+1)\psi} R_\psi}_{=R_{-k\psi}} z_k + \sigma R_{-(k+1)\psi} \eta_{k+1} = y_k + \sigma \tilde{\eta}_{k+1},$$

and by Lemma 4.2 each $\tilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$; they stay independent because each is a fixed linear map of a different independent η . Next, U_ψ is block diagonal with orthogonal blocks, so $U_\psi^\top U_\psi = \text{diag}(I_2, \dots, I_2) = I_{2L}$ and $\det U_\psi = \prod_k \det R_{-k\psi} = 1$. Applying U_ψ to (4.8) gives $U_\psi x = e^{-t} U_\psi a + \sqrt{\Delta_t} U_\psi \xi$ with $U_\psi \xi \sim \mathcal{N}(0, I_{2L})$, so $U_\psi x$ has exactly the law of the noised non-rotating trajectory: the de-rotation commutes with the channel. For the densities, $z = U_\psi^\top y$ and $P_t(z | \psi) = P_t^{(0)}(U_\psi z) |\det U_\psi|$, the Jacobian factor being one. Differentiating that identity componentwise, with $\partial y_b / \partial z_c = (U_\psi)_{bc}$,

$$\frac{\partial}{\partial z_c} \log P_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{bc} \left(\nabla \log P_t^{(0)} \right)_b (U_\psi z) = \left[U_\psi^\top S^{(0)}(U_\psi z, t) \right]_c.$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$ while a point transforms with U_ψ . $\square$

4.6.3 The clean law after the gauge

Iterating (4.13) and letting the intermediate terms cancel,

$$y_k - y_0 = \sum_{r=1}^k (y_r - y_{r-1}) = \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad \text{so} \quad y_k = A + \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad (4.20)$$

with $A := y_0 \sim p_\lambda$ and the empty sum equal to zero. Every frame contains the *same* random anchor, drawn once on the ring, plus a partial sum of independent innovations; the anchor is the only non-Gaussian ingredient of the model.

The clean law factorises over the ring prior and the one-step transitions, so $\log P_0(y) = -(\|y_0\| - 1)^2/2\lambda - (2\sigma^2)^{-1} \sum_{k=0}^{L-2} \|y_{k+1} - y_k\|^2 + \text{const.}$ Differentiating is a counting exercise: an interior y_k appears in exactly two summands, the edge arriving at it and the edge leaving it, whose derivatives are $-(y_k - y_{k-1})/\sigma^2$ and $+(y_{k+1} - y_k)/\sigma^2$. Adding them, and using $\nabla_y \|y\| = y/\|y\|$ for the ring term,

$$\begin{aligned} S_0(y, 0) &= \frac{1 - \|y_0\|}{\lambda} \frac{y_0}{\|y_0\|} + \frac{y_1 - y_0}{\sigma^2}, \\ S_k(y, 0) &= \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}, \quad 1 \leq k \leq L-2, \\ S_{L-1}(y, 0) &= \frac{y_{L-2} - y_{L-1}}{\sigma^2}. \end{aligned} \quad (4.21)$$

The interior clean score is a discrete second derivative along internal time, that is a discrete Laplacian: each block measures how far frame k departs from the average of its neighbours. It is strictly local, and it vanishes identically on the model's own noiseless solution, because a trajectory rotating at radius exactly one maps under the gauge to a constant y_k , for which every second difference vanishes and, with $\|y_0\| = 1$, so does the ring term. That vanishing is verified to machine zero (Table 4.1).

4.6.4 The exact surrogate score

Proof of Proposition 4.4.

Proof. Expanding the exponent of (4.16) in powers of α , with the correlated covariance

$C_t \otimes I_2$, gives (4.17). Writing $\log P_t(x) = \text{const} - \frac{1}{2}x^\top (C_t^{-1} \otimes I_2)x + \log Z(h(x))$, the gradient of the first term is $-(C_t^{-1}x)_k$ in block k , while $\partial h / \partial x_k = e^{-t} q_k I_2$ and $\nabla_h \log Z = \mathbb{E}[A | x]$, the first cumulant identity for the exponential family in h . Differentiating once more, so that the second derivative of $\log Z$ returns $\text{Cov}(A | x)$, gives the response. $\square$

4.7 Limitations

1. **One parameter set for the geometry.** The ring width, walk noise and trajectory length are held at a single baseline. Only the noise level and the data budget are swept, and nothing is claimed about scaling in L .
2. **Radius positivity.** The radial process is formally supported on all of $\mathbb{R}$. At the baseline the probability of a negative radius is negligible, but a strict model would reflect at the origin or use a positive diffusion.
3. **Known rotation in the gauge argument.** Lemma 4.3 assumes ψ known; §4.5 is what happens when it is not, and the two-species case is the smallest version of the speciation phenomenon discussed in Section 2.7.
4. **Grid approximation.** The radial integral is numerically converged at these parameters but is a quadrature, not a symbolic closed form. The validation of §4.3 bounds the error; it does not remove it.
5. **The marginal-visible control is not measured.** Target (a) of §4.5 specifies the contrast but no result for it is reported, so the claim that the two targets behave differently rests on the theorem and not on a matched pair of measurements. Nothing else in the chapter depends on it.

The immediate extensions are to vary L systematically, to infer the angular phase rather than use an oracle basis, to add an unknown rotation class, and to test whether a time-dependent local architecture with receptive field $L_\epsilon(t)$ can match the full score.

Chapter 5

The Gaussian Chain by Linear Algebra

The stationary Gaussian AR(1) chain is the one model in this thesis whose joint score can be written down in closed form and checked by hand. This chapter does that with matrices: the clean law, its dense covariance and sparse precision, the score in two equivalent forms, and the lifecycle the coupling structure passes through as the diffusion time grows. Chapter 6 then recomputes the same object by message passing and shows the two agree. Everything here is exact unless marked as a numerical check, and each boxed formula is covered by a named check of the audit suite (Appendix A).

5.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \dots, a_{L-1})$ of the stationary AR(1) chain (3.2) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (3.5) for a time t ,

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (5.1)$$

and the object of study is the joint score of the noisy sequence,

$$S(x, t) = \nabla_x \log P_t(x) \in \mathbb{R}^L, \quad P_t(x) = \int_{\mathbb{R}^L} P_0(a) \prod_{k=0}^{L-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t) da. \quad (5.2)$$

The gradient is taken with respect to all L noisy frames simultaneously, under the law of the whole noisy trajectory (Section 2.4).

5.2 The clean law: dense covariance, sparse precision

The chain factorisation (3.3) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{L-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2) = \mathcal{N}(a; 0, \Sigma_0). \quad (5.3)$$

Proposition 5.1 (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals, $\gamma(d) = \alpha^{|d|}$.

Proposition 5.2 (Tridiagonal precision). $Q_0 := \Sigma_0^{-1}$ is exactly tridiagonal:

$$(Q_0)_{kk} = \begin{cases} \frac{1}{\sigma_\eta^2}, & k \in \{0, L-1\}, \\ \frac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < L-1, \end{cases} \quad (Q_0)_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}, \quad (5.4)$$

and every entry at distance ≥ 2 vanishes identically.

Toolbox 5.1 — Deriving the tridiagonal precision from the log-density

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (5.3),

$$-2 \log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{L-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \quad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. The only term coupling a_k and a_{k+1} is the transition $(a_{k+1} - \alpha a_k)^2 / \sigma_\eta^2$, with cross coefficient $-2\alpha / \sigma_\eta^2$, hence precision entry $-\alpha / \sigma_\eta^2$; no term couples variables at distance ≥ 2 , so those entries are exactly zero. An

interior a_k appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with α^2/σ_η^2 , summing to $(1 + \alpha^2)/\sigma_\eta^2$. At the left boundary the prior term $1/\sigma_0^2 = (1 - \alpha^2)/\sigma_\eta^2$ plus the right transition α^2/σ_η^2 give $1/\sigma_\eta^2$; the right boundary is symmetric.

□

Remark 5.3 (Tridiagonality is Markovianity, not Gaussianity). A zero in the precision matrix is a conditional independence given all other variables; by the Hammersley–Clifford theorem (Section 3.6), factorisation over the cliques of a graph and conditional independence with respect to that graph are the same property. Gaussianity entered the derivation only to identify the precision with the Hessian of $-\log P_0$; the support of that Hessian, nearest neighbours only, comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern; the Gaussian case makes the entries constants.

The covariance is dense (every frame correlates with every other) while the precision is tridiagonal: correlation is global, conditional dependence is local. The score is governed by the second object.

5.3 The score, two ways

5.3.1 Linear algebra

Proposition 5.4 (Noised law). $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ with

$$\boxed{\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I_L} \quad (5.5)$$

This is (3.6). The diagonal is $e^{-2t} + \Delta_t = 1$ at every t (variance preservation); only the off-diagonal entries, $\alpha^{|i-j|} e^{-2t}$, depend on the diffusion time, because the noise of different frames is independent. Since $\log P_t(x) = -\frac{1}{2} x^\top \Sigma_t^{-1} x + \text{const}$,

$$\boxed{S(x, t) = -Q_t x, \quad Q_t := \Sigma_t^{-1}.} \quad (5.6)$$

The joint score is linear in x , the hallmark of Gaussianity, and its entire structure is the noisy precision matrix Q_t : which frames couple, how strongly, and at which t , are questions about this one matrix, answered in Sections 5.4 and 6.3.

5.3.2 Bayesian inference

The score–posterior identity (2.10) expresses the same score through the posterior mean of the clean chain,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x] - x_k}{\Delta_t}, \quad (5.7)$$

and the posterior is Gaussian with parameters used throughout:

Proposition 5.5 (Posterior of the clean chain). $P(a | x) = \mathcal{N}(a; m, J^{-1})$ with

$$\boxed{J = \frac{e^{-2t}}{\Delta_t} I + Q_0, \quad J m = h := \frac{e^{-t}}{\Delta_t} x.} \quad (5.8)$$

Toolbox 5.2 — Posterior precision and field, by completing the square

Bayes' rule with the likelihood (5.1):

$$\begin{aligned} -2 \log P(a|x) &= a^\top Q_0 a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^\top \left(Q_0 + \frac{e^{-2t}}{\Delta_t} I \right) a - 2 \frac{e^{-t}}{\Delta_t} x^\top a + \text{const}. \end{aligned}$$

Matching against the generic form $a^\top J a - 2h^\top a$ gives (5.8). Two structural observations. (i) The likelihood is diagonal in a : observing each frame through its own channel adds the channel precision e^{-2t}/Δ_t to every diagonal entry of the prior precision and touches nothing else. (ii) Consequently J is tridiagonal at every t : the *posterior* of the clean chain remains a nearest-neighbour Markov field; what fills up with t is the precision Q_t of the noisy marginal law. Keeping J (tridiagonal, posterior) and Q_t (dense, noisy marginal) distinct prevents much confusion. $\square$

Substituting $m = J^{-1}h$ into (5.7) must return $-Q_t x$; the operator identity behind

this is Woodbury's (Appendix 3.8), and the audit suite checks the two routes agree to 2.7×10^{-15} across sweeps of (L, α, t) . The agreement is an end-to-end check of two computations built from different formulas.

5.3.3 The smallest example, $L = 2$

For $L = 2$, with $r := \alpha e^{-2t}$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \implies S_0(x, t) = -\frac{x_0 - r x_1}{1 - r^2}, \quad S_1(x, t) = -\frac{x_1 - r x_0}{1 - r^2}. \quad (5.9)$$

By contrast, each x_k is marginally $\mathcal{N}(0, 1)$ at every t (variance preservation), so the per-frame marginal score is $-x_k$: independent of t and of the other frame, it carries no information about the dynamics. The joint score contains the cross term $-rx_1/(1 - r^2)$: the denoiser of frame 0 uses evidence from frame 1 with a weight that equals α at $t = 0$ and decays as e^{-2t} . This is the coupling structure whose full lifecycle the next section derives, in miniature.

5.4 The spectral form and the lifecycle of the precision

Between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$ lies the whole lifecycle of the coupling structure. The analysis rests on the spectral theorem and its affine-family corollary: affine images of a symmetric matrix share its eigenbasis, $(cA + dI)u_i = (c\omega_i + d)u_i$. Applying this to the forward map (5.5), which is affine in Σ_0 , gives the representation from which everything in this section follows.

Theorem 5.6 (Spectral form). *Let $\Sigma_0 = U \operatorname{diag}(\omega_i) U^\top$. Then for every $t \geq 0$,*

$$\Sigma_t = U \operatorname{diag}(\lambda_i(t)) U^\top, \quad Q_t = U \operatorname{diag}\left(\frac{1}{\lambda_i(t)}\right) U^\top, \quad \lambda_i(t) = e^{-2t} \omega_i + \Delta_t. \quad (5.10)$$

Diffusion never rotates the problem: the eigenvectors of Σ_t and Q_t are those of Σ_0 , frozen for all t , while each eigenvalue flows along the affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, monotonically towards 1. Because Σ_0 is symmetric Toeplitz, its eigenvectors are asymptotically the Fourier modes of the sequence : in frequency space the denoising problem is L independent scalar problems at every diffusion time. Two computational corollaries are recorded for later use: factoring Δ_t out of (5.5) gives the SNR form

$$Q_t = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}, \quad \gamma_t := e^{-2t}/\Delta_t, \quad (5.11)$$

and multiplying (5.5) by Q_0 and inverting gives the resolvent form

$$Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}, \quad c_t := e^{2t} - 1, \quad (5.12)$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small t .

5.4.1 Losing tridiagonality: when, how, and how much

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each t .

When: immediately. By (5.10) every entry of Q_t is analytic in t . The distance- d entries, identically zero at $t = 0$ for $d \geq 2$, are non-zero for every $t > 0$: no exact zero survives any positive diffusion time.

How: one diagonal per order in t . The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on.

Theorem 5.7 (Band-fill leading order). *For small t and frame distance $d \geq 1$,*

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (5.13)$$

Toolbox 5.3 — Band fill from the Neumann series

In the resolvent form (5.12), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = (1 + O(t)) \left[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \dots \right].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m : the entry at distance d receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1} (2t)^{d-1} (1 + O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t : the band fills outward, one diagonal per order. $\square$

Remark 5.8 (The prefactor carries no factorial, and why that is checked). Equation (5.13) has no $1/(d-1)!$: the leading distance- d term is a single Neumann term, with no combinatorial sum behind it. The distinction is easy to miss and worth verifying numerically, because slope fits in $\log t$ are blind to constant prefactors, a wrong constant survives every check on the exponent. The audit therefore tests the coefficient itself, and it separates the two candidates cleanly: the no-factorial form matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \dots, 5\}$ (at $\alpha = 0.9$, $L = 25$), while a factorial version would be wrong by 86% already at $d = 3$. Every prefactor in this thesis is audited on the same grounds, not only every exponent.

How much: the off-tridiagonal mass. Define the off-tridiagonal Frobenius mass of Q_t : the norm of everything outside the band that Markov structure would allow. At small t the band-fill law makes it rise from zero, dominated by the $d = 2$ diagonal, linearly in t . At large t , expanding the inverse of $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ gives

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \quad (5.14)$$

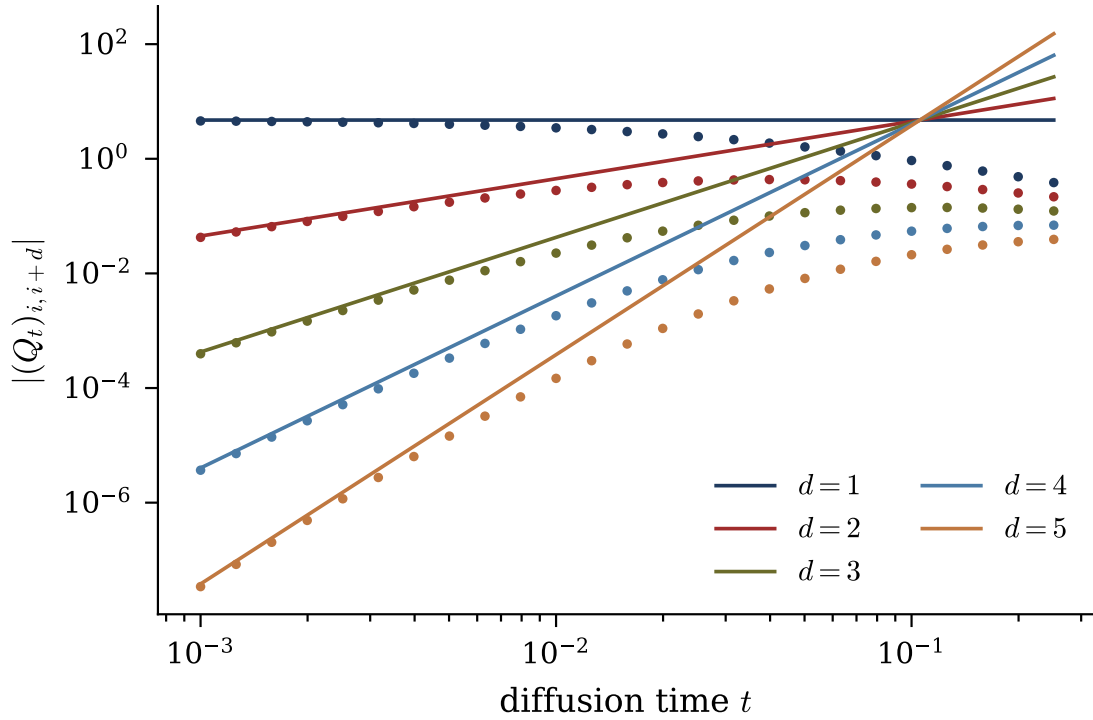


Figure 5.1: Band-fill law. Measured $|(Q_t)_{i,i+d}|$ against t (dots) and the leading order (5.13) (solid lines, no fitted parameters), for $\alpha = 0.9$, $L = 25$, $d = 1, \dots, 5$. The slopes are $d - 1$; agreement at small t is at the percent level for every d .

so $\|Q_t - I\|_F \simeq e^{-2t} \|\Sigma_0 - I\|_F$: all structure, off-tridiagonal and tridiagonal alike, decays at the rate e^{-2t} , and the score approaches the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 5.2): the noisy sequence is most non-Markov at intermediate diffusion times. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d ; maximally coupled at intermediate t ; isotropic as $t \rightarrow \infty$ at rate e^{-2t} .

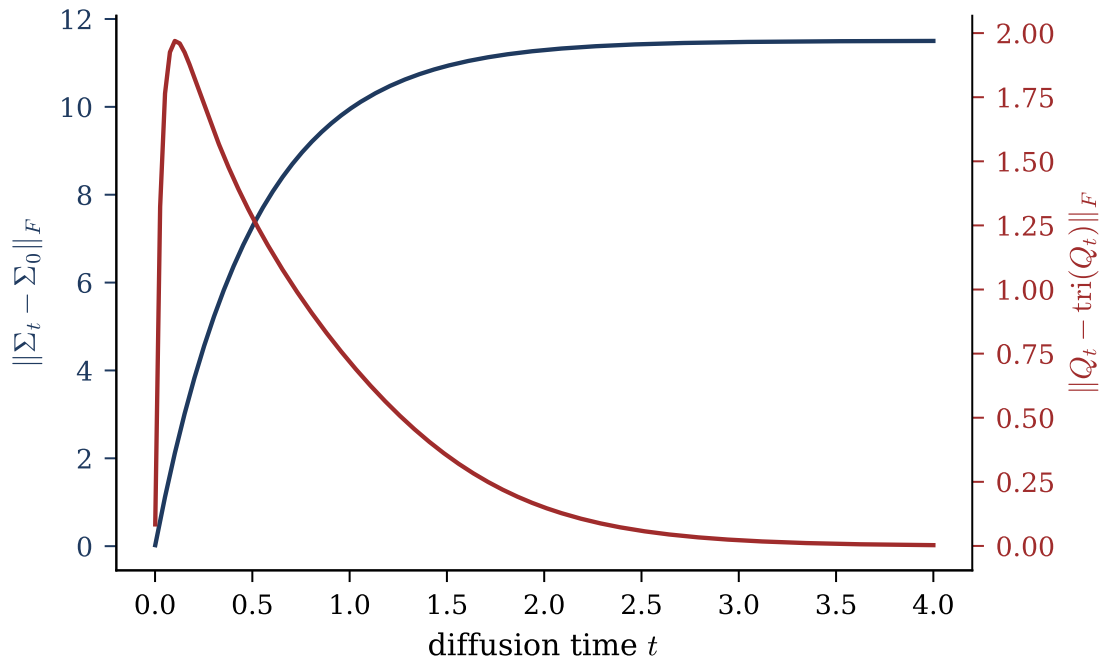


Figure 5.2: Structure across diffusion time ($\alpha = 0.8$, $L = 40$). $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t , and decays to zero: the band fills, then empties.

Chapter 6

The Gaussian Chain by Belief Propagation

Chapter 5 obtained the joint score by inverting a matrix. This chapter obtains the same score by passing messages along the chain, which costs $O(L)$ updates instead of an $O(L^3)$ inversion and, unlike the matrix route, survives the loss of Gaussianity in Chapter 7. Three things follow: the recursion is exactly the Kalman filter and the Rauch–Tung–Striebel smoother, update for update; the posterior has a closed form in the bulk; and the influence of distant evidence decays geometrically, which is what makes a local approximation of the score possible at all.

6.1 Belief propagation on the chain

The matrix route inverts an $L \times L$ matrix, a global computation over the whole sequence. The score–posterior identity says the same object is a posterior expectation, and the posterior has a structure that supports local computation. This section develops that computation once, in full.

6.1.1 The factor graph of the posterior

For any Markov-chain prior and per-frame OU evidence, Bayes' rule gives

$$p(a | x) \propto \psi_{\text{pr}}(a_0) \prod_{k=0}^{L-2} \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \prod_{k=0}^{L-1} \psi_{\text{ob}}^{(k)}(a_k), \quad \psi_{\text{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t} a_k, \Delta_t), \quad (6.1)$$

a hidden Markov model (Rabiner, 1989): a latent chain observed through independent per-frame noise. By (2.10), the k -th score component requires exactly the smoothing marginal $\mathbb{E}[a_k | x]$ of (6.1).

The factor graph of this posterior has L variable nodes, $2L$ factor nodes (one prior, $L-1$ transitions, L observations; the data x_k are constants clamped inside their factors) and $3L - 1$ edges. One fewer edge than nodes, on a connected graph, is exactly the tree condition of Section 3.6.

Proposition 6.1. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a | x)$ factorises over a tree-shaped factor graph, and all L smoothing marginals are computed exactly by one forward and one backward sweep of sum-product messages.*

Proposition 6.1 counts sweeps, not work: it says nothing about how the messages are represented, and for continuous variables each message is a function (Section 3.7). Keeping tree-exactness of the algorithm (a graph property) separate from representability of the messages (a property of the model's function class) is the conceptual hinge of this thesis; the Gaussian case closes the gap, and Chapter 7 measures what happens when it no longer does.

6.1.2 The sum-product recursion, with the bookkeeping explicit

Specialising the sum-product rules (3.12) to this tree, the schedule collapses into one forward and one backward sweep (Theorem 3.5), and the bookkeeping deserves care.

Proposition 6.2 (Chain sweeps, Convention A). *Define messages*

$$\mu_{\rightarrow 0}(a_0) := \psi_{\text{pr}}(a_0), \quad \mu_{\rightarrow k+1}(a_{k+1}) := \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) \, da_k, \quad (6.2)$$

$$\mu_{\leftarrow L-1}(a_{L-1}) := 1, \quad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) \, da_k. \quad (6.3)$$

Then for every k ,

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) : \quad (6.4)$$

the forward message into a_k carries the evidence $x_0, \dots, x_{k-1}$ only, the backward message carries $x_{k+1}, \dots, x_{L-1}$ only, and the local evidence enters once, at combination time.

The proof is direct marginalisation: integrating (6.1) over all a_j , $j \neq k$, splits at a_k by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions.

Remark 6.3 (The double-counting trap). An alternative convention absorbs the local evidence into the forward message, $\mu_{\rightarrow k}^{\text{B}} = \mu_{\rightarrow k} \cdot \psi_{\text{ob}}^{(k)}$; the product (6.4) then counts x_k twice and must be corrected by dividing one factor back out. The cost of getting this wrong is not subtle: under Convention A the BP score matches the matrix score to 10^{-14} , and under the naive alternative the disagreement is of order one. Every message in this thesis is therefore stated with its evidence content explicit.

6.2 The Gaussian closure and the Kalman smoother

For the Gaussian chain, two facts close the recursion, and they are the only ones needed: a product of Gaussians in the same variable is Gaussian (3.14), and an affine map of a Gaussian, marginalised, is Gaussian (3.15). Together they say that the recursions (6.2)–(6.3) perform exactly one product and one affine marginalisation per step. The local evidence, read as a function of a_k , is itself Gaussian, $\psi_{\text{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$: a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t , the pinning strength that appears throughout this chapter.

Theorem 6.4 (Gaussian closure of BP on the chain). *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every (L, α, t) . Each message is carried by two numbers, each update is a few arithmetic operations, and the two sweeps compute the exact posterior marginals, hence the exact joint score. Here, and only here, the sweep count of Proposition 6.1 translates into an actual computation of fixed, small cost per frame.*

Derivation. Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the zero-precision limit of the family). Each step of (6.2)–(6.3) is one application of the product identity (3.14) (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (6.4) is Lemma 1 twice. Exactness follows from tree exactness of sum–product (Theorem 3.5). $\square$

Remark 6.5 (Closure is algebra, not a central limit theorem). The closure just proved holds at node degree two exactly as it would at any degree: it is algebra, not an aggregation effect, as discussed in the companion compendium. The central-limit argument matters only for the per-node closures of statistical physics (recorded in the companion compendium), and on the chain those fail in a way that can be made exact, with a breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ (the companion compendium).

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, update by update, the classical algorithms of linear estimation.

Toolbox 6.1 — The Kalman filter, derived from the BP sweep

Write the forward message in Convention A as $\mu_{\rightarrow k} = \mathcal{N}(a_k; m_{k|k-1}, P_{k|k-1})$, the predicted state given past observations. One BP step does two things.

Update (absorb the evidence; Lemma 1). Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the filtered Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\frac{1}{P_{k|k}} = \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t},$$

$$m_{k|k} = P_{k|k} \left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t} x_k}{\Delta_t} \right) = m_{k|k-1} + G_k(e^t x_k - m_{k|k-1}),$$

where $G_k = P_{k|k} e^{-2t} / \Delta_t \in (0, 1)$ is the Kalman gain: the filtered mean is the prediction corrected by a gain-weighted innovation (Kalman, 1960).

Predict (push through the dynamics; Lemma 2).

$$m_{k+1|k} = \alpha m_{k|k}, \quad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

The forward BP sweep is the Kalman filter, message by message.

Smoothing (combine both directions). The belief (6.4) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given all observations, $\mathcal{N}(a_k; m_{k|L-1}, P_{k|L-1})$; carrying out the same algebra for the backward recursion yields the fixed-interval smoother of Rauch et al. (1965). The two-sweep BP schedule on the chain is the Kalman filter followed by the RTS smoother, exactly as the generic correspondence recorded in the companion compendium predicts. $\square$

Combining with the score-posterior identity answers RQ2: the k -th component of the joint diffusion score is an affine function of the Kalman-smoothed estimate of frame k ,

$$S_k(x, t) = \frac{e^{-t} m_{k|L-1} - x_k}{\Delta_t}. \quad (6.5)$$

For Markov sequences, score computation is a smoothing problem, and the classical theory of linear smoothing applies directly.

Numerical agreement of the two routes. The audit suite runs BP against the matrix route over 80 configurations ($L \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x): maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores, i.e. double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix A). One qualitative detail from a worked $L = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is positive

although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

6.3 The bulk in closed form, and the price of locality

For a long chain the interior (bulk) is translation invariant and can be solved exactly. By Proposition 5.5 the posterior precision in the bulk is tridiagonal Toeplitz with

$$\boxed{J_d = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad \beta = -\frac{\alpha}{\sigma_\eta^2}, \quad \sigma_\eta^2 = 1 - \alpha^2.} \quad (6.6)$$

The diagonal J_d is the sum of the evidence precision e^{-2t}/Δ_t (divergent as $1/2t$ at small t , vanishing at large t) and the constant prior precision; the coupling β is pure prior. The dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

Theorem 6.6 (Exact bulk variance and geometric covariance). *In the bulk of a long chain, the posterior variance and covariance are*

$$\boxed{V = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (J^{-1})_{i,i+d} = q^d V, \quad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2|\beta|} \in (0, 1).} \quad (6.7)$$

Toolbox 6.2 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^d$ for $d \geq 0$, symmetric in $\pm d$.

The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta Cq^{d-1} + J_d Cq^d + \beta Cq^{d+1} = 0 \quad \Longleftrightarrow \quad \beta q^2 + J_d q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$ all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta Cq = 1$, and since $2\beta q =$

$-(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which always holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} > 0$$

for every $|\alpha| < 1$ and $t > 0$. This positivity margin is also what guarantees the belief-propagation fixed point exists for every coupling and diffusion time (the companion compendium). $\square$

The number $q(\alpha, t)$ measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$. Its limits are

$$\begin{aligned} t \rightarrow 0 : \quad q &\approx \frac{|\beta|}{J_d} \rightarrow 0 \quad (\text{each frame pinned by its own observation}); \\ t \rightarrow \infty : \quad q &\rightarrow |\alpha| \quad (\text{posterior} \rightarrow \text{prior}), \end{aligned} \tag{6.8}$$

the second because $q \rightarrow ((1 + \alpha^2) - (1 - \alpha^2))/(2|\alpha|) = |\alpha|$ when the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from 0 to the prior's own $1/\log(1/|\alpha|)$.

6.3.1 The locality law

RQ4 asks how rapidly the influence of remote observations decays. Define the local estimator of radius r : the posterior mean of a_k given only the observations within distance r , $m_k^{(r)} = \mathbb{E}[a_k | x_{k-r}, \dots, x_{k+r}]$. Two facts make the comparison clean. Any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact on its window; the only approximation is the truncation of range. And both estimators are linear in x , so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\text{RMS}_r = \sqrt{e^\top \Sigma_t e}$, a deterministic number with no sampling involved.

Proposition 6.7 (Geometric locality: influence coefficients). *In the bulk, the influence of the observation at distance d on the exact posterior mean is $(J^{-1})_{k, k \pm d} \propto q(\alpha, t)^d$, exactly,*

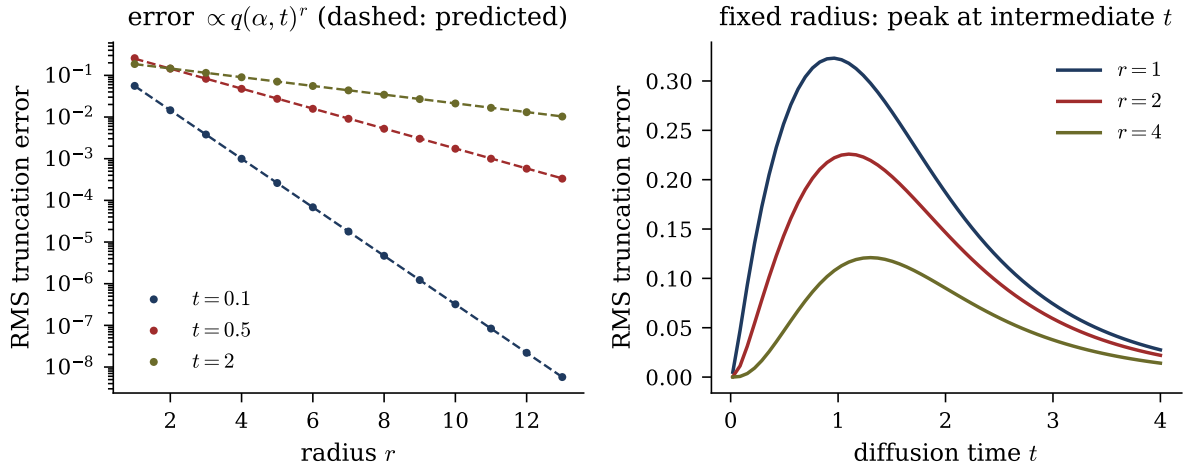


Figure 6.1: Locality of the joint score ($\alpha = 0.8$, $L = 121$, deterministic RMS over the data law). Left: exact truncation error of the radius- r local estimator (dots) at three diffusion times; the dashed lines have the predicted slopes $q(\alpha, t)^r$ from (6.7), with no fitting. Right: the same error at fixed radius as a function of t : it vanishes at both ends and peaks at intermediate t , where the band of Q_t is fullest.

by Theorem 6.6. Truncating the evidence at radius r therefore discards a leading term of order q^{r+1} .

What Proposition 6.7 controls is the size of the *discarded* evidence, and that part is exact. It does not by itself prove that the radius- r estimator's error inherits the rate: $m_k^{(r)}$ is the posterior mean of a *re-normalised* model on the window, so truncation changes the boundary conditions as well as removing the leading coefficient, and bounding the difference between those two effects is not done here.

The rate is instead confirmed numerically, and closely. Fitted slopes of $\log \text{RMS}_r$ over $r = 2, \dots, 13$ at the centre of an $L = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 6.1, where the dashed lines carry the predicted slope with no fitting. The claim this thesis makes is therefore: an exact geometric law for the influence coefficients, and a measured geometric law with the same rate for the windowed estimator's error, over the stated range of r . A two-sided bound establishing the second from the first is left open.

6.3.2 Truncating the matrix versus truncating the inference

A truncated score can be built in two inequivalent ways. *Truncate the answer*: band the matrix, $\hat{S} = -Q_t^{(b)}x$ with entries beyond $|i - j| \leq b$ zeroed. *Truncate the algorithm*: let messages travel at most r hops, i.e. run exact inference on the window $x_{k-r}, \dots, x_{k+r}$ (the local estimator above). Both are linear in x , so their RMS errors are again deterministic. Table 6.1 compares them at equal locality budget ($b = r = 1$, $L = 40$, $\alpha = 0.8$):

Table 6.1: Truncating the answer against truncating the algorithm, at equal locality budget ($b = r = 1$, $L = 40$, $\alpha = 0.8$). Both estimators are linear in x , so both errors are deterministic rather than sampled. The cost of locality peaks at intermediate diffusion times and the ordering reverses nowhere.

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
banded matrix ($b = 1$)	0.210	0.302	0.135	0.0035
truncated inference ($r = 1$)	0.123	0.235	0.130	0.0035

Two observations. The cost of locality is non-monotone in t and peaks at intermediate times, consistent with Section 5.4.1. And at small t truncating the algorithm beats truncating the answer (12.3% against 21.0% at $t = 0.05$): the windowed inference solves a correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large t , where the off-diagonal content vanishes. Either error is measured to decay geometrically in the allowed range at the rate $q(\alpha, t)$ of Proposition 6.7.

Chapter 7

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 5 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable, it is a minimal step outside the Gaussian family, so every deviation we find is attributable to non-Gaussianity alone. The chapter first establishes what remains exact (the score-posterior identity, the tree structure, one boundary integral, and a differential characterisation of the $L = 1$ marginal), then locates precisely where closed forms stop, and finally measures, against a validated numerical reference, what a Gaussian message family costs.

7.1 What changes, and what is preserved

The Laplace AR(1) chain replaces the Gaussian innovations of (3.2) with Laplace ones, keeping the initial law and the second-order structure fixed:

$$a_0 \sim \mathcal{N}(0, 1), \quad a_k = \alpha a_{k-1} + \varepsilon_k, \quad \varepsilon_k \sim \text{Laplace}(0, b) \text{ i.i.d.}, \quad k = 1, \dots, L-1, \quad (7.1)$$

observed through the same per-frame OU channel. The innovation scale is *not* free: $b = \sqrt{(1 - \alpha^2)/2}$, so that $\text{Var}(\varepsilon_k) = 1 - \alpha^2$ and the chain has unit marginal variance and $\text{Cov}(a_j, a_k) = \alpha^{|j-k|}$, exactly as in the Gaussian case. This is what makes the comparison in this chapter a comparison: the Gaussian and Laplace chains agree on every second moment and differ *only* beyond them, the innovation’s excess kurtosis (0 against +3) being the single non-Gaussianity knob. The same convention, Gaussian initial state, variance-matched innovations, is used for every non-Gaussian family in this thesis, and carries forward unchanged to the estimated chain of (8.1). Three structural facts survive unchanged, because none of them used Gaussianity of the prior. The score-posterior identity (2.10) holds for any p_0 . The posterior factorises over the same tree-shaped factor graph (Proposition 6.1), so the sum-product recursion (6.2)–(6.3) still computes exact smoothing marginals in one forward and one backward sweep. And the OU likelihood factors are still Gaussian in a_k . What is lost is representability: the messages are no longer Gaussian, no finite-dimensional family we know of is closed under the updates, and the cost of an update is no longer bounded. The score also ceases to be linear in x , which is the first thing to make precise.

7.2 One frame: the score becomes nonlinear

For $L = 1$ there is no chain; the model isolates the ingredient “non-Gaussian prior meets Gaussian channel”. Let $a \sim \text{Laplace}(0, b)$, $p_0(a) = e^{-|a|/b}/2b$, and write $\mu := e^{-t}$. The

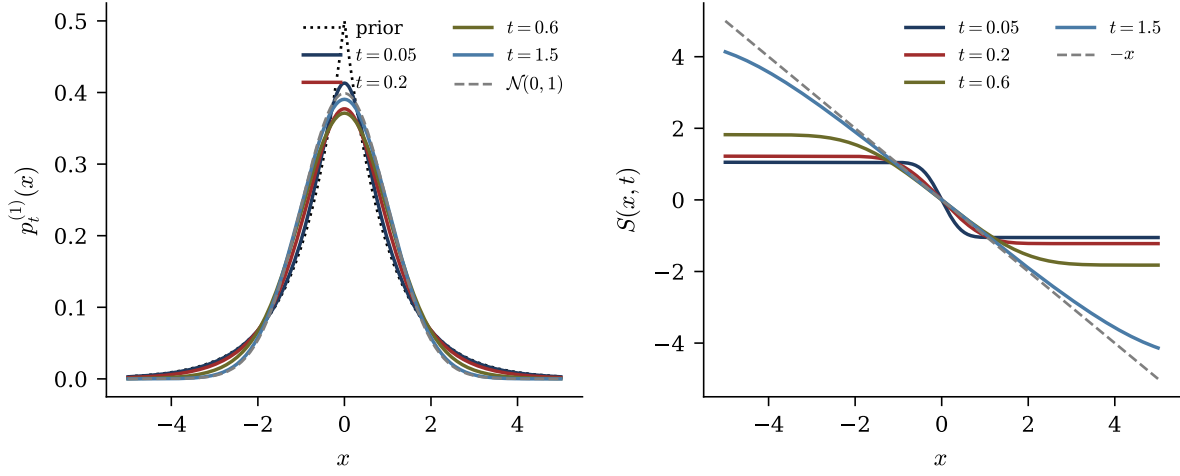


Figure 7.1: The Laplace $L = 1$ problem ($b = 1$). Left: noised marginal $p_t^{(1)}$ at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed). Right: the score at the same times; the dashed line is the attractor $-x$. The nonlinearity lives near $x = 0$ and flattens as t grows; the limits are the exact statements (7.3). Curves computed by deterministic quadrature.

noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a) \mathcal{N}(x; \mu a, \Delta_t) da. \quad (7.2)$$

As $t \rightarrow 0$ the Gaussian kernel concentrates and $p_t^{(1)} \rightarrow p_0$; as $t \rightarrow \infty$, $p_t^{(1)} \rightarrow \mathcal{N}(0, 1)$, the OU attractor. By the score-posterior identity, $S(x, t) = (\mu \mathbb{E}[a|x] - x)/\Delta_t$; for a Laplace prior the conditional mean is a nonlinear function of x , and the score inherits the nonlinearity. Its limits are exact: for every fixed $x \neq 0$,

$$t \rightarrow 0 : \quad S(x, t) \longrightarrow \partial_x \log p_0(x) = -\frac{\text{sign}(x)}{b}, \quad t \rightarrow \infty : \quad S(x, t) \longrightarrow -x. \quad (7.3)$$

At small t the score converges pointwise to the sign-shaped score of the prior, with the nonlinearity concentrated in the kink near $x = 0$; at large t it flattens into the linear attractor score $-x$ (Figure 7.1). The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement 1.19×10^{-7} over a grid of (b, t, x) , at the truncation floor of the finite difference.

7.2.1 A differential characterisation of the noised marginal

The channel writes x as a sum of independent variables, so (7.2) is a convolution, which factorises in Fourier space:

$$\hat{p}_t^{(1)}(k) = \frac{1}{1 + b^2 \mu^2 k^2} e^{-\Delta_t k^2/2}, \quad (7.4)$$

the product of the (rescaled) Laplace and Gaussian characteristic functions.

Theorem 7.1 (Screened Poisson equation). *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{(1 - b^2 \mu^2 \partial_x^2) p_t^{(1)}(x) = G_{\Delta_t}(x)}, \quad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \quad (7.5)$$

The derivation is Toolbox 7.1.

Toolbox 7.1 — From the characteristic function to the screened Poisson equation

Multiply (7.4) by $(1 + b^2 \mu^2 k^2)$:

$$(1 + b^2 \mu^2 k^2) \hat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is the characteristic function of G_{Δ_t} . Multiplication by k^2 in Fourier space is $-\partial_x^2$ in real space, so the polynomial $(1 + b^2 \mu^2 k^2)$ becomes the operator $(1 - b^2 \mu^2 \partial_x^2)$, and (7.5) follows by inverse transformation. The audit checks (7.4) against direct quadrature of the characteristic function to 2.9×10^{-6} , the floor of oscillatory quadrature. $\square$

Equation (7.5) states that the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$: the memory of the prior's tails shrinks as e^{-t} . The deviation from Gaussianity is governed by a single local differential operator whose strength dies with diffusion time.

One further exact consequence identifies what a Gaussian approximation must get

wrong. Differentiating the score-posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t} \left(1 - \frac{\mu^2}{\Delta_t} \text{Var}[a|x] \right). \quad (7.6)$$

For a Gaussian prior the posterior variance is independent of x and H_t is constant. For the Laplace prior $\text{Var}[a|x]$ depends on x and H_t is a nontrivial function: the breakdown of a constant curvature is the second-order form of the breakdown of score linearity. A Gaussian message is exactly the object that cannot represent an x -dependent curvature, so (7.6) states what the Gaussian closure of Section 7.5 will miss, before it is measured.

7.3 The chain: where closed forms stop

Theorem 7.2 (Joint score from chain messages). *For any chain prior, in particular (7.1), and the OU likelihood, the joint score is, componentwise,*

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k|x] - x_k}{\Delta_t}, \quad \mathbb{E}[a_k|x] = \frac{\int a_k \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}{\int \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}, \quad (7.7)$$

with the messages of Proposition 6.2.

This is a complexity statement: the L -dimensional posterior integral is replaced by two one-dimensional message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity; what does not survive is the collapse of those one-dimensional integrals into closed-form arithmetic. Exactly one such integral has a closed form.

Proposition 7.3 (One-step boundary integral). *For the Laplace transition $M(a'|a) = \frac{1}{2b} e^{-|a' - \alpha a|/b}$ and OU evidence $\mathcal{N}(x'; \mu a', \Delta_t)$,*

$$\int_{\mathbb{R}} M(a'|a) \mathcal{N}(x'; \mu a', \Delta_t) da' = p_t^{(1)}(x' - \mu \alpha a) : \quad (7.8)$$

the first backward message of the chain is the $L = 1$ noised marginal evaluated at a shifted

argument.

Proof. The innovation substitution $r := a' - \alpha a$, which turns the transition into the bare Laplace density at r and the evidence into $\mathcal{N}(x' - \mu\alpha a; \mu r, \Delta_t)$; the integral is then the definition (7.2). The shift $\mu\alpha a$ is what frame a predicts for the next noisy observation, and the whole $L = 1$ analysis, screened Poisson equation included, recycles as the chain's boundary building block.

Remark 7.4 (Why iteration destroys the closed form). Equation (7.8) works because the integrand contains exactly one Laplace kink times exactly one Gaussian. Every subsequent BP step violates this: the incoming message already carries the propagated structure of several frames, so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely, and the clean form does not survive one iteration. In the Gaussian chain every step is Gaussian-times-Gaussian and closure propagates forever; this is the precise structural location of the difference between the two models, and the reason the Gaussian projection below is an approximation with an error to be measured, not an identity.

A change of coordinates shows the obstruction is intrinsic, not an artefact of the frame basis. In innovation coordinates $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ (unit Jacobian), the prior factorises into independent variables, r_0 Gaussian and $r_{k \geq 1}$ i.i.d. Laplace, but the likelihood becomes $p(x|r) \propto \exp(-\frac{\mu^2}{2\Delta_t} r^\top B^\top B r + \frac{\mu}{\Delta_t} x^\top B r)$ with B the unit lower-triangular map $a = Br$, and $B^\top B$ is dense. The coupling does not disappear; it migrates from the prior to the likelihood. There is no basis in which both are diagonal at once.

7.4 A validated numerical reference: grid BP

Since functional messages admit no closed form, the exact recursion is approximated numerically: messages are represented by their values on N_g equispaced points in $[-A, A]$ and the update integrals are carried out by trapezoidal quadrature, each update touching

every pair of grid points. This *grid BP* is a numerical approximation of the functional-message recursion, and it can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain. There the trapezoid rule on analytic, rapidly decaying integrands converges spectrally, so failure is abrupt rather than gradual: at $N_g = 101$, $A = 8$ the score error sits at the 10^{-15} floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $N_g = 401$, $A = 8$, with a reference solution at $N_g = 801$ and the resolution constraint $\sqrt{2t} \gtrsim 3h$ on the grid spacing $h = 2A/(N_g - 1)$ checked automatically; grid self-convergence error remains 10^4 to 10^7 times smaller than every effect measured below. Within this regime we treat grid BP as the reference score; outside it no claim is made.

Remark 7.5 (Symbols that are easy to swap). Three pairs are worth fixing before the numerical sections use them. The grid resolution is N_g and the number of training *sequences* is N ; they are independent, and every configuration states both. The chain length is L and its internal index is u , never t , which is reserved for diffusion time throughout. And q always means the geometric decay rate $q(\alpha, t)$ of Section 6.3; the innovation variance is σ_η^2 , never q . The companion paper writes the sequence count M where this thesis writes N ; the grid symbol is the same in both.

All score errors are amplified posterior-mean errors: for any two solvers with means $\widehat{m}$ and m_{ref} , the score-posterior identity gives, exactly,

$$\widehat{s} - s_{\text{ref}} = \frac{e^{-t}}{\Delta_t} (\widehat{m} - m_{\text{ref}}), \quad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0). \quad (7.9)$$

The numerical residual of this identity is logged in every experiment and stays at 10^{-15} to 10^{-12} . Equation (7.9) explains in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

7.5 The Gaussian-message approximation, measured

The object under test is Gaussian-projected BP: the same sweeps, with every message forced into the Gaussian family by moment matching. This is the algorithm one would deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of statistical physics (the companion compendium).

Before measuring its error, identify exactly what it computes, because the answer turns this from an implementation-specific approximation study into a precise question. Write $\Pi_G f$ for the Gaussian density matching the mean and variance of $f/\int f$.

Theorem 7.6 (Gaussian closure is the LMMSE estimator). *Let moment-projected sum-product be the forward-backward recursion of (6.2)–(6.3) that applies Π_G to each message right after every transition integral, multiplying by the site's unary likelihood before the next. Take the chain (7.1) with $\alpha \neq 0$, Gaussian unary likelihoods, $\mathbb{E}[a_0] = 0$, an innovation law with mean zero and finite variance σ_η^2 , and an initial message that is Gaussian or projected by Π_G . Then its posterior means are exactly those of the covariance-matched Gaussian model:*

$$\hat{a}_{\text{LMMSE}} = \text{Cov}(a, x) \text{Cov}(x)^{-1} x = e^{-t} \Sigma_0 \left(e^{-2t} \Sigma_0 + \Delta_t I \right)^{-1} x, \quad (7.10)$$

and it returns the same estimator for every innovation law sharing the moments $(0, \sigma_\eta^2)$.

7.5.1 The hypotheses

Four hypotheses are load-bearing and are worth separating, because three of them are easy to read as technical and are not.

Finite second moments are what make Π_G defined at all.

Zero mean is not cosmetic. The general LMMSE estimator is

$$\mathbb{E}[a] + \text{Cov}(a, x) \text{Cov}(x)^{-1} (x - \mathbb{E}[x]),$$

and (7.10) is its centred special case.

$\alpha \neq 0$ is required because the backward map of the closure recursion divides by α . At $\alpha = 0$ the chain has no edges, the sites are independent, and the claim holds trivially and separately, each site's posterior mean being the scalar Gaussian $e^{-t}\Sigma_0(e^{-2t}\Sigma_0 + \Delta_t)^{-1}x_i$.

Where the projection is applied matters because Π_G does not commute with multiplication by the likelihood. Theorem 7.6 fixes the order — project immediately after each transition integral, then multiply by the site's unary likelihood — and a different order is a different algorithm with a different fixed point.

One further point of scope. The statement concerns exact message *functions* on $\mathbb{R}$ and an analytic projection operator. **The grid of Section 7.4 plays no part in it.** Discretisation enters only when the *unrestricted* non-Gaussian messages are represented numerically, which is what the closure is compared against. Conflating the two would turn a theorem about a closure into a claim about quadrature.

7.5.2 Proof of Theorem 7.6

Proof. Write $\vec{m}_i$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood at that site. We show by induction along the chain that every projected message coincides with the corresponding exact message of the covariance-matched Gaussian chain.

Base. The initial message is Gaussian by hypothesis, or is projected to the Gaussian with the same first two moments. Either way it equals the Gaussian chain's initial message, which has those same moments by construction.

Forward step. Suppose $\vec{m}_{i-1}$ agrees with the Gaussian chain's message, so it is Gaussian with mean m and variance v . Multiplying by ℓ_{i-1} is a product of two Gaussian densities and yields a Gaussian, with mean and variance given by the usual precision-weighted combination (3.14); this is identical in both chains, because the likelihood is the same object in both. Taking the transition integral against $a_i = \alpha a_{i-1} + \varepsilon_i$ convolves with the innovation law and scales by α , which maps

$$(m, v) \longmapsto (\alpha m, \alpha^2 v + \sigma_\eta^2). \quad (7.11)$$

The mean and variance of a convolution depend on the innovation law *only* through its own mean and variance. So (7.11) is the same map for every innovation with moments $(0, \sigma_\eta^2)$, and in particular the same as for the Gaussian chain. In the Gaussian chain the convolution of two Gaussians is already Gaussian and no projection occurs; here Π_G is applied, and since the pre-projection function has exactly the moments just computed, the projection returns precisely the Gaussian chain's message.

Backward step. The same argument runs with the transition map inverted,

$$(m, v) \mapsto (m/\alpha, (v + \sigma_\eta^2)/\alpha^2), \quad (7.12)$$

which is where $\alpha \neq 0$ is used. Its base case needs one word of care. The terminal backward message is $\overleftarrow{m}_{L-1} \equiv 1$, which is not a normalisable density and so is not in the domain of Π_G . The induction therefore starts one step earlier, at the terminal unary likelihood ℓ_{L-1} , which is Gaussian and normalisable; every subsequent backward message is a likelihood-weighted transition integral of a normalisable Gaussian and stays in the domain. Nothing is lost by starting there, because $\overleftarrow{m}_{L-1} \equiv 1$ contributes no factor: the first object the recursion actually projects is ℓ_{L-1} .

Conclusion. Every projected belief therefore equals the corresponding belief of the covariance-matched Gaussian chain, and the returned posterior mean is that chain's posterior mean. For jointly Gaussian zero-mean (a, x) the conditional mean is linear in x and so coincides with the best linear estimator. Since $x = e^{-t}a + \sqrt{\Delta_t}z$ with z independent of a ,

$$\text{Cov}(a, x) = e^{-t}\Sigma_0, \quad \text{Cov}(x) = e^{-2t}\Sigma_0 + \Delta_t I, \quad (7.13)$$

which substituted into $\text{Cov}(a, x) \text{Cov}(x)^{-1}x$ gives exactly (7.10). Finally, the recursion never read the innovation law beyond its moments $(0, \sigma_\eta^2)$ — neither (7.11) nor (7.12) involves anything else — so the same estimator is returned for every innovation law with those moments. $\square$

Remark 7.7 (What the theorem does not say). It does not say the closure is a good

approximation. It says what the closure *computes*, which is a different and more useful kind of statement: the gap measured in Section 7.5 is the distance between full non-Gaussian posterior inference and the best second-order linear estimator of the matched Gaussian model. Whether that distance is small is an empirical question, answered there and not here.

Theorem 7.6 fixes the meaning of every number below. The comparison is not “grid BP against one particular approximation”: it is *full non-Gaussian posterior inference against the best second-order linear estimator*, and the measured gap is precisely the price of discarding everything beyond second moments. It also explains why the errors reported here are attributable to non-Gaussianity alone, the projected arm is literally blind to the only thing that distinguishes the two chains.

Table 7.1 reports the median relative score error of the Gaussian closure on the Laplace chain:

Table 7.1: The Gaussian-message approximation against validated grid BP on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t , reference $N_g = 801$). By Theorem 7.6 the first row is the price of discarding everything beyond second moments, not the error of one particular implementation. The second row is the share of trials exceeding 10% relative error, which reaches zero an order of magnitude in t before the median does.

t	0.05	0.32	0.98	2.07	3.0
median relative score error	0.24	4.6×10^{-2}	6.8×10^{-3}	2.5×10^{-4}	7.3×10^{-6}
$\mathbb{P}(\text{err} > 0.1)$	1.00	0.08	0.00	0.00	0.00

(Figure 7.2). Two thresholds are worth separating, because they are different statistics and sit an order of magnitude apart in t . The probability of a trial exceeding 10% relative error first reaches zero at $t \approx 0.98$; the *median* error first falls below 10^{-3} only at $t \approx 2.1$. In between, around $t \approx 1$, where the failure probability has already vanished, the typical error is still $\sim 7 \times 10^{-3}$.

The closure error is of order tens of percent at small diffusion time and decays as diffusion noise grows. The natural explanation is that the tilted posteriors approach Gaussianity, and it is consistent with the measurement without being established by it: the Tweedie prefactor e^{-t}/Δ_t also attenuates with t , so a falling score error does not by

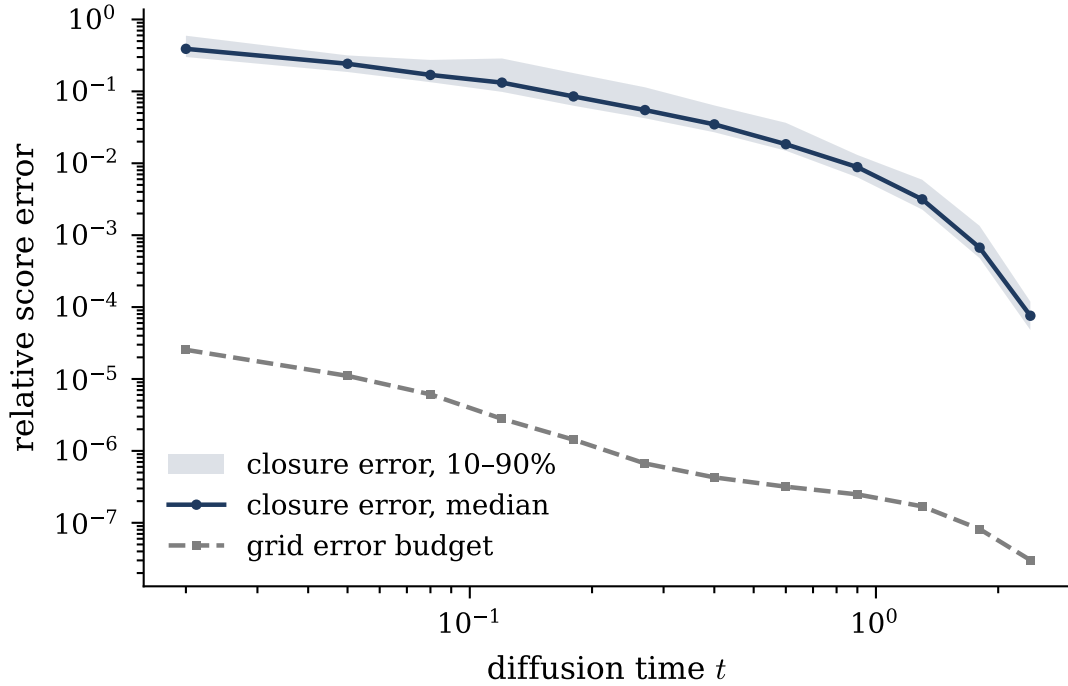


Figure 7.2: Gaussian-closure score error on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t ; median and 10–90% band) against the grid error budget measured on identical data. The closure error exceeds the reference error by 10^4 to 10^7 at every t , so the measurement is trustworthy where it is used.

itself separate a posterior becoming more Gaussian from a prefactor damping the effect of any given posterior-mean discrepancy. Posterior non-Gaussianity is not measured directly here. Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian, occasionally bimodal; the single Gaussian commits to a compromise mean; and (7.9) amplifies the shift. The score *direction* is much more robust than its magnitude: the median cosine similarity between projected and reference scores is 0.976 at $t = 0.05$ (where the magnitude error is 24%), exceeds 0.99 for all $t \geq 0.11$, and the worst single trial of the sweep stays above 0.879. These are numerical observations for the stated parameter ranges, not theorems.

7.5.3 Varying the innovation law

Sweeping six innovation families with identical covariance $\alpha^{|i-j|}$ (Laplace, Student- t with $\nu = 5, 8$, and symmetric Gaussian mixtures of increasing separation) isolates the shape of the innovation law as the only moving part. Three observations, all within the tested grids. First, the noise level dominates: every family's error decays to about 10^{-3} or below by $t = 2.4$; all substantial closure error lives at $t \lesssim 0.4$. Second, bimodality is the worst case: at $t = 0.05$, $\alpha = 0.5$, the median error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student- $t_{\nu=5}$, and 0.07 for a mildly bimodal mixture; a single Gaussian cannot represent two posterior modes and commits to the gap between them. Third, stronger chain correlation helps rather than hurts: for Laplace at $t = 0.05$ the error is 0.43, 0.27, 0.13 at $\alpha = 0.5, 0.85, 0.95$; informative neighbours act like additional Gaussian observations, which is consistent with the local tilted posteriors becoming more Gaussian, again as an interpretation rather than a measurement.

Chapter 8

Learning the Parameters: EM on the Chain

Everything so far assumed the transition kernel was given. This chapter drops that assumption. The data are still a Markov chain observed through the Ornstein–Uhlenbeck channel, but the kernel that generated them is unknown and has to be recovered from the noised sequences alone, no clean data, no supervision, and no differentiation through the inference.

The tool is expectation–maximisation, and the reason it is cheap here is Fisher’s identity: one complete forward–backward pass of belief propagation already contains the exact gradient of the marginal log-likelihood with respect to the parameters. The E-step is the BP recursion of Chapter 6 with the kernel held fixed; the M-step moves the kernel. Nothing new has to be implemented, and nothing has to be back-propagated through.

8.1 Model: an estimated mixture innovation

The clean chain is the one of (7.1), in the index convention of this chapter,

$$a_1 \sim \mathcal{N}(0, 1), \quad a_i = \alpha a_{i-1} + \varepsilon_i, \quad i = 2, \dots, n, \quad (8.1)$$

with innovations ε_i i.i.d., mean zero, variance $\sigma_\eta^2 = 1 - \alpha^2$, the same Gaussian initial state and the same variance-matching that Chapter 7 used, but now drawn from a member of a *parametric family* rather than a fixed law. Forward noising is the coordinatewise Ornstein–Uhlenbeck channel of Section 3.2, $x = e^{-t}a + \sqrt{\Delta_t}\xi$, unchanged from every earlier chapter.

The estimated family is a C -component Gaussian mixture on the innovation. It is the transition kernel written $M(a' | a)$ in Chapters 3 and 7, carrying its parameters explicitly because they are now what is being estimated:

$$K_\theta(a' | a) = \sum_{c=1}^C \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad \theta = \{\alpha, (\pi_c, \nu_c, s_c)_{c=1}^C\}, \quad (8.2)$$

so both the autoregressive coefficient α and the shape of the innovation density are free parameters, estimated jointly by maximum marginal likelihood from noised data alone. At $C = 1$ this collapses to a single Gaussian, recovering the model of Chapter 5; larger C can represent skew, heavy tails, and multimodality to the resolution of the grid. The initial law $a_1 \sim \mathcal{N}(0, 1)$ is *held fixed* at estimation time, it is the value used to generate every family compared in Chapter 10, so there is no misspecification in it, but neither is there direct evidence it could itself be recovered from noised data; that question is not addressed here.

Two assumptions carry the argument and, consistent with the practice of this thesis, are stated rather than tested by the estimator itself: the clean process is exactly first-order Markov, and the forward noise is exactly coordinatewise. Section 10.4 of Chapter 10 measures what happens when the first of these is false by a controlled amount; the second is never relaxed anywhere in this thesis.

Remark 8.1 (Covariance-stationary, not strictly stationary). Choosing $\sigma_\eta^2 = 1 - \alpha^2$ makes the chain covariance-stationary — $\text{Var}(a_i) = 1$ and $\text{Cov}(a_i, a_j) = \alpha^{|i-j|}$ hold exactly for *every* innovation law, by the same second-moment recursion as the Gaussian case. It does not make the chain strictly stationary: for a non-Gaussian innovation the invariant marginal law of the chain is not $\mathcal{N}(0, 1)$, so fixing $a_1 \sim \mathcal{N}(0, 1)$ starts every family off its

own invariant distribution. Nothing compared in Chapter 10 depends on the distinction, because every family compared shares the same a_1 and the same covariance structure. The claim available here is covariance-stationarity, and it is the one used.

8.2 What remains exact

Two facts from Section 7.4 carry over unchanged, and everything the estimator does rests on them, so they are worth restating here.

Proposition 8.2 (The posterior factor graph is still the prior’s chain). *Coordinatewise forward noising contributes exactly one unary factor per site, $x_i \mapsto \phi_i(a_i) \propto \exp\left(-(x_i - e^{-t}a_i)^2/2\Delta_t\right)$. It reweights node potentials and creates no new edges among the latent a_i , for any innovation law, mixture or otherwise. The posterior over the clean chain given the noised observation therefore factorises on exactly the same chain graph as the prior.*

Proposition 8.3 (The E-step is exact). *Since the posterior graph is a tree, one forward and one backward sum-product sweep return the exact single-site and pairwise marginals of the posterior, at the level of functional messages, for any innovation law. Consequently the E-step of an EM procedure for θ requires no variational approximation and no sampling: the sufficient statistics EM needs are exact conditional expectations under the true posterior, mixture label included.*

Proof. Both are consequences of the chain being a tree (Theorem 3.5): the observation factors are unary, so they reweight nodes without joining any pair, and sum-product on a tree returns exact marginals whatever the factors are. Neither step assumes the innovation law admits a closed-form message update. $\square$

What is *not* exact is the representation of a functional message on a computer, which is where Section 8.4 and the grid-BP machinery of Section 7.4 enter.

8.3 The EM algorithm

The latent variables are the clean states a and the mixture labels $c = (c_i)$, so write the complete-data kernel with its label exposed,

$$\widetilde{K}_\theta(a', c | a) = \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad K_\theta(a' | a) = \sum_c \widetilde{K}_\theta(a', c | a). \quad (8.3)$$

Fisher's identity then gives the gradient of the *marginal* log-likelihood $\mathcal{L}(\theta)$ as a posterior expectation of the complete-data score,

$$\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}_{a,c|x,\theta} \left[\sum_{i=2}^n \nabla_\theta \log \widetilde{K}_\theta(a_i, c_i | a_{i-1}) \right] = \langle \Xi(\theta), \nabla_\theta \log \widetilde{K}_\theta \rangle, \quad (8.4)$$

where $\Xi(\theta)$ collects the exact posterior sufficient statistics BP supplies at the current parameter — the conditional expectations of the pairwise sums, the label responsibilities, and the second moments that (8.3) requires — and the pairing is the sum over edges of the corresponding expectations. The initial law and the observation channel carry no θ , so neither contributes a term. The practical content of (8.4) is that belief propagation is never differentiated through: BP supplies the conditional expectation $\Xi(\theta)$ at the current parameter, and the gradient with respect to θ is then an ordinary derivative of the kernel's log-density. This is what makes the E-step of Proposition 8.3 directly usable inside an optimiser.

The M-step for the mixture kernel is *not* an exact maximisation. Given $\Xi(\theta^{(k)})$, updating $(\pi_c, \nu_c, s_c)_{c=1}^C$ exactly would require solving a mixture-likelihood problem with its own latent label, so the implementation alternates closed-form *conditional* maximisers, the mixture block and α , until the per-edge gain in Q falls below a tolerance or a cap of sixteen sweeps is reached, recording on the returned kernel which of the two occurred. The algorithm is therefore generalised EM, or ECM (Meng and Rubin, 1993), not exact EM: a sweep ascends Q without maximising it.

The cap of sixteen is set by measurement, not convention. Varying only this budget at $C = 8$, $n = 256$, and a fixed outer count of 40, the fitted innovation excess kurtosis moves

by 28% between four sweeps and sixteen ($1.68 \rightarrow 2.14$, against a true value of 3.0) while α moves in the fourth decimal, the same rate asymmetry between shape and correlation that Section 9.1 finds in the outer loop, appearing one level down. The effect is confined to short outer runs: by 60 outer iterations the two loops substitute for one another and the inner budget stops mattering. Sixteen removes the interaction rather than requiring a reader to establish which regime a given cell is in.

Monotone ascent of $\mathcal{L}$ is guaranteed by the standard EM argument applied to the surrogate $Q(\theta \mid \theta^{(k)}) = \langle \Xi(\theta^{(k)}), \log \widetilde{K}_\theta \rangle$, ascending Q is all monotonicity requires, and is checked directly: across an executable audit of the implementation, the minimum increase in Q over twelve consecutive steps was +2.61 for a Gaussian kernel, +4.34 for Laplace, and +4.18 for an eight-component mixture, with zero decreases recorded in any logged run. The returned kernel’s evidence equals `trace.log_evidence[-1]` by construction, with a regression test asserting it, so the reported convergence trace always describes the model actually returned.

8.4 The numerical representation: grid BP, again

Messages remain functions on $\mathbb{R}$, so they are represented exactly as in Section 7.4: values on N_g equispaced points in $[-A, A]$, updates by trapezoidal quadrature. Two error sources are tracked separately here as well. Truncation error, the probability mass the finite domain $[-A, A]$ misses, is diagnosed by the maximum boundary mass over the sites of a chain. That is a statistic of a sampled trajectory rather than a bound, so it is recorded over 256 chains per configuration: the worst chain falls from 2.3×10^{-3} at $A = 4$ to 1.2×10^{-6} at $A = 8$, the 90th percentile from 2.7×10^{-4} to 1.5×10^{-8} , and neither depends on the grid resolution N_g . The gap between the two rows matters, because a figure read off a single trajectory sits near the median and so understates the tail by two orders of magnitude. Quadrature error, from the finite spacing $h = 2A/(N_g - 1)$, the grid holds N_g points including both endpoints, on an otherwise exact interior update, falls as $O(h^2)$: a factor of four per halving of h , matching the trapezoidal rate on a smooth

integrand. At the working configuration used for every experiment in this chapter and in Chapters 9 and 10, $N_g = 401$, $A = 8$, both sources sit far below any effect subsequently measured: the discretised recursion reproduces the closed-form Gaussian posterior mean to 9.2×10^{-15} , and agrees with an independently coded brute-force enumeration, sharing no code with the BP recursion, to 1.6×10^{-14} on posterior means and 1.8×10^{-15} on the log-evidence.

Which finite-grid model is being optimised. “Exact for the discretised chain” needs one more specification than it has been given so far, because there are two conventions available and they lead to different objectives. The transition array used here is the continuum density $K_\theta(a' | a)$ evaluated at the grid points and combined with the trapezoidal weights, with *no* per-column renormalisation. BP is therefore exact, up to floating point, for the chain-shaped *factor model* that array defines, and that model is not literally a normalised finite-state transition kernel on the truncated domain. The distance between the two is precisely the column-mass residual already reported as the quadrature diagnostic. At the working configuration, the worst column with $|a| \leq A/2$ departs from unit mass by 9.6×10^{-4} ; such a column sits far enough inside the domain that its transition density has no tail outside the grid, so its residual measures quadrature alone. The worst column anywhere departs by 1.9×10^{-2} , at the domain edge where the density is truncated.

The alternative convention, renormalising each parent-conditioned column under the quadrature weights, would make the array a genuine finite-state kernel. Its normaliser would then depend on θ , most strongly near the boundary, and that dependence would have to be carried: the derivative of $\log Z_\theta$ would enter both Fisher’s identity and the M-step, and the surrogate could no longer be written as $Q(\theta | \theta^{(k)}) = \langle \Xi(\theta^{(k)}), \log \widetilde{K}_\theta \rangle$ with K_θ the raw density. Because no renormalisation is applied, that term is absent by construction and not by oversight, and the surrogate above is the correct one for the model actually fitted. The price is that the fitted object is the continuum kernel read through a quadrature rule, which is what the column-mass residual measures and what the resolution gate of Section 8.4 bounds.

Figure 8.1 separates the two error sources the paragraphs above keep apart: resolution and truncation are refined one at a time, so that neither diagnostic can be mistaken for a certificate of the other.

Remark 8.4 (The boundary-mass diagnostic is empirical, not a bound). A boundary-mass figure computed from one sampled chain does not bound the truncation error for every run, and the gap is large enough to matter. Running the same recursion on one chain and on 256 of them, at the working configuration and over all 12 noise levels, the batch maximum exceeds the single-chain reading by a median factor of 86 and by as much as 3,281; at the mildest level it is only 1. A single draw therefore lands somewhere in the body of a distribution with a very long tail, and where it lands is not knowable from the draw. This is why the figures above are reported as a maximum and a 90th percentile over chains rather than as one number. The diagnostic is an empirical estimate of typical truncation loss, not a certified bound. What it does *not* affect is the separation of truncation from quadrature error reported above, which is a pure function of (A, N_g) and the prior and involves no sampled chain at all, that comparison reproduces to 0.000 relative difference under reseeding and is exactly the result Section 7.4 already relies on.

One further identity is worth restating from Section 7.4 because it governs every result in Chapter 10 exactly as it governed Chapter 7: whatever an approximation, Gaussian closure, grid truncation, or an under-converged fit, does to the posterior mean is amplified by $e^{-t}/\Delta_t \sim 1/2t$ in the score, so small diffusion times remain the regime where any deficiency is most visible.

Figure 8.2 collects the optimisation-side checks: monotone ascent of the marginal likelihood, recovery of α from initialisations far from the truth, and the $n^{-1/2}$ decay of the recovery error. The first two are what make the procedure an EM rather than a heuristic; the third is what makes it an estimator.

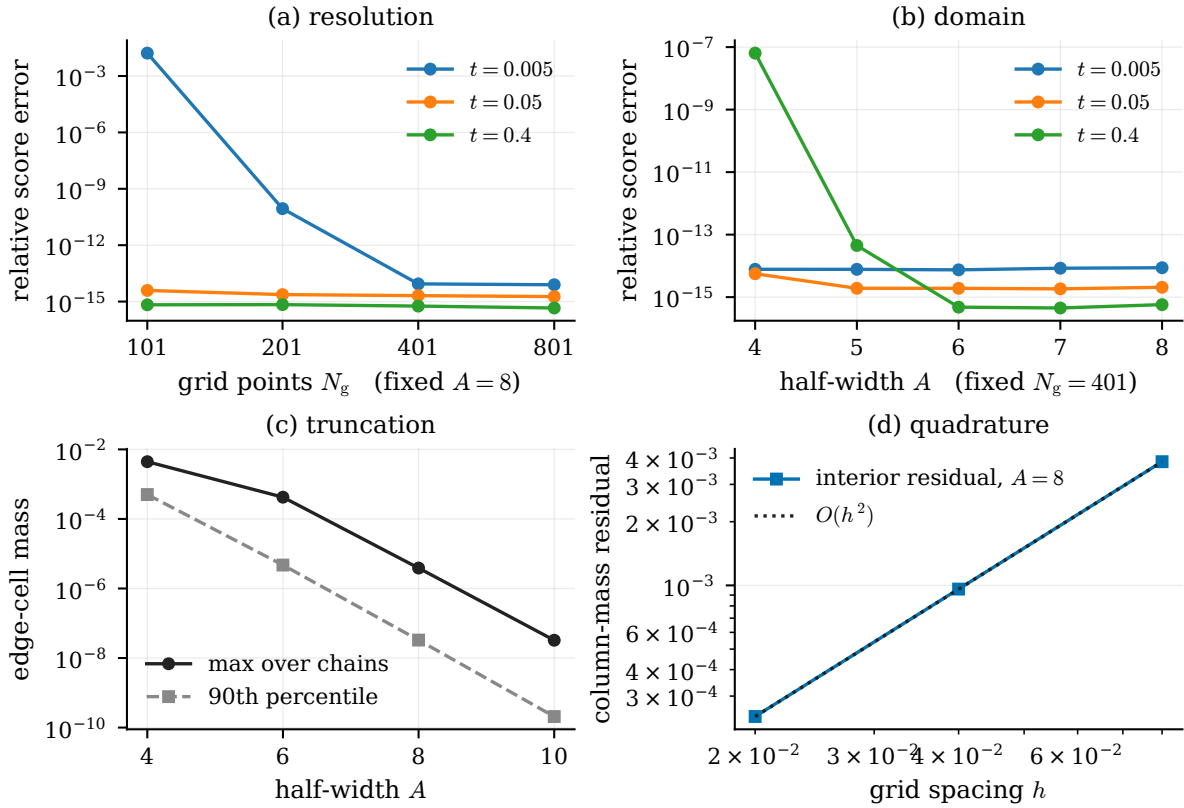


Figure 8.1: Grid and domain control on a Laplace chain ($\alpha = 0.85$, $L = 32$ sites), four panels. (a) Relative score error against a high-resolution reference as grid resolution N_g grows at fixed half-width $A = 8$; (b) the same as A grows at fixed $N_g = 401$ — both saturate at the floating-point floor well inside the working configuration. (c) Forward-message boundary mass against A , the truncation diagnostic, drawn as *both* the maximum and the 90th percentile over 256 chains, which differ by roughly two orders of magnitude for the reason given in Remark 8.4. (d) Interior column-mass residual against spacing at $A = 8$, the quadrature diagnostic, with an $O(h^2)$ reference. Neither diagnostic alone certifies convergence. Source: `outputs/frozen/exp_01/` and `outputs/frozen/exp_18/`.

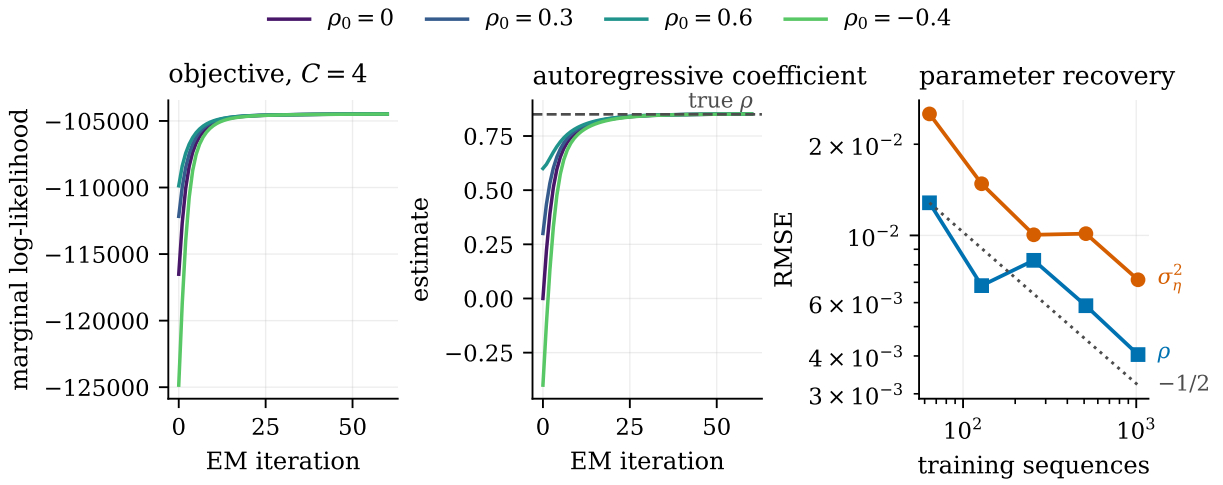


Figure 8.2: Optimisation diagnostics for the EM procedure of Section 8.3. Left: marginal log-likelihood by iteration from four initialisations of α ; the largest decrease across iterations is zero in every run. Centre: the estimated α over the same runs, converging to $[0.8517, 0.8520]$ against a truth of 0.85 from initialisations spanning $[-0.4, 0.6]$ — direct evidence that α is estimated, not supplied. Right: RMSE of the recovered α and innovation variance against the number of training sequences, with an $n^{-1/2}$ reference line, at the same truth of 0.85 and over 16 seeds. Appendix A maps every panel to the run that produced it.

Chapter 9

Learning the Kernel, and What Convergence Costs

Recovering two numbers is one problem; recovering a whole transition law is another. This chapter reports what happens when the innovation is represented nonparametrically, and the finding is not the recovery itself but its *schedule*: the correlation and the innovation shape converge at very different rates, and a stopping rule read off the first certifies a kernel that has not converged in the second. Every non-Gaussian claim in this thesis depends on the coordinate that settles last, so this is a prerequisite for Chapter 10 rather than a diagnostic beside it.

9.1 What the channel costs the E-step

What this chapter reports is not a theorem but a measurement, and it is the one that most directly qualifies the results of Chapter 10: how many EM iterations the algorithm of Section 8.3 actually needs.

Table 9.1 shows what the channel costs, in two parts: the through-channel fit at four updates, and where a clean and a channel arm each finish.

Table 9.1: What the observation channel costs in rate rather than in where the fit lands. Above: the through-channel fit at $N = 512$, $C = 8$, as a median over 16 seeds at four updates — the correlation is essentially in place by 30 while the innovation shape is still moving at 120. Below: where the two arms finish, fit on clean transition pairs and on the same chains once through the channel. Generating values are $\alpha = 0.85$ and excess kurtosis 3.0; a finite mixture does not contain the generating law exactly, so what is approached is the best fit within the family.

update	α	innovation excess kurtosis
1	0.4402	0.078
30	0.8412	1.275
120	0.8499	3.019
600	0.8504	2.980
<i>where each arm finishes</i>		
clean transition pairs	0.8509	2.939
through the channel	0.8526	2.729

The two halves say different things. The trace shows the coordinates separating: the correlation is essentially in place by thirty updates while the innovation shape is still climbing at a hundred and twenty, because the mixture label is a latent variable of its own and the channel compounds the missing information, the mechanism of [Dempster et al. \(1977\)](#) appearing in the convergence rate, where the theory places it.

The channel has two distinct effects, and they should not be collapsed into one. First, it changes the local rate of EM through the fraction of missing information, which is the effect the trace above displays. Second, at finite sample size it reduces the observed Fisher information and can therefore increase estimation uncertainty, an effect measured directly, by coordinate, in [Section 10.5](#). Under identifiability and correct specification it need not move the population optimum, and that is the limited sense in which the clean and channel arms may approach the same endpoint while doing so at different rates. The endpoints here are consistent with that reading: the two arms finish within 0.002 of each other on the correlation and within a fifth of a unit on excess kurtosis, both close to the generating values. This is evidence of similar fitted endpoints in this configuration, not

of an absent information or accuracy penalty.

9.2 The convergence rates of the two coordinates

That is one configuration. A dedicated sweep measured the cost across sixteen seeds, recording for each fit the first update after which a coordinate stays within a fixed relative tolerance of its value at the end of the run. At $N = 512$, $C = 8$ and the uniform noise design, over 16 seeds run to 800 updates, the correlation coefficient settles at a median of 80 updates and the innovation shape at 229, the largest observed value being 638. Forming the ratio within each fit and then taking the median over fits gives 2.8; the quotient of the two medians is 2.9, and the paired figure is the one reported, because both coordinates are measured on the same fit.

Two things about what that measures. “Settled” is trace stability, the first update after which a coordinate stays within 10^{-3} of its end-of-run value for α , or 2×10^{-2} for the excess kurtosis, and not distance to an optimiser; a coordinate can sit still while the density it belongs to moves along a shallow direction. And the run has to be long enough for the statistic to mean anything: the sweep was run at two lengths, and these medians come from the 800-update shard, the only one in which no configuration settles near its own cap. Against them, the fixed budget of 40 iterations that several experiments in Chapter 10 used as a default falls short of the shape coordinate’s median by about a factor of six, and leaves 105 of the sweep’s 112 configurations unsettled in it. That count pools both shards, which is conservative: censoring can only shorten a measured settling time, so it cannot inflate the count.

9.3 The consequence for every non-Gaussian claim

Remark 9.1 (Fixed budgets compare convergence rates, not estimators). This is the single methodological lesson of the chapter, and it recurs throughout the next one. Whatever coordinate of a fit converges slower looks worse under a fixed iteration count, and a slower-

converging configuration under-iterated by the same budget produces what looks like a genuine deficit rather than symmetric noise, which is precisely the signature that makes it easy to mistake for a real effect. Compared correctly, at eight seeds and a fixed budget of 200 updates, no channel penalty is resolved in the reported shape summary: the paired clean – noised difference at that operating point is -0.052 ± 0.217 , consistent with zero. The operating point is not a converged one, the chapter’s own median settling time for the shape coordinate is 229 updates, so this is a statement about the paired difference at 200 updates, not about the converged estimators. A single non-converged draw would not have supported that comparison in either direction. Where Chapter 10 reports a result obtained at a fixed, possibly insufficient budget, it says so explicitly and states which quantities are and are not affected.

Chapter 10

What Structure Buys, and Where It Stops Paying

Chapter 8 built an estimator that is exact in its inference step and approximate only in its numerical representation and its optimisation budget. This chapter asks what that estimator is worth. Five questions organise it, in the order the evidence answers them most cleanly.

How much does the structured estimator reduce denoising error at a given amount of data; how much of that reduction survives against a baseline that is given the locality but not the model; how much mixture capacity does the estimator need; does the pointwise advantage survive when the estimated score is used to *generate*, by integrating the reverse process; and how far can the Markov assumption itself be wrong before the structure stops helping?

Three are answered outright. The capacity question is answered for the estimator as implemented, and in the opposite direction to the one a smaller design suggested, but not for mixture capacity in the abstract, and Section 10.3 says exactly where that line falls. The generative-fidelity question is not answered at all, and the section that addresses it says why the experiment as run cannot settle it. The summary at the end of the chapter states each of the five at the strength its evidence supports.

Every comparison below fits the mixture kernel of (8.2) by the EM procedure of Chapter 8 and compares it against neural denoisers trained by ordinary denoising score matching on the same noised data, with no access to the Markov factorisation, the linear kernel form, or homogeneity across sites — three structural facts the estimator is given and the networks are not. Two network architectures recur throughout: a fully connected MLP with no locality bias, and a weight-shared local network with a fixed receptive field. The second is a *locality-respecting* baseline, not one matched to the estimator: EM–BP smooths over the whole chain and has no radius of its own to match. The comparison measures the value of the structure supplied, not a general claim about neural denoisers.

10.1 Pointwise data efficiency

10.1.1 What is being measured

Every number in this chapter is a ratio of two relative L^2 score errors, and because the ratio is what the conclusions rest on, the definition is written out here rather than deferred.

Fix a diffusion time t and a test set $\{x^{(j)}\}_{j=1}^J$ of noised sequences drawn from the model of Section 8.1. Let $S^*(x, t) = \nabla_x \log p_t(x)$ be the exact joint score, available in closed form for the Gaussian chain and from the validated grid reference of Section 7.4 otherwise. For an estimator $\hat{S}$, the error on that test set is

$$\mathcal{E}[\hat{S}] = \frac{\left(\sum_{j=1}^J \left\| \hat{S}(x^{(j)}, t) - S^*(x^{(j)}, t) \right\|_2^2 \right)^{1/2}}{\left(\sum_{j=1}^J \left\| S^*(x^{(j)}, t) \right\|_2^2 \right)^{1/2}}, \quad (10.1)$$

the ratio of two Euclidean norms over the whole test set, so that a value of 0.1 means the estimate is wrong by a tenth of the signal’s own size. This is `metrics.rel_12` applied to the stacked residual, and it is scale-free by construction: multiplying the data by a constant leaves it unchanged.

The two arms are the EM–BP estimator of Chapter 8, which is given the Markov factorisation and estimates σ_η^2 and α from the same N noised sequences the network sees,

and a denoising-score-matching network trained on those sequences by (2.12) and given no structural information at all. Writing $\mathcal{E}_{\text{net}}$ and $\mathcal{E}_{\text{EM}}$ for their errors on a common test set, the reported quantity is

$$R = \frac{\mathcal{E}_{\text{net}}}{\mathcal{E}_{\text{EM}}}, \quad R > 1 \text{ means the structured estimator is more accurate,} \quad (10.2)$$

evaluated cell by cell, where a cell is one (training-set size, seed) pair. Both arms are re-fitted independently in every cell.

Two aggregation choices remain, and they matter because R is a ratio of means and not a mean of ratios. This chapter reports the seed-first estimand: errors are averaged over seeds at fixed size, and the ratio is formed from those averages. Appendix B recomputes the same frozen cells under four alternatives; the direction and the order of magnitude are the same in all of them, and the estimand quoted here is systematically the largest, which is stated wherever the number appears.

10.1.2 The measured advantage

Under the structural assumptions of Section 8.1, at the data budgets tested, the structured estimator attains lower relative denoising error than the global MLP by a factor of 7.0–17.6 across the 8 training-set sizes (Table 10.1, Figure 10.1). The factor is *not* monotone in the budget, and it is the largest of several defensible aggregations of the same cells — Appendix B gives the other three, with a paired interval on the one quoted here. These figures are generated by `tools/make_tab_efficiency.py` from the frozen outputs rather than transcribed, so the table, the figure and this sentence cannot drift apart.

That first comparison is against a network with neither a locality prior nor weight sharing. When those two biases are supplied through the window-head baseline, the observed ratio narrows to 2.3–6.1 $\times$, measured at the same headline protocol and reported with its own scoping in Section 10.2. This change does not identify a causal decomposition: architecture, parameter sharing, capacity and optimisation change together across the two baselines. It shows that the generic-MLP comparison overstates what can be attributed

specifically to the supplied Markov model. The relevant locality result is the influence decay of Chapter 5.

Table 10.1: Relative denoising error against the deterministic grid-BP reference under the true kernel, averaged over the noise schedule; 16 seeds $\pm$ one standard error, aggregated per seed. Both arms select their optimisation budget on a disjoint validation bundle. EM-BP is more accurate in 1,518 of 1,520 cells. The ratio column is the mean of the per-cell ratios and not the quotient of the two columns beside it, which is a smaller number; Appendix B gives both, and three further aggregations. Protocol, cell exclusions and the budget calibration are in Appendix A.

sequences N	network	EM-BP (24 free)	ratio
32	0.4031 ± 0.0059	0.0589 ± 0.0026	7.0
64	0.3285 ± 0.0026	0.0501 ± 0.0041	7.8
128	0.2759 ± 0.0024	0.0343 ± 0.0017	8.8
256	0.2268 ± 0.0010	0.0258 ± 0.0014	9.7
512	0.1878 ± 0.0006	0.0219 ± 0.0011	9.0
1024	0.1569 ± 0.0006	0.0162 ± 0.0012	12.5
2048	0.1326 ± 0.0005	0.0132 ± 0.0008	15.5
4096	0.1171 ± 0.0004	0.0094 ± 0.0007	17.6

Every tunable quantity either arm is allowed to choose is chosen on a validation bundle disjoint from both the training set and the test set, and the two arms are given the same privilege: the network selects its output parameterisation, predicting the noise or the clean signal, *and* its training length, EM-BP its iteration count. That choice coincides with the test-set optimum in 81.1% of cells for the network and 77.6% for EM-BP. A dedicated audit of the selection rule, over 1,344 cells at 16 seeds, puts a size on what the remainder costs: the mean advantage is $13.07\times$ under validation selection and $13.06\times$ when each arm is instead given its test-set optimum, a difference far smaller than the spread across seeds. The protocol is the defensible one independently of the size of the correction it produces.

10.2 How much of the margin is the architecture?

A ratio against a fully connected network cannot separate knowing the Markov structure from sharing weights across sites, and the second is the cheaper explanation. This section

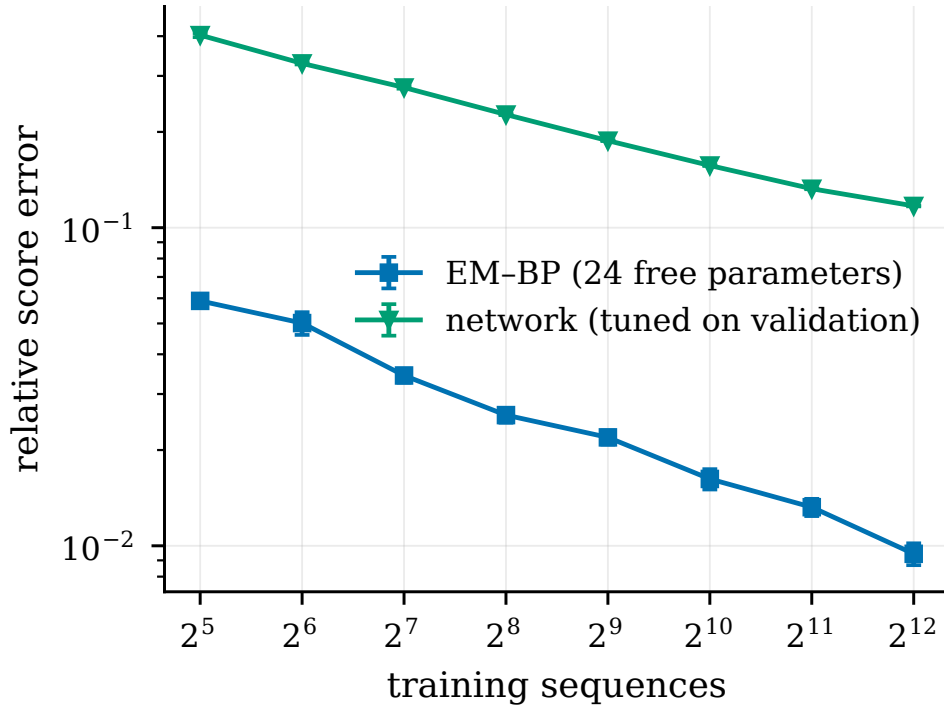


Figure 10.1: Relative score error against training-set size, mean over 16 replicates with standard errors, for the EM-BP estimator and the validation-tuned MLP. The window-head baseline of Section 10.2 scores every site rather than the aggregate plotted here and is reported as a ratio table (Table 10.3) rather than superimposed on this figure. Drawn from the same outputs, over the same size grid and with the same seed-first aggregation, as Table 10.1.

measures against a structure-aware baseline at the headline protocol: the same twelve noise levels and the same sixteen seeds as Table 10.1, every site scored rather than the centre alone, the baseline’s architecture and radius fixed by a screen on separate development seeds, and each arm’s checkpoint chosen on a validation bundle disjoint from test. The deployment and provenance conditions are in Appendix A.

Which architecture, and under what search. Three candidates were screened on 4 development seeds, disjoint from the 16 used below: a weight-shared window head (which cannot propagate past its radius however much data it sees), a dilated convolutional stack, and a bidirectional message-passing network (both able, in principle, to propagate across the whole chain). Each was run over its own hyperparameter grid, a shared learning-rate

grid, both output parameterisations, and a common checkpoint ladder from 500 to 64000 steps, 32 configurations in all, scored on the bulk region. **The window head won**, at radius 4 and width 64, with the bidirectional message passing network next (0.137 against 0.172).

Table 10.2: The architecture screen that selected Table 10.3’s baseline. Each candidate architecture was run over its own hyperparameter grid on 4 development seeds (disjoint from the confirmatory seeds), at $N \in \{32, 2048\}$, with a shared learning-rate grid, both output parameterisations, and a shared checkpoint ladder; the reported risk is the mean over development seeds on the interior slice used for selection. ★ marks the winner carried into the confirmatory run. The dilated stack carries an order of magnitude more parameters than the winner and finishes last, so the outcome is not an artefact of one arm being given more capacity than another.

architecture	configurations	parameters	selection risk
weight-shared window head ★	16	4.8k–21.4k	0.137
dilated convolutional stack	8	66.4k–264k	0.205
bidirectional message passing	8	9.0k–34.3k	0.172

Figure 10.2 shows the whole search rather than its winner, and it is the more useful object: the three architectures separate by parameter count far more cleanly than by risk, and the two arms built to propagate further occupy the expensive half of the axis without buying anything there.

This is a model-selection fact and not a claim about architectures: under this search space, protocol and development seeds, the simplest candidate scored best. It is not evidence that long-range propagation is useless here, that radius 4 is population-optimal, or that either other architecture could not overtake it under different tuning. What it does rule out is that the long-range arms were starved: the dilated stack carries 66.4k–264k parameters against the window head’s 4.8k–21.4k, an order of magnitude more, and finished last. Parameter count alone is not proof of adequate optimisation, so this bounds one explanation without eliminating the class. The comparison below is therefore against the best candidate in this screen, and that scoping travels with every claim that follows.

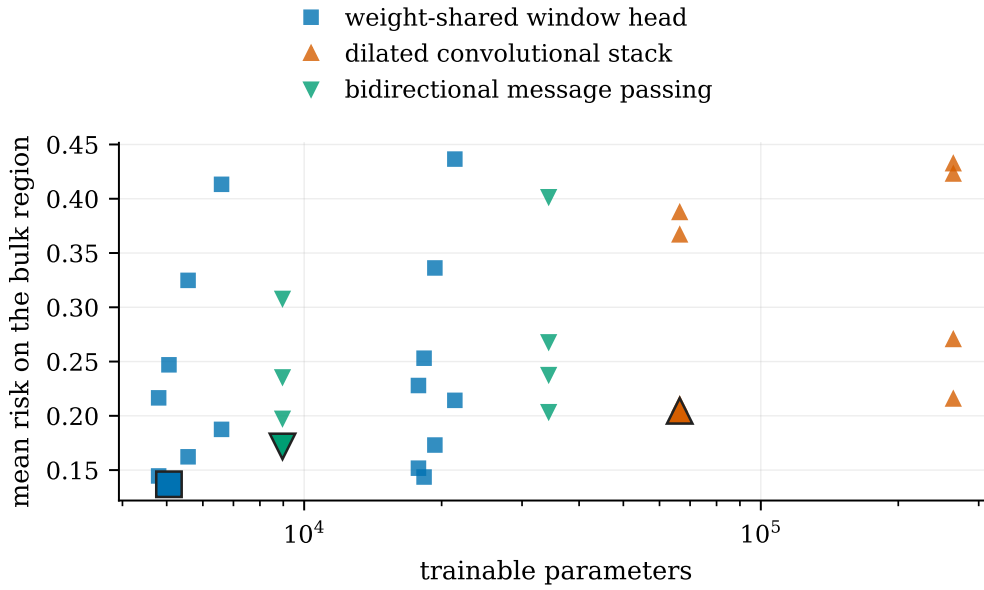


Figure 10.2: Every configuration of the screen, scored on the bulk region and averaged over the 4 development seeds. One marker per architecture; the outlined marker is each architecture’s best configuration. The horizontal axis is what makes the comparison readable: the dilated stack sits an order of magnitude to the right and does not reach the window head’s risk anywhere in the screen. Drawn from the same artifact as Table 10.2.

Table 10.3: EM–BP against the screened structure-aware baseline, at the headline protocol throughout. The baseline is a weight-shared window head whose architecture and hyperparameters were fixed by the screen of Table 10.2 on development seeds; only the checkpoint is chosen here, on a validation bundle disjoint from test, and EM–BP’s iteration count is chosen the same way. Risks are all-site and averaged over the 16 seeds; the ratio is $\text{risk}(\text{window})/\text{risk}(\text{EM–BP})$ formed within each seed and then averaged, $\pm$ one s.e. *All* scores every site, and is the headline; *centre* is a single-site diagnostic. A third region, the predeclared interior slice, is scored in the stored output and used for architecture selection but not reported here.

N	risk, all sites		ratio	
	EM–BP	window	all sites	centre site
32	0.1005	0.2006	2.34 ± 0.26	2.04 ± 0.24
128	0.0652	0.1594	2.82 ± 0.30	2.35 ± 0.26
512	0.0394	0.1353	3.58 ± 0.18	3.25 ± 0.18
2048	0.0224	0.1267	6.14 ± 0.45	5.43 ± 0.39

A ratio against a fully connected network is not comparable to this one: a large fraction of that headline margin disappears once locality and weight sharing are supplied. What

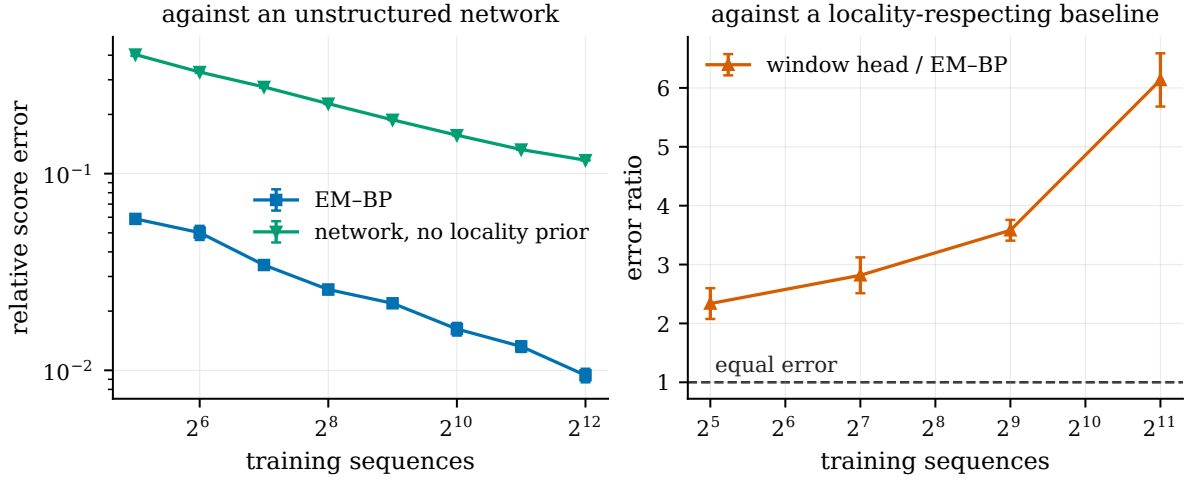


Figure 10.3: The two comparisons together. *Left:* relative score error against training-set size for EM-BP and a network given no locality prior, with one standard error over the 16 seeds. *Right:* the paired per-seed ratio of window-head risk to EM-BP risk, against a baseline that *does* have locality and weight sharing, with the break-even line drawn. The left gap is the headline 7.0–17.6 $\times$; the right one is 2.3–6.1 $\times$, and the difference between the two panels is how much of the headline margin does not survive supplying those two structural biases. Both ratios grow across the tested sizes. That is a description of four measured sizes, not a mechanism: different statistical rates and different optimisation limits produce the same signature, and no design here separates architecture, parameter sharing, capacity and optimisation, which change together when the baseline is swapped. Drawn from the same certified cells as Tables 10.1 and 10.3.

survives against a baseline that has them, the window head, is 2.3–6.1 $\times$, all-site scored, across the same four sizes as Table 10.1. EM-BP is more accurate in every one of the 192 scored cells; that count is 16 seeds $\times$ 4 sizes $\times$ 3 scoring regions (all, bulk, centre), of which the table displays two, and it is a descriptive diagnostic rather than a replication count, the regions and sizes share fitted models and training data, so the inferential unit behind each displayed row is the 16 paired seeds.

The ratio increases across the four sizes tested, from 2.3 $\times$ to 6.1 $\times$. That is a description of four measurements and nothing more is claimed from it. In particular it is not evidence that the residual is non-architectural. An architecture with an irreducible approximation floor b^2 would give $R_{\text{window}}/R_{\text{EM}} = (b^2 + c_w/N)/(c_e/N)$, which grows linearly in N for an entirely architectural reason. The raw risks in Table 10.3 are consistent with exactly that: EM-BP’s risk falls by a factor of about four and a half across the range,

while the window head's falls by less than two. Different statistical rates or different optimisation limits would produce the same signature. The direction is measured; the mechanism is not.

Both arms choose their budget symmetrically, on a validation bundle disjoint from test, *within the budget grids on offer*: 22% of EM–BP cells and 27% of window-head cells select the last rung of their respective ladder. Selecting the last rung is not proof of convergence, it is a sign that validation error was still falling where the grid ran out, and that a wider grid might have improved that arm further. So the reported values compare the two procedures at stated, validation-selected computational budgets, not at converged optima. Appendix B reports the same ratio under 4 aggregations of the same cells; none of them, at any size, falls below $1.98\times$.

10.3 How much mixture capacity is worth having

Enlarging C in (8.2) should help until the mixture can represent the true innovation law and stop helping after. The design that tests this pairs each capacity against $C = 1$ — a single Gaussian innovation, the kernel the estimator would use if it did not model the innovation law at all — on identical training and evaluation data within each cell, at 16 seeds and two training sizes, with three initialisations per cell and the reported one selected on validation. The fits are run to their plateau and not to a fixed iteration count, because Section 9.1 measures the fitted shape as settling far later than the correlation coefficient does, and shape is what capacity moves.

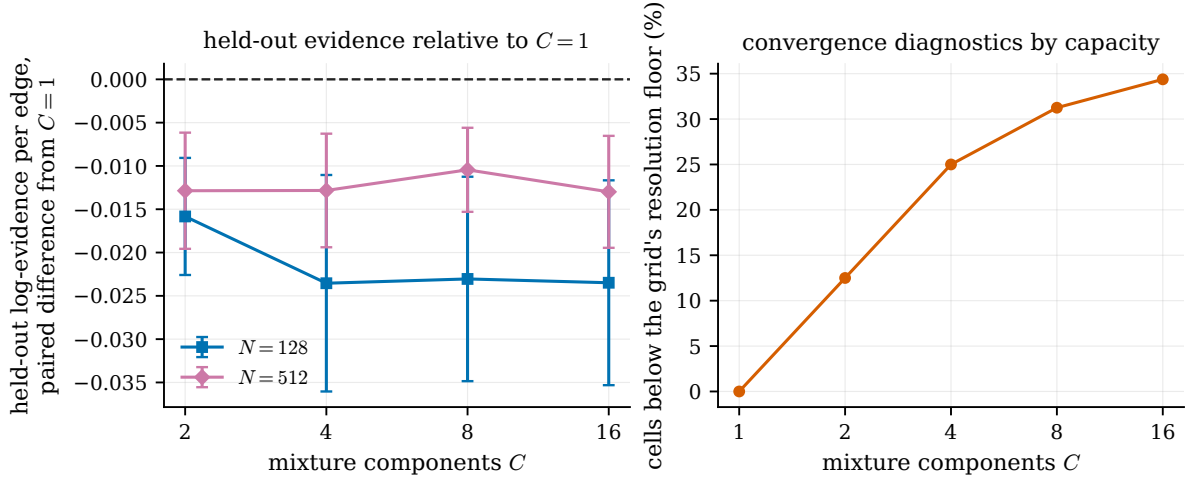


Figure 10.4: Left: the paired difference in held-out log-evidence per edge against $C = 1$, with one standard error over the 16 seeds. Every interval lies below zero at both training sizes, which is the result of this section. Right: the share of fits whose narrowest component falls under the grid’s resolution floor of 2 spacings, which climbs monotonically with C — the second caveat of Remark 10.1, and the reason the left panel is a statement about this estimator on this grid. Every cell at $C \geq 4$ reached the 1200-iteration cap rather than the convergence tolerance. Both panels are drawn from the cells of Table 10.4.

Table 10.4: Held-out log-evidence per edge at each mixture capacity, paired within seed against the single-Gaussian innovation $C = 1$ on identical training and evaluation data, 16 seeds, fits run to their plateau rather than to a fixed iteration count. Positive would mean the mixture buys something. No capacity does, at either size, and the sign count is the more honest summary of the two columns: the mean is dominated by a few large-magnitude seeds while the direction is consistent across most of them. The last column counts cells, over both sizes, whose narrowest fitted component falls below the grid’s resolution floor of 2 spacings — a share that climbs with C , and a reason not to read the table as a pure statement about capacity. Means and standard errors are in units of 10^{-2} .

C	$N = 128$			$N = 512$			under-resolved
	mean	s.e.	seeds better	mean	s.e.	seeds better	
2	-1.58	0.68	4/16	-1.29	0.67	3/16	4/32
4	-2.35	1.25	3/16	-1.28	0.66	2/16	8/32
8	-2.30	1.18	3/16	-1.04	0.49	2/16	10/32
16	-2.35	1.18	3/16	-1.30	0.65	2/16	11/32

The result is that **under this procedure — this grid, this generalised M-step, this validation rule and this iteration cap — no mixture capacity improves on a**

single Gaussian innovation at either training size. The qualification is quantitative. Every cell at $C \geq 4$ stops at the cap rather than at the tolerance, and the share of fits the grid cannot resolve grows with C , so the design cannot separate a statistical disadvantage of larger mixtures from an optimisation or a discretisation one. What it can say, it says at every seed. The means are dominated by a few large-magnitude seeds, so the sign count carries the claim: at $C = 8$ the mixture is ahead on only 2 to 3 of 16 paired seeds, with median paired differences of -5.73×10^{-3} and -9.89×10^{-3} per edge. The same holds on schedule-level score risk.

The predeclared equivalence test sharpens this rather than softening it. The region — $|\Delta(\text{evidence/edge})| < 10^{-4}$ together with $|\Delta R|/R < 1\%$ — was fixed before the contrast was computed, precisely so that “ $C = 8$ and $C = 16$ are equivalent” could be asserted or refused rather than inferred from a failure to reject. It is refused. At $N = 128$ the $C = 8 \rightarrow C = 16$ step is resolved against the region and in the direction of $C = 16$ being *worse*: evidence -4.4×10^{-4} with interval $[-9.2 \times 10^{-4}, -1.1 \times 10^{-4}]$, risk $+5.0 \times 10^{-3}$ with interval $[+9.7 \times 10^{-4}, +1.1 \times 10^{-2}]$, both lying wholly outside it. At $N = 512$ it is unresolved. So neither “capacity saturates by eight” nor “capacity keeps paying past eight” is supported. What is measured is narrower: at $N = 128$, the observed $C = 8 \rightarrow C = 16$ contrast is resolved outside the predeclared equivalence region, in the direction of worse held-out evidence and worse score risk for $C = 16$.

Remark 10.1 (Two things this does not isolate). Both point the same way, and both are reasons to read the table as a statement about this estimator on this grid and not about mixture capacity as such. First, every cell at $C \geq 4$ stopped at the 1200-iteration cap rather than at the convergence tolerance (96 of 96). The cap was placed at the measured practical plateau. A representative $C = 16$ fit taken to 2000 iterations has per-edge gain 6.9×10^{-8} at 400, about 6×10^{-9} by 800, and then does not move: 6.08×10^{-9} at 1200 against 6.09×10^{-9} at 2000, never reaching the strict tolerance at any practical count. The residual asymmetry is therefore small, but it is not zero, and it disfavors the higher capacities. Second, the share of fits whose narrowest component falls below the grid’s resolution floor of 2 spacings climbs with capacity, from 0/32 at $C = 1$ to 11/32

at $C = 16$. Part of what the larger mixtures are fitting is therefore narrower than the grid can represent, which is a property of the discretisation and not of the innovation law. Separating the two would need a finer grid at fixed capacity, which is not run here.

A smaller design — six seeds, one training size, one initialisation and a fixed 40-iteration budget — reported the opposite sign, and the reversal is worth recording rather than quietly dropping: it does not reproduce at sixteen seeds and a plateau-stopped budget. Which of the two readings the budget produced, and by what mechanism, is not separately tested here; the companion compendium carries the comparison.

Remark 10.2 (Why the iteration budget, and not the capacity, has to be controlled). The reason the design above runs to a plateau is that a fixed budget cannot distinguish the two. Section 9.1 measures the innovation shape as settling at a median of 229 updates against 80 for the correlation coefficient — a factor of 2.8 formed within each fit and then aggregated, with a largest observed value of 638. Against a fixed 40-iteration budget, 105 of the 112 configurations in that sweep had not settled in the shape coordinate. Because a larger C has more shape parameters to move within the same budget, under-iteration alone produces a curve that looks exactly like a capacity effect whether or not capacity is doing anything. This is why any quantity resting on fitted or generated *shape* at a fixed budget, the withdrawn reverse-dynamics comparison included, is evidence about an under-converged operating point and not about the converged estimator.

Per-iteration runtime changes little over the tested capacity range, $13.5 \rightarrow 14.3$ seconds from $C = 2$ to $C = 16$ while the number of free parameters grows $6 \rightarrow 48$, because the E-step's $O(L N_g^2)$ cost dominates the component-dependent M-step. This shows that increasing C adds little cost to one iteration. It does not remove the confounds above: the higher-capacity fits are precisely the ones for which the iteration-cap and grid-resolution diagnostics become less favourable.

10.4 Where the Markov assumption stops paying

Every result so far assumes the clean chain is exactly first-order Markov. Two controlled violations were measured against an exact reference (linear Gaussian algebra for the contaminated model, so the estimator being wrong is not confused with the reference being wrong): a rank-one global latent shared across all sites at strength β , and a long-range coupling at strength γ added on top of the chain. This experiment has its own noise schedule and its own training size, so its ratios measure how the advantage responds to misspecification and are not on the same scale as Table 10.1's; only the direction and the crossing point are read from them. Both arms train at the headline protocol's budget.

The measurement is reported for two innovation laws. The Gaussian arm carries both mechanisms, since its contaminated model has an exact reference in closed form; the Laplace arm carries the rank-one mechanism, where the same contamination can be applied to a non-Gaussian chain and the reference computed on the grid. Reading them together separates what is a property of the chain structure from what is a property of Gaussianity. On the rank-one mechanism the two agree in shape, the same ordering, the same smooth decay, no crossing, and differ at the margin. At $\beta = 1$ the Gaussian arm is still ahead at every noise level against both baselines, worst ratio 1.06; the Laplace arm is still ahead against the convolutional baseline, worst ratio 1.32, but reaches the break-even line against the fully connected one at the largest diffusion time, at 0.998. The ordering carries across both innovation laws tested; the size of the margin it leaves does not quite. Whether it carries beyond these two families is not established here.

Table 10.5: Two controlled violations of the Markov assumption, each evaluated against the appropriate reference: the Gaussian rows against an exact linear-algebra reference for the contaminated model, the Laplace rank-one rows against the validated grid-BP reference. The first two rows of each block are the baseline’s relative score error divided by EM–BP’s, averaged over the noise schedule, so a value above 1 means the structured estimator wins and a bold value means it loses. The third row is the fitted correlation minus the Chow–Liu correlation of the true contaminated law: the distance between the chain that was fitted and the best chain there is. The fourth is EM–BP’s own relative score error at its worst noise level, which says how much of a change in the ratio is the estimator degrading rather than the baseline improving.

	global latent, rank-one, strength β				
	0.00	0.10	0.25	0.50	1.00
Gaussian, vs CNN	16.41	14.84	6.68	3.21	2.16
Gaussian, vs MLP	29.34	22.48	11.28	4.00	1.67
Laplace, vs CNN	11.14	10.92	6.21	2.96	2.09
Laplace, vs MLP	16.51	17.64	10.13	3.67	1.62
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	−0.0025	−0.0008	+0.0008	+0.0051	+0.0144
Gaussian: EM–BP error, worst t	0.019	0.025	0.055	0.134	0.167
Laplace: EM–BP error, worst t	0.022	0.022	0.047	0.132	0.183
	long-range precision coupling, strength γ				
	0.00	0.05	0.10	0.20	0.40
Gaussian, vs CNN	16.07	1.24	0.97	0.88	0.75
Gaussian, vs MLP	27.84	1.66	1.19	0.84	0.70
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	+0.0018	+0.0376	+0.0557	+0.0806	+0.0976
Gaussian: EM–BP error, worst t	0.016	0.298	0.402	0.490	0.579

The two violations behave qualitatively differently, and the difference is principled rather than incidental. A global latent shared across every site makes the score residual *exactly* rank one, by a Woodbury identity, and over the range tested a chain-structured estimator accommodates that correction within its own fitted coefficients: the fitted correlation climbs from 0.848 on the clean chain to 0.939 at $\beta = 1$: the fitted chain partially accommodates the rank-one perturbation by changing its effective lag-one correlation. This is

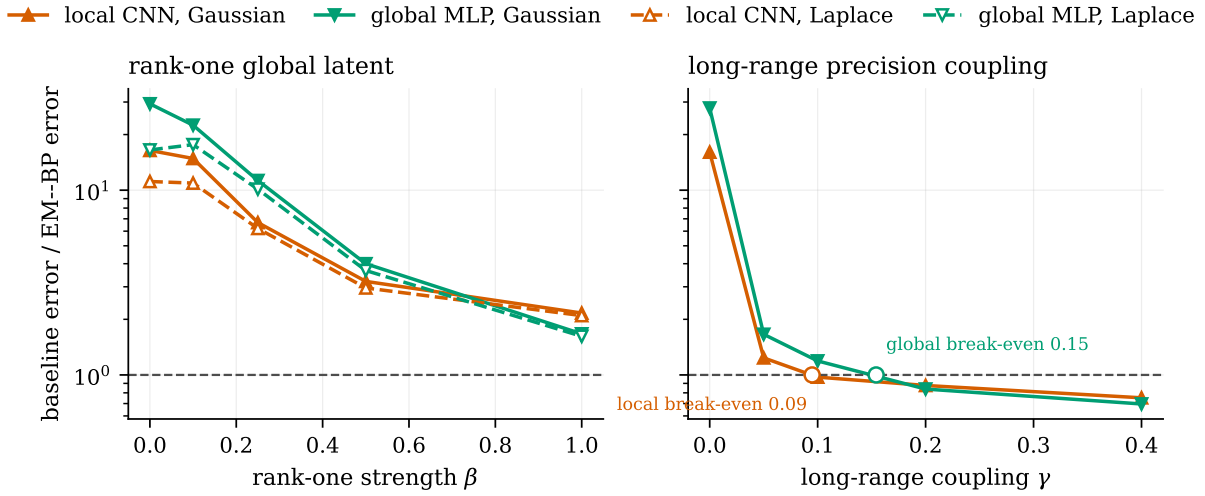


Figure 10.5: The same cells as Table 10.5, on a logarithmic vertical axis with the break-even line drawn; solid is the Gaussian innovation, dashed with open markers the Laplace one. The two mechanisms are not two amounts of the same thing: under a rank-one global latent the advantage decays smoothly and does not reach the line, even where half the marginal variance is a shared constant; under long-range coupling it crosses within the first tested strength and stays below. The Laplace curves track the Gaussian ones across the whole rank-one sweep, which shows that the rank-one result is not confined to the Gaussian experiment; its generality beyond these two innovation families remains open. The long-range panel is Gaussian only, since the contaminated Laplace model has no closed-form reference. Every curve is a mean over the noise schedule, so it shows the direction rather than the worst case, the tie reported in the text at $\beta = 1$ is at one noise level and is not visible here. All fall because the estimator degrades, not because the baselines improve: the last row of each block in Table 10.5 carries EM-BP’s own error.

partial accommodation over a measured range, not an exact representation, a first-order Markov family does not contain the contaminated law, but the residual it cannot represent stays small enough that the advantage never inverts: at $\beta = 1$, where half the marginal variance is a shared global constant, the ratio at its worst noise level is still 1.06 against the weaker of the two baselines.

Long-range coupling is only *approximately* low rank, and a chain family cannot represent it at all. The advantage against the convolutional baseline is gone by $\gamma = 0.1$, having survived $\gamma = 0.05$, and by $\gamma = 0.2$ the estimator loses to both baselines and not to one; at $\gamma = 0.4$ the ratio at its worst noise level is 0.45, where Table 10.5 reports the larger schedule-averaged value, and its own relative score error reaches 58%.

Two rows of the table separate the mechanism from the outcome. The estimator’s own

error is what moves in both blocks, the baselines barely change, so neither column is a story about the networks improving. The last row of each block is what distinguishes the two violations. Under rank-one contamination the fitted chain stays within 0.014 of the Chow–Liu correlation, the best first-order approximation of the contaminated truth, so its fitted lag-one correlation tracks the corresponding first-order projection. That is one scalar agreeing, not the whole fitted law certified. Under long-range coupling the same gap is 7 times larger, which is what it looks like when the family can no longer represent what generated the data. The uncontaminated controls of the two blocks are the same configuration reached from two directions, and they agree to within 2% (16.41 and 16.07), which is what makes the rest of the table readable across blocks.

Remark 10.3. This extends the lesson of Chapter 7: there, what broke under a harder innovation law was the *representability of messages* in a finite-dimensional family, while the tree, the two-sweep schedule, and the score–posterior identity survived unconditionally. Here, the further question is what breaks when the graph is not a tree to begin with. The answer is not "misspecification degrades the advantage" uniformly, it is that a chain-structured estimator tolerates the one violation it can absorb into its own parameters and fails on the one it structurally cannot represent, and the two are not interchangeable amounts of "how wrong the model is": they are different in kind.

10.5 The channel removes shape information faster than correlation information

Section 9.1 found that the innovation shape converges far slower than the autoregressive coefficient under EM. A direct information-theoretic measurement runs alongside that finding.

This calculation complements the convergence experiment but does not derive its iteration count. The table below measures how much statistical information about each coordinate survives the observation channel. Local EM convergence is governed instead

by the fraction of missing information, equivalently by the Jacobian of the EM update near its fixed point. The two measurements point in the same direction here, but they answer different questions, and a flatter observed likelihood does not by itself entail a given number of updates.

What is computed. The innovation law is a generalised Gaussian $f_p(\varepsilon) \propto \exp(-|\varepsilon/s|^p)$, with s fixed by $\text{Var}(\varepsilon) = \sigma_\eta^2 = 1 - \alpha^2$ at every p , so p moves the fourth moment at *exactly fixed* variance and is a genuine shape coordinate rather than a reparameterised scale. For a three-parameter vector $\theta = (\alpha, \sigma_\eta^2, p)$, Fisher’s identity (8.4) supplies the exact per-chain score of the marginal log-likelihood from one forward–backward pass, and the information matrix is estimated in *outer-product* form, $J = D^{-1} \sum_d g_d g_d^\top$ over $D = 512$ chains of $L = 32$ sites, at $N_g = 401$, $A = 8$. This is the outer-product estimator, not the observed negative Hessian, and it is an estimate at the true parameter rather than at a fitted one. The reported quantity is the *efficient* information $I_{p \cdot (\alpha, \sigma_\eta^2)} = 1/(J^{-1})_{pp}$, which is what remains about p once the nuisance coordinates must be estimated too — equal to the diagonal entry only when the coordinates are orthogonal and strictly smaller otherwise. The gap between J_{pp} and this quantity is recorded per cell as the nuisance cost, and the condition number of J is recorded alongside it, since the inversion is where a near-singular J would show up.

What it shows. The efficient information for p decays two to three orders of magnitude faster with diffusion noise than the information for α itself. Table 10.6 reads the loss between the smallest and largest diffusion time tested:

Table 10.6: How much Fisher information about each coordinate the channel destroys, read between the smallest and largest diffusion time tested ($t = 0.05$ against $t = 1.6$). The shape coordinate loses two to three orders of magnitude more than the correlation does, which is a property of the noised likelihood rather than of any optimiser. The reported quantity is the efficient information, so the nuisance coordinates are estimated rather than known.

true p	info loss, α	info loss, σ_η^2	info loss, p (shape)
1.0	$78\times$	$235\times$	8.7×10^3
1.5	$83\times$	$459\times$	6.4×10^4
3.0	$56\times$	$522\times$	6.8×10^4

This is a property of the noised likelihood itself, independent of any optimiser: at any fixed sample size, and at every diffusion time past the smallest ones, there is less information about the shape coordinate in noised data than about the correlation. That is consistent with the shape coordinate being the slow one, though as noted above it does not derive the rate. It is also the reason the capacity sweep of Section 10.3 is exactly the kind of measurement where an insufficient iteration budget is dangerous: the coordinate that is hardest to move is also the one with the least signal per observation.

Two limits on how far these figures should be pushed. They are ratios of outer-product estimates at $D = 512$ chains, so only the order is meaningful and the entries are printed to two significant digits for that reason. And the comparison is across the tested noise grid rather than a limit: the decay is measured between $t = 0.05$ and $t = 1.6$, and how it continues past the schedule's end is not established here. What the table supports is the direction and the order, shape information collapses far faster than correlation information, which is the claim the rest of the chapter rests on.

Chapter 11

Discussion and Conclusions

11.1 Answers to the research questions

The derivations are in the chapters named; the answers are stated here without repeating them.

RQ1: the joint score and its couplings. Linear and known in closed form, $S(x, t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$. Its couplings have a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one further diagonal per order in t , maximally coupled at intermediate times, returning to the identity at rate e^{-2t} . A single bulk quantity $q(\alpha, t) \in (0, |\alpha|]$ sets the posterior correlation length. This is a consequence of the model's linearity and not of the chain structure alone (Chapter 5).

RQ2: belief propagation and the Kalman smoother. They are the same computation. The channel corrupts each coordinate independently, so every observation enters through a unary factor, the posterior factor graph stays a tree, and one forward and one backward sweep give all smoothing marginals. On the Gaussian chain the messages close algebraically and the recursion is the Kalman filter followed by RTS smoothing, update by update; the two routes agree to 10^{-15} – 10^{-14} across the tested grid, reproduced by an independently written implementation. Tree exactness holds for any innovation law; closure

in a finite-dimensional family holds only because the model is Gaussian (Chapter 6).

RQ3: what Gaussian messages cost. Median relative score error 0.24 at $t = 0.05$ on Laplace chains, below 10^{-2} by $t \approx 1$ and below 10^{-3} by $t \approx 2.1$, with direction preserved far better than magnitude. Theorem 7.6 says what that number is: moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model, so the measurement is full posterior inference against the best second-order linear estimator. What the number measures is therefore fixed by the model class and not by the implementation, and it is the same for every innovation law with matching first two moments (Chapter 7).

RQ4: how fast remote evidence stops mattering. The influence of evidence at distance d decays exactly as $q(\alpha, t)^d$, and the error of the radius- r windowed estimator is measured to decay at the same rate over $r = 2, \dots, 13$. These stay two claims: windowing changes the recursion's boundary conditions in a way the influence argument does not cover, so the second is a numerical law over the tested range. At equal locality budget, truncating the inference beats truncating the score matrix at small t (12.3% against 21.0% at $t = 0.05$) (Chapter 6).

RQ5: recovering an unknown transition. Fisher's identity turns one forward-backward pass into the exact gradient of the marginal log-likelihood, so the E-step is structurally exact for any innovation law and nothing is differentiated through the recursion. The checks are part of the answer, not a caveat on it: the channel acts unevenly across coordinates, and the fitted innovation shape settles at a median of 229 updates against 80 for the correlation. A stopping rule read off the correlation trace therefore certifies a kernel that has not converged in the coordinate every non-Gaussian claim depends on, so a fixed iteration budget compares convergence rates and not estimators (Chapters 8 and 9).

RQ6: what supplying the Markov assumption is worth. At equal training data, $7.0\text{--}17.6\times$ lower pointwise error than networks trained by denoising score matching, across 8 sizes; against a baseline given locality and weight sharing, $2.3\text{--}6.1\times$. Three scope conditions travel with those numbers. The arms are asymmetric in supplied information by construction, so this measures what a correct structural assumption is worth where it holds exactly, and does not rank learning algorithms. The gap between the two ratios is not a decomposition into architecture and estimation terms, because architecture, capacity, sharing and optimisation change together when the baseline is swapped. And both are ratios of errors at fixed data, not of the data each method needs for fixed error; the curves do not cross in the measured range, so no data-equivalence factor follows (Chapter 10).

11.2 What is new

The exact structure of a chain diffusion score. The joint score of a noised Markov chain is characterised completely for the Gaussian case: its closed form, its eigenbasis at every diffusion time, the law by which its couplings fill and then decay, and the single parameter that controls posterior influence range. The identification of chain BP with the Kalman–RTS recursion makes the two classical literatures one object rather than two analogous ones.

An exact reading of what Gaussian closure computes. Where previous practice measured the error of moment-projected message passing as the error of an approximation, Theorem 7.6 identifies the estimator it returns. The measured Laplace gap is therefore the price of discarding everything beyond second moments, a property of the model class, shared by every innovation law with the same first two moments, and not a property of one implementation. The validated grid reference that makes the measurement possible is itself a contribution, since the functional recursion has no closed form.

Transition learning, and the measured value of structure. The transition law is estimated from noised sequences alone by exact chain inference inside EM, and the result-

ing estimator is compared against trained networks at equal data under a protocol that selects both arms on held-out validation. Two boundaries of the advantage are mapped rather than asserted: mixture capacity beyond a single Gaussian innovation buys nothing for this estimator under the grid and cap implemented here, and a chain-structured estimator partially accommodates a rank-one violation of the Markov assumption while failing on a long-range one at a coupling strength of roughly one tenth. The convergence measurement that separates rate from estimator is a methodological result in its own right, and it is what withdrew an earlier reading of the generative comparison.

11.3 Limitations

1. **Model class.** The exact solutions concern a one-dimensional state per frame and linear AR(1) dynamics, extended to a two-dimensional nonlinear state only in Chapter 4. The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited, and no low-rank structure is asserted for the Laplace $L = 2$ posterior Hessian beyond the narrow regime where it holds.
2. **The numerical reference.** The non-Gaussian results are numerical. Grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3h$ on the grid spacing h), and the Laplace closure error is a measured curve rather than a formula.
3. **Optimisation and finite seeds.** The learning results are statistical rather than exact: estimates over a finite number of seeds, reported with paired uncertainty, several resolved at only one of the two training sizes tested. Every capacity cell above $C = 4$ stops at an iteration cap rather than at tolerance, so a statistical disadvantage of larger mixtures is not separated from an optimisation or a discretisation one. The misspecification study probes exactly two violations of the Markov assumption; it establishes that the two are not interchangeable in effect, not a general theory of how much non-Markov structure a chain estimator tolerates.
4. **Generative fidelity, and transfer.** Whether pointwise accuracy and generative

fidelity can move in opposite directions within one estimator is unresolved here: the comparison that addresses it evaluates procedures whose fitted shape coordinates are unequally settled, on the single coordinate the generative metric scores, so its reading is withdrawn rather than reversed. More broadly, these are properties of one controlled model. Whether trained score networks on real sequence data inherit them is untested here. The image and video extensions explored in the underlying research package are outside the scope of this thesis and are not reported.

11.4 Future work

Four directions, in the order in which they would most change what this thesis can claim.

The reverse-generation comparison, with converged fits. The withdrawn study integrated the reverse SDE under the estimated and the trained scores, but on fits the convergence analysis shows to be unequally settled. Selecting both arms on a convergence criterion rather than a fixed iteration count would answer the question the experiment was built for, and close the one question this thesis leaves open.

A grid-by-capacity design. The capacity result leaves two mechanisms entangled. A finer grid at fixed capacity would separate fits falling below the grid’s resolution from genuine overfitting, since only the second survives mesh refinement; a budget ladder at fixed capacity would do the same for the iteration cap.

Richer departures from the Markov assumption. Two violations at fixed strengths are probed here. A paired design across strengths, and further violation types, would test whether the finding that the two mechanisms are not interchangeable generalises. Discrete chains are untouched: with a finite alphabet the messages are finite vectors and inference is exact with no closure at all, giving a second solvable benchmark alongside the Gaussian one.

Spatial as well as temporal structure. Whether the structural results survive longer sequences, other innovation laws, and data with spatial structure is the question that decides how far any of this carries beyond the solvable setting. The reading of U-Nets

as belief propagation (Section 2.6) suggests where to look: on a hierarchical rather than a chain graph, the same inference argument should apply, with the tree-exactness that makes it work here replaced by an approximation whose cost would have to be measured the way Chapter 7 measures the Gaussian closure.

Appendix A

Reproducibility

A.1 Figure-to-code map

Every figure is committed as a vector PDF under `overleaf/shared/figures/`, one directory for all four documents so that two of them cannot render different pictures under the same name. All are white background; randomness, where used, is seeded. Three packages compute the panels, and Table A.1 says which computes which.

The learning figures come from `tools/make_figures.py` in the `nongaussian-bp` package, which reads the frozen outputs directly and is the same script that writes the generated tables. The Gaussian-chain and AMP figures come from `code/experiments.py` in `gaussian-bp`, which recomputes each panel from the closed forms of that package’s audit; the same panels appear there under `figures/` as PNG, with the same file names. The ring figures come from `code/make_figures.py` in the `rotating-ring` package, whose closed-form panels are recomputed at build time and whose measured panels read that package’s curated CSVs; they are carried here under the `fig_ring_*` names rather than that script’s numbered ones.

One caveat belongs with this. The restyling step that turned the older packages’ outputs into the uniform vector PDFs used here was never committed and no longer exists, so those figures cannot be regenerated in their present style; this is why they, alone among the figures here, still carry Type 3 fonts. The computations behind every panel

remain in the packages named above and are re-runnable, so what was lost is a styling pass rather than a result, and no quantitative claim depends on it: every number is generated, and none is read off a figure.

Table A.1: Every figure in this thesis, the file it is committed as, and the package whose code computes the numbers in it. The styling caveat above applies to the figures from the older packages; the two schematics are drawn in $\text{\TeX}$ and compute nothing.

Figure	File	Numbers computed by
1.1	drawn in $\text{\TeX}$ (<i>TikZ</i>)	schematic; no computed quantity
3.2	drawn in $\text{\TeX}$ (<i>TikZ</i>)	schematic; no computed quantity
3.1	<code>fig_toymodel_score.pdf</code>	<code>nongaussian-bp</code> , closed forms
5.1	<code>fig_band_fill.pdf</code>	<code>gaussian-bp</code> , closed forms
5.2	<code>fig_tridiag_loss.pdf</code>	<code>gaussian-bp</code> , closed forms
6.1	<code>fig_local_vs_full.pdf</code>	<code>gaussian-bp</code> , closed forms
7.1	<code>fig_laplace_k1.pdf</code>	<code>unified-note</code> , quadrature
7.2	<code>fig_laplace_closure.pdf</code>	<code>nongaussian-bp</code> exp. 02, per-trial CSV
8.1	<code>fig_grid_domain.pdf</code>	<code>nongaussian-bp</code> exp. 01 and 18
8.2	<code>fig_em_diagnostics.pdf</code>	<code>nongaussian-bp</code> exp. 06 rate16 and 18, frozen
10.1	<code>fig_sample_efficiency.pdf</code>	<code>nongaussian-bp</code> exp. 07, frozen
10.3	<code>fig_value_of_structure.pdf</code>	<code>nongaussian-bp</code> exp. 07 and 31, frozen
10.2	<code>fig_screening.pdf</code>	<code>nongaussian-bp</code> exp. 31 screen, frozen
10.4	<code>fig_capacity.pdf</code>	<code>nongaussian-bp</code> exp. 32, frozen
10.5	<code>fig_nonmarkov.pdf</code>	<code>nongaussian-bp</code> exp. 21, clean rerun
4.1	<code>fig_ring_model.pdf</code>	ring package, closed forms and simulation
4.2	<code>fig_ring_gauge.pdf</code>	ring package, closed forms and simulation
4.3	<code>fig_ring_surrogate_response.pdf</code>	ring package, exact surrogate score

A.2 The learning experiments of Chapters 8 and 10

The closed-form and quadrature studies above are deterministic and run on a laptop. The estimator comparisons are neither, and they are gated differently.

Provenance. Runs deployed by `hpc/deploy_clean.sh` are certified: the script refuses to build from a dirty tree, ships a `git archive` of the commit rather than the working tree, records the SHA-256 of that tarball, and extracts on the cluster under the commit id so that a run cannot overwrite or edit its own source. The motivating failure is preserved in that script’s header: an earlier two-step stamp-then-rsync path produced `outputs/exp_12_scaled/`, whose recorded commit cannot have run the recorded command, and whose numbers are therefore not recoverable and are not cited in this thesis. `tools/provenance_gate.py` intersects a run’s recorded dirty-file list with the import closure of its own experiment module, so that uncommitted work elsewhere in the tree is distinguished from uncommitted work on the code path that produced the numbers. That distinction does the work below, because the two are not equally serious.

How each run is certified. Not every experiment here was run after `deploy_clean.sh` existed, so three routes to certification appear. Nothing cited in this thesis is uncertified. The development record behind each route, which deployments failed, and why each check was added, is in the companion compendium.

Certified. `exp_31`, the architecture screen and the confirmatory comparison, and `exp_32`, capacity, were deployed by the script and pass `require_clean` unmodified. Every number in Sections 10.2 and 10.3 comes from them.

Legacy digest, empty intersection. `exp_07`, the headline efficiency comparison and its validation-selection audit, with `exp_27` behind the settling times of Section 9.1, and `exp_18`, predate the script. They record a commit, a host, a job id and a resolved configuration, but no archive digest, so the plain gate refuses them. What makes them citable is narrower and checkable: the intersection of each run’s recorded dirty list with its own import closure is *empty*, so no uncommitted edit could have reached the code that produced the numbers. `make_tab_efficiency.py` recomputes that intersection on every run and refuses if it is ever non-empty, not resting on a check performed once by hand. The one `exp_07` row where the intersection was *not* empty, $N = 8192$, whose dirty list contained the experiment, the frozen configuration and `src/em.py`, is withdrawn and

appears nowhere in this thesis.

Certified after rerun. `exp_21`, behind Table 10.5 and Figure 10.5, is reported from `outputs/frozen/exp_21_clean`: fifteen cells from two git-archive deployments, every one with an empty dirty list. It replaces an earlier run whose recorded commit did not reconstruct the program, and the Gaussian arm reproduces that run on every number quoted here, to the precision printed. Because the Laplace and Gaussian arms were deployed separately, the rerun spans two commits, so `make_tab_nonmarkov.py` refuses unless no single cell pools commits *and* every file in the experiment’s sixteen-file import closure is byte-identical between them. Both hold, and the check runs on every regeneration.

Generated numbers. No efficiency, structured-baseline or aggregation number in this thesis is typed. Each is defined once as a macro in `overleaf/shared/sections/`, written there by one of ten generators under `research/nongaussian-bp/tools/`, each reading the recorded outputs directly:

```
make_tab_efficiency  make_aggregation_robustness
make_tab_structured  make_aggregation_structured
make_tab_screening   make_convergence_numbers
make_tab_capacity     make_grid_diagnostics
make_tab_nonmarkov   make_figures
```

The prose then cites the macro rather than the value, so tables, figures and text share a size grid and an aggregation rule by construction and not by inspection.

What Table 10.1 covers. Four qualifications, kept here and not in the caption, which had grown into a methods subsection set in small type. Both arms select their optimisation budget on a disjoint validation bundle, the network its parameterisation *and* training length, EM–BP its iteration count, and that choice agrees with the test-set optimum in 81.1% and 77.6% of cells respectively, so the protocol is symmetric and the selection is close to an oracle without being one. The $N = 4096$ row uses a raised budget, the caps having been set for $N \leq 2048$; rerunning $N = 2048$ at that budget on the same seeds

moves its ratio by -0.16 ± 0.18 , so the caps are selected without binding. The resolution gate excludes 16 of 1,536 cells, because in those the narrowest fitted mixture component is under two grid cells wide; including them changes no ratio by more than 0.08. The network wins in 2 of the 1,520 reported cells, both at the smallest sizes — Appendix B resolves where the advantage is largest.

What Table 10.3 covers. The confirmatory run scores three regions per cell — all, bulk, centre — of which the table displays two; the third is the predeclared interior slice, and it is the one the architecture screen selects on. That is why the 192 scored cells are $16 \times 4 \times 3$ and not the $16 \times 4 \times 2$ a two-column table suggests. The count is a descriptive diagnostic and not a replication count: regions are a scoring choice applied after fitting, and sizes share nothing but the seed, so the inferential unit behind each displayed row is the 16 paired seeds. Both arms are checkpointed and selected on a validation bundle disjoint from the test bundle, and neither bundle is shared with training.

The architecture screen. Table 10.2 reports it in full. Its source is `screening.csv` under `outputs/frozen/exp_31_screen/` at commit 569a67c, which is the artifact the confirmatory run names in its own recorded configuration (`winners=auto:` that path) rather than a reconstruction. 32 configurations over 4 development seeds, at two training sizes, with learning rates 0.001, 0.003, both output parameterisations, and a checkpoint ladder from 500 to 64000 steps shared across all three architectures. Selection is on mean risk over the development seeds, scored on the bulk region; the development seeds are numbered from 10000 and share no seed with the confirmatory set, so the reported interval is not a selection artefact. The winner is the window head at radius 4, width 64, learning rate 0.003, eps parameterisation. Parameter counts across the grids are 4.8k–21.4k (window), 9.0k–34.3k (message passing) and 66.4k–264k (dilated convolution): the arm with the most capacity placed last, which is the relevant check on whether the comparison was tuned evenly.

The convergence sweep, and the one exception to the gate above. The settling times of Section 9.1 come from `outputs/frozen/exp_27_*`, run on 16 August 2026 under SLURM job 628601 and its shards. That sweep predates `hpc/deploy_clean.sh`, so its records carry the job id, host and resolved configuration but no source-archive digest, and `provenance_gate` refuses it by default; `tools/make_convergence_numbers.py` passes `allow_legacy` and this paragraph is the disclosure that goes with it. The quoted numbers come from the 800-iteration shard, the only run length in the sweep at which no configuration settles at its own cap; the 400-iteration shard is censored by its cap and its settling times are lower bounds, so it is used only for the count of configurations that miss a 40-iteration budget, which censoring cannot inflate. The measurement replaces a figure of “on the order of 2000 iterations” that appeared in earlier drafts and was never measured: it had been carried over from a diagnostic that ran a fit *out to* 2000 iterations to show its per-edge gain had plateaued.

The transition array’s normalisation. Stated in Section 8.4 and repeated here because it determines which probabilistic object the M-step optimises: columns are *not* renormalised under the quadrature weights, so the fitted object is the continuum kernel read through a quadrature rule and the surrogate carries no θ -dependent log-normaliser. The column-mass residual is the measured distance to a normalised finite-state kernel.

Excluded inputs. $N = 8192$ is excluded from both Table 10.1 and Figure 10.1: the intersection of its dirty list with `exp_07`’s import closure contains the experiment, the frozen configuration and `src/em.py` themselves, so the run is not recoverable. The $N = 4096$ rows are retained: their intersection is empty, so no edit in that tree could have moved those numbers.

A.3 Environment

Python ≥ 3.10 with NumPy, SciPy, pandas and Matplotlib. The neural baselines of Chapter 10 are implemented in pure NumPy (`src/nnet.py` for the fully connected de-

noiser, `src/seq_nets.py` and `src/local_head.py` for the locality-respecting architectures) with an explicit `numpy.random.Generator` driving initialisation and minibatch order, so training is stochastic but reproducible from the recorded seed. Two optional dependencies are used and neither is on the exactness path: CuPy, selected by the `BP_DEVICE` environment variable, runs the same single BP implementation on a GPU, `tests/test_backend_parity.py` asserts the two devices agree, so a device change fails a test rather than moving a number quietly, and PyTorch, imported lazily in `src/fid.py` alone, for the generative metric. Arithmetic is float64 throughout.

The sweeps of Chapters 8 and 10 run under SLURM from the batch scripts in `hpc/`, one per experiment. The document compiles with a single run of `tectonic main.tex`.

Appendix B

How the Reported Ratios Were Aggregated

A ratio is a nonlinear function of two means, so it depends on the order in which the averaging is done: $\mathbb{E}[X]/\mathbb{E}[Y]$ is not $\mathbb{E}[X/Y]$, and the gap between them grows with the spread. Both ratios reported in this thesis therefore need the same question asked of them, namely whether the number is a property of the estimator or of the estimand chosen to summarise it.

Both were recomputed under four alternative aggregations of the same frozen cells, refitting nothing. The outcome, in one sentence each. For the unstructured comparison the direction is the same under every aggregation and the estimand reported in Chapter 10 is systematically the largest of them, which is stated wherever that number appears. For the structure-aware comparison the reported estimand is *not* uniformly the largest, but the floor across all four aggregations is $1.98\times$ with a spread of 0.48, so the reading of a low single-digit multiple at the smallest training set growing to roughly six at the largest survives every one of them.

The full recomputation, both tables and the protocol behind them, is Chapter 19 of the companion compendium. It is generated by `tools/make_aggregation_robustness.py` from the same frozen outputs that produce the headline table, so the two cannot drift apart.

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